

Central Bank Independence- Measuring its Effectiveness during Ordinary
and Extraordinary Times: Evidence from Selected Developing and
Developed Countries

A Doctoral Dissertation

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Abstract

The idea of central bank independence has been widely accepted over the last several decades by many countries around the world, both developed and developing. As such, many countries around the world granted autonomy to their central banks during the 1980s and 1990s. The majority of past studies have examined primarily the impact of central bank independence on inflation. However, the additional theoretical benefits are much more far reaching, the result of a more stable and prosperous macroeconomic environment. Additionally, there is only now sufficient data to empirically determine whether many of these claims are true. This study examines central bank independence in developing countries of Latin America and Asia as well as selected developed countries to determine what actual impact an autonomous central bank has had. It also examines such phenomena as financial crises (including the current global crisis of 2007-2009), inflation targeting, legal systems, country development and fiscal policy to determine the effects of these items on not only inflation, but the broad spectrum of macroeconomic outcomes.

Keywords:

Central Bank Independence, Currency Crisis, Latin America, Asia

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Abbreviations and Nomenclature

CB	Central Bank
CBI	Central Bank Independence
CBGov	Central Bank Governor Turnover
Cons	Government Consumption (as a % of GDP)
DD	Development Dummy
FCD	Financial Crisis Dummy
FXD	Exchange Rate Regime Dummy
GMT	Grilli, Masciandaro and Tabellini Index
IID	International Inflation Dummy
IT	Inflation Targeting
LCBI	Legal Central Bank Independence
MCI	Modified Cukierman Index
OCBI	Operational Central Bank Independence
TaxRev	Tax Revenue (as a % of GDP)

Chapter 1- Introduction

The term “central bank independence” (or abbreviated, CBI) can be broadly defined as the degree of freedom of the central bank to pursue monetary policy without interference from political considerations (Sirivedhin and Hataiseree, 2000). In other words, independence implies that a nation’s central bank can follow a trajectory outside the political realm and does not have to abide by the government’s request to follow a certain monetary policy, such as printing money to pay for federal deficits (Gruben and Welch, 1993). Since being first written about academically in the late 1980s, many countries have come to adopt this policy and many governments have come to recognize this as standard procedure. From the late 1980s to the mid-1990s, countries as diverse as Malaysia, New Zealand, Chile, Mexico, Argentina, Spain, France and Brazil granted their respective central banks more autonomy from the government (Cukierman, 1994). However, despite this proliferation, highly independent central banks in many countries of the developing world are a relatively new phenomenon (Boylan, 2001).

The idea of granting a nation’s central bank autonomy from the government, although a relatively simple one at first glance, is powerful in the sense that if a central bank is independent in its decision making abilities, the government in power at a given time will be unable to turn back monetary decisions that have been made or, in a similar vein, force actions to be undertaken. Thus, in theory, an independent and autonomous central bank should lead to a more stable economic environment in the respective country (Maxfield, 1997). It represents a prominent transformation, because it not only removes a key aspect of economic decision making from political control but has kept many

governmental powers from overturning growth-oriented central bank policies (Boylan, 2001).

The primary motive behind the granting of such autonomy was the belief that an independent and sovereign central bank would have more power to diminish inflation and price volatility and thusly, economic growth would be stimulated. According to Cukierman (1994), there is widespread consensus now that central banks should be independent because it helps to achieve price stability. This belief stems not only from monetary theory but also from empirical studies on the subject, both in developed and developing countries. The traditional argument for central bank autonomy is that the power to spend money (the government) should be separate from the power to print money (the central bank).

Although there has been much study done on the theoretical rationale behind it, few empirical studies, other than those examining the effects of inflation, have been conducted to assess accurately whether central bank independence actually leads to desirable macroeconomic outcomes. That is, although many countries have granted autonomy to their respective central bank with the hope of not only taming inflation and inflation volatility, but also achieving a broader spectrum of macroeconomic goals, such as low unemployment, sustained GDP growth, increased investment, less volatile interest rates and further financial market liberalization, many of these hopes have not been proven empirically, especially in the developing world.

The focus of this study will be to examine a select group of countries in three defined regions (Asia, Latin America, and the Developed West) and identify whether macroeconomic variables such as inflation, unemployment, interest rates, exchange rates,

direct investment, exports, savings rates and economic growth have responded positively since the granting of central bank autonomy. Some of these variables have been examined indirectly in previous studies, but many of the results, in particular regarding developing countries, have been inconclusive. Thus, one of the primary contributions of this study is that it will directly examine the effects of central bank independence on these principal macroeconomic variables over a longer term period (since 1960 in many cases), analyzing both the pre- and post-central bank independence periods. Another important contribution is that it also recognizes the heteroscedastic (variance in errors) nature of the countries' time series data and utilizes a different form of regression model, that of Generalized Autoregressive Conditional Heteroskedasticity (i.e. GARCH), that is designed to better treat time series of this nature versus Ordinary Least Squares (or some variant of OLS). Once correlation is established, causality (which is a stronger relationship) will be examined as well.

Another of the unique contributions of this paper is that it will examine the phenomena of financial crises as they relate to central bank independence, including the current global crisis of 2008-2009. It will not only examine the effects that financial crises have had on countries, but will also examine countries such as Mexico that experienced this type of crisis both before and after their central bank was made independent in order to determine what effect (if any) this independence had on the lessening of severity of a given crisis. It is important to attempt to determine what impact, if any, central bank independence has had on lessening the severity of crises, as it is perhaps during these types of extraordinary circumstances that an independent central bank is most needed.

Again, it will use a relatively long time series and utilize some of the most current data available (through the first quarter of 2009) to assess accurately the impacts of central bank independence. Once more, many countries only granted some degree of autonomy to their central banks within the last 10 to 15 years, and it is only now possible to measure the potential gains the effects that central bank independence has afforded with some degree of statistical certainty. Additionally, the effects of central bank independence may not be immediate (Jácome and Vazquez, 2005); therefore a certain amount of post-independence time may be necessary to evaluate accurately the effects.

The countries that will be studied (Mexico, Argentina, Brazil, China, India, Thailand, South Korea, Malaysia, England and the United States, among others) were chosen because they are representative of the countries in different regions of the world that have undergone central bank independence to varying degrees and also because they have undergone financial crises during the 1990s or early 2000s. China and India were selected specifically because they are considered emerging economic powers, are the most populous countries in the world, and have not been studied in any depth in this area. England was selected for study because it recently (1997) granted autonomy to its central bank, is one of the world's most advanced economies, and has not yet had its impact studied. The United States' Federal Reserve (its central bank) was chosen because it is arguably the most influential central bank in the world and has been highly operationally independent for some time now.

The following table illustrates the major contributions of this study.

Table 1.1- Summary of Major Contributions

1. It will examine not only inflation, but also the broader spectrum of macroeconomic outcomes.
2. It will examine representative countries from three regions: Asia, Latin America and the Developed West (as well as an Overall Sample of all countries represented).
3. It will observe in detail both the pre- and post-central bank independence periods.
4. It will observe financial crises as they relate to central bank independence, including the current global financial crisis of 2008-2009.
5. It will examine such institutions as legal systems, corruption, the system of “checks and balances” and fiscal policy and quantify their effects on central bank independence.
6. It will include the U.S., England, China, and India (among others), which are unique cases, major global influences and have been studied relatively little up to this point.
7. It will use a very detailed time series type analysis (versus purely cross sectional).
8. It will use a more adequate, newer form of regression model (GARCH) to analyze the time series data, which takes into account the shocks and variances over time.
9. It will establish not only correlation, but also causality between independent and dependent variables.
10. It will use a relatively long time series (almost 50 years) and will take advantage of some of the most current data available (through Q1: 2009).

Given the topic background and this author’s potential contributions, the primary research question is the following:

Table 1.2- Research Question

To what degree is a responsible, non-partisan central bank essential for the macroeconomic well-being of an economy? Does it have to be independent (legally and/or operationally) in order to achieve the broad spectrum of desired economic outcomes for the respective country? How do other legal, political and institutional circumstances in the country influence a central bank’s performance?
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This paper is organized as follows: Chapter Two will provide an extensive review of the literature on and theory behind the subject of central bank independence. Chapter Three will provide hypotheses. Chapter Four will cover methodology and data. Chapter Five will provide the results from the previous section. Lastly, Chapter Six will provide conclusions. In the Appendices, there is information related to the dates of central bank independence in the sample countries, years in which financial crises were experienced, and tables listing country groupings for different categories used in hypotheses.

Chapter 2- Literature Review

2.1 Central Bank Autonomy- An Overview

2.1.1 Motivation for Autonomy

Among both developed and developing countries, there has been significant controversy regarding central bank independence. However, despite this debate, between 1990 and 1995 at least 30 countries all over the world (including five from Latin America) granted independence to their respective central banks. This is due to most countries' central banks and finance ministries understanding and accepting that high inflation is detrimental to growth and that price stability is one of the central banks' primary functions (Maxfield, 1997). Central bank reforms have been focused in several particular areas. One is the rewriting of legal statuses regarding central banks' operations and relations with other branches of the government. Another is the issue of central bank accountability, and the third is the issue of increased central bank transparency (Crowe and Meade, 2008).

Overall, one of the main factors motivating central bank reforms and the granting of autonomy by their respective countries' governments has been the belief that central banks' independence may be an institutional mechanism that leads to price stability, thus increasing the credibility of policymakers' commitments to stable prices (Diana et al, 2004). Furthermore, an independent central bank has the freedom to implement monetary and exchange rate policy in order to achieve its objective. Additionally, a clear definition and prioritization of the objectives of the central bank enhances its credibility both domestically and internationally (Gutiérrez, 2003). However, it should be noted that central bank autonomy is in many cases part of a broader "package" of reforms that many

times includes judicial reform, privatization, removal of trade barriers, etc. (Acemoglu et al, 2008).

2.1.1.1 Early History and Global Events

Although the idea of central bank independence may seem new, perhaps it is not. Independent central banks have been around since the interwar period of the earlier part of the 20th Century, although not much was studied about them nor were the independent central banks of the time considered exceptional. Beginning in the postwar era of the early 1950s, more began to be written about them, and the idea had its supporters and detractors (Forder, 2005). However, it was only in 1988 that Alesina stated, “Independent central banks have been associated with a lower average inflation rate and may have been responsible for reducing politically induced volatility of monetary policy and inflation” and that the idea began to be accepted as common policy (Alesina, 1988). One of the early success stories was that of New Zealand, which granted its central bank autonomy in 1989 and achieved a considerable decrease in inflation rates (Carlstrom and Fuerst, 2006). By 1997, with the granting of independence to the Bank of England, the transition could be considered complete and Alesina’s suggestion just ten years prior could be considered an established finding.

There are several specific reasons that the concept of central bank independence only began to take hold beginning in the late 1980s and into the 1990s. First, the highly independent central banks such as the German Bundesbank and the Swiss National Bank produced very good long-term results in achieving price stability. Their success in controlling inflation for relatively long periods of time gave initial impetus to other countries to grant autonomy to their own central banks. Second, the rampant inflation of

the 1970s in many countries around the world led many political and monetary policy leaders to search for a solution to the ravaging effects of high inflation. The episodes of hyperinflation in Latin America in the 1980s (greater than 1000% per year in many cases) caused these countries to look for mechanisms that would quell the recurrence of this monetary malady, and a more independent central bank was seen as a viable alternative. Third, the desire of the breakaway republics of the former Soviet Union to transform their economies into market style economies also led to further interest in the granting of autonomy to respective country central banks (Cukierman, 1994).

2.1.1.2 Monetary Policy and Factors

Central banks formulate monetary policy, affect exchange rates, act as the gatekeepers of financial stability and are responsible for economic variables crucial for growth and development. Monetary policy, which describes effort to control the demand and supply of money (through interest rates), can influence economic variables such as inflation, unemployment, and growth. In fact, this type of policy is one the greatest influences on inflation and interest rates (Lin et al, 1994). There is now widespread consensus that the main objectives of monetary policy should be price stability, an interest rate set without political interference, and transparent procedures on the part of the central bank (Cukierman and Walsh, 2002). Moreover, one of the most widely accepted tenets of monetary theory is that persistent inflation is a monetary phenomenon (Cukierman, 1986). Additionally, central bank independence tends to grow when countries realize that the central bank is efficient and effective at helping to shape monetary policy (Grilli et al, 1991).

It also appears that for many developing economies, the nature and magnitude of central bank independence is influenced to a great extent by how much the respective government is relying on its central bank for monetary stabilization (Bagheri and Habibi, 1998). The commonly held view that unstable macroeconomic policies inhibit economic growth evolved from the experiences of Latin American economies in the 1970s and 1980s, and more recently by the countries of Eastern Europe and the former Soviet Union. This view has been tested by adding inflation or inflation volatility as regressors in growth equations and is found to have a significant negative impact on growth (Cukierman et al, 1993).

2.1.1.3 Fiscal Policy and Factors

Fiscal policy is the level, composition and change of a government's taxation and expenditure (Stewart and Venugopal, 2009). It can essentially take two forms—expansionary or contractionary. Expansionary fiscal policy can be represented by an increase in government demand, a cut in taxes, or a combination of the two that raises aggregate demand in the economy. This is usually put in place by a nation's central bank to stimulate the economy and would tend to cause a currency appreciation and a rise in output. On the other hand, a contractionary policy would be represented by a decrease in government demand or increase in taxes (or some combination of the two) and would be done when the economy, in the central bank's opinion, is in danger of “overheating” (Krugman and Obstfeld, 1994).

Supporters of expansionary fiscal policy state that an increase in government spending could stimulate long term growth through ensuring efficiency in resource allocation, regulation of markets and stabilization of the economy as well as the

government's role in promoting accumulation of knowledge, research and development, productive public investment, human capital development, and law and order (all of which can generate growth both in the short- and long-run). Opponents hold the view that government operations are inherently bureaucratic and inefficient and therefore suppress rather than promote growth. (M'Amanja and Morrissey, 2005). In practical terms, it is much easier to change monetary policy than fiscal policy, as fiscal policy requires going through lengthy legislative delays. Monetary policy, on the other hand, is easily changed by the central bank and can thus be used when more rapid changes are required, even when a change in fiscal policy may be more appropriate (Krugman and Obstfeld, 1994).

There are two macroeconomic reasons why government spending can undermine economic performance. The first reason is "resource displacement". In other words, when a government spends money, it is using capital that is then no longer available for private sector use. The second reason is the "financing cost." When a government taxes, it not only takes money from the productive sector, but it also raises revenue by means of a tax system that generally reduces incentives to work, save, and invest (Mitchell, 2005). Although taxation is particularly important for addressing economic inequalities through raising additional revenues for programs designed to reduce these inequalities, tax policies in developing countries are many times weak in terms of revenue mobilization and frequently fail in dealing with income disparity (Stewart and Venugopal, 2009).

Brazil is a good example of the difficulties involved in the trade-offs between government consumption and tax revenue and the complexities involved. It has one of the highest levels of taxation in the developing world and the highest in Latin America. It originally increased taxes as a result of the doubling in public consumption as a share of

GDP in the 1980s. Despite this extremely high level of taxation, public investment is one of the lowest of developing countries, with pensions and public debt interest consuming much of the government budget. One of the more feasible solutions to this problem would be to increase the domestic savings rate and gradually reduce the long term interest rates through a normally functioning credit market. However, the process is difficult, the tradeoffs are many and any negative consequences would be difficult to reverse (Hausmann, 2008).

Central bank independence, as a credible mechanism to price stability, may contribute to monetary and fiscal discipline by constraining the spending decision of the government (Neyapti and Ozgur, 2007). However, binding commitments to monetary rules are not necessarily welfare improving if monetary and fiscal policies are not coordinated (Alesina and Tabellini, 1987).

2.1.1.4 Institutional Factors

Institutions play a fundamental role in determining the course of economic policy. However, strong institutions that are free from political influence such as a country's central bank have traditionally been absent in much of the world (Edwards, 1995). In general, independence for central banks consists of three areas: the ability of central bank governors to refuse to monetize public debt, the security of tenure of the central bank governors and the board members (which helps to impede government demands of a specific policy at a given time) and price stability in policymaking (Posen, 1998). Additionally, central bank independence is shown to be positively correlated with political freedom and political stability (Bagheri and Habibi, 1998).

An independent central bank can follow non-inflationary policies regardless of what the government's mandates and objectives are. Although inflationary policies can sometimes stimulate growth in the short-run, in the long-run they are detrimental to a nation's economy due not only to rising prices but also to price volatility (Gruben and Welch, 1993). In practice, central bank independence tends to follow one of two models--*goal* independence, in which the central bank has the autonomy to follow its own prerogatives, or *instrument* independence, in which the central bank only has the autonomy to set an instrument into place to pursue the mandate that the government has deemed appropriate (Crowe and Meade, 2008).

2.1.1.5 Credibility

Central bank independence sends a strong signal of creditworthiness to potential investors (Eijffinger and de Haan, 1996). For a central bank, attaining credibility is eased if the public can easily determine the financial strength of the central bank, yet for a variety of reasons this is often difficult. Banking crises elevate to an even greater degree the value of a central bank being able to credibly demonstrate its power to foster financial stability. There is a wide variation in the financial strength of central banks around the world. However, the European Union has stressed the importance of financial strength and independence as a prerequisite to any country wanting to join the EU and adopt the Euro. Credibility essentially involves three elements: adopting the right objective function, enabling the attainment of the objective, and being transparent (Stella, 2002).

Central bank independence positively influences the speed of disinflation and thusly a perception that government is committed to inflation reforms (Diana et al, 2004). Therefore, the government's degree of openness in identifying the objectives of the

central bank along with the degree of power afforded it establishes the extent of the government's commitment to a given policy rule (Eijffinger and de Haan, 1996). Additionally, the granting of autonomy to a nation's central bank will likely decrease inflation by increasing the credibility of a central bank's commitment to price stability. Under this assumption, if a central bank's credibility regarding its price stabilization commitment increases, then decision makers in the private sector have less reason to alter their behavior to protect themselves from government induced inflation surprises. In other words, enhanced credibility should reduce the distance between expectations and reality (Posen, 1998).

Interestingly, as a country becomes more open to the international economy, it is possible that a leftist government is more likely to grant autonomy to the respective country's central bank. This is because central bank independence is regarded as a sign of credibility, and leftist governments many times need to establish credibility in the international community and demonstrate their commitment to neoliberal economic policies. In other words, promises made by central bank governors from independent central banks are more likely to be believed by international investors and the international financial community, which is even more important if there is a left-leaning government in power at the time (Hicks, 2004). Many of the countries examined in this study have been or currently are run by leftist governments, thus making central bank sovereignty all the more significant.

2.1.1.6 Transparency and Corruption

Transparency of central banks has been defined as the degree of genuine understanding of the monetary policy process and policy decisions by the public

(Winkler, 2000). One of the major benefits of increasing the autonomy of central banks is that its actions become more transparent to those observing them. Compared with past decades, there is substantially more transparency now in central bank operations as they relate to monetary policy. To illustrate, in most central banks price stability is by far the most important objective with all other objectives (such as employment, growth, etc.) being secondary to varying degrees (as was the case with the Bank of England Charter of 1997). This represents a vast increase in transparency over previous years because for much of the 20th Century central banks operated with no clear indication of priorities (Cukierman and Gerlach, 2003). Although central banks have traditionally been associated with secrecy, as a central bank becomes more independent, it needs to be more accountable for its actions (Fraser, 1994).

2.1.2 Empirical Studies

2.1.2.1 Inflation and the Money Supply

Most of the 20th Century was characterized by positive inflation rates in most countries of the world. The most common rationale for this phenomenon is the interaction between policymakers and a rational public within the context of the Phillips curve. This concept is based on the idea that monetary policymakers are concerned with both employment and price levels, but that their preferred level of employment is above the natural level. This is most likely due to the political pressures of maintaining relatively high employment and the shorter term perspective that this would prompt, which tends to cause an inflation bias (Cukierman and Gerlach, 2003).

Some scholars, therefore, have argued that governments and central banks are more tempted to impart an inflationary bias to the economy, thus sacrificing long-term

economic benefit to short-term political gain. However, this bias can be overcome by establishing an autonomous central bank that does not respond to myopic political pressures (Eijffinger et al, 1996). Additionally, the more authority a central bank has in setting interest rates and money supply growth, the less likely it will be for politicians to be able to use these mechanisms for their own means and not of the overall goal of achieving price stability. The central bank, as the fiscal agent for the federal government, has the varying ability (depending on its independence) to impose fiscal restraint on the government (through its ability to make/refuse to make unsecured loans from its reserves and print/refuse to print currency notes). The extent of independence of the central bank has its greatest impact on its ability to control the money supply, and thereby limit the rise of inflation. In other words, it is difficult to control the money supply if the government makes excessive demands on the central bank to print money that the bank feels it cannot reject. Thus, the argument is that an independent central bank would have more power and be more secure in its ability to refuse these types of actions (Maxfield, 1997).

Moreover, the theoretical literature has emphasized three primary arguments to explain why the delegation of monetary authority to an autonomous central bank would assist in lowering inflation, *ceteris paribus*. The first argument is that an independent central bank will be able to resist government pressure to relax monetary conditions. The second is that an independent central bank will not as easily yield to government pressure to finance government deficits by printing money and will most likely urge the government to actually reduce the deficit. The third argument is that an independent central bank will be more likely to take a longer term view of the economy compared to

the government (Gutiérrez, 2003). To put it in few words, central bank autonomy acts as a constraint of the fiscal authority's behavior (Carlstrom and Fuerst, 2006).

There were several earlier empirical studies, such as Bade and Parkin (1985) and Alesina (1988, 1989) that successfully linked central bank independence with lower levels of inflation. However, Bade and Parkin (1988) later examined central bank independence and inflation levels for ten countries but found no relationship. Cukierman et al (1992) and Cukierman et al (1993), however, did find a negative relationship between central bank independence and inflation and inflation volatility in developed countries but not in developing ones. The common rationale for this is that the rule of law is traditionally not respected to as great a degree in developing countries. However, Gutiérrez (2003) proposes that the constitution of a country is more likely to be respected than ordinary statutes governing central bank independence. Some countries not only granted autonomy to their central banks but also went so far as to entrench it in the constitution. The study concludes that those countries that have central bank independence entrenched in their respective constitutions (as do many in Latin America) have better inflation performance than those that do not.

More recently, Kuttner and Posen (2001) and De Souza (2002) measure the impact of central bank autonomy on inflation and exchange rates and determine that autonomy is associated with lower inflation. Jácome and Vasquez (2005) note that higher central bank independence is related to lower inflation in Latin American and Caribbean countries. Hicks (2004) confirms that central bank independence is associated with lower levels of inflation, at least in developed countries. In Ahsan et al (2007), the authors use

an index of central bank independence and governance and find there is a strong relationship between inflation and central bank independence in the Asia Pacific region.

Although many empirical studies from the past few years have found a statistically significant correlation between CBI and inflation, proving causality (i.e. that CBI actually causes a decrease in inflation and/or inflation volatility) is much more difficult. To illustrate, Jácome and Vazquez (2005), using panel regressions, find a negative relationship between legal central bank independence and inflation. The results hold even after controlling for international inflation, banking crises, and exchange rate regimes. However, the authors are unable to find a causal relationship running from CBI to inflation. Furthermore, the effects of central bank reform on inflation may not be immediate, since the implementation of a new institutional environment usually requires the enactment of more specific regulations and time for the newly reformed central bank to develop a reputation with market participants (Jácome and Vazquez, 2005).

Thus, there is now a commonly held view that an independent central bank should bring about lower inflation rates along with more inflation stability. However, the greater debate focuses on whether this inflation benefit comes at a cost. It is possible that an independent central bank does indeed lower inflation, but that greater output variability occurs as a result, as demonstrated by Cukierman and Gerlach (2003). However, Alesina and Gatti (1995) had previously refuted this. They prove that the institution of an independent and inflation-averse central bank has two benefits--it reduces inflation and eliminates politically induced output variability, since monetary policy is not under the direct control of governments with changing preferences. In other words, an independent

central bank can achieve both lower average inflation and lower output variability (Alesina and Gatti, 1995).

Lastly, central bank independence seems to perform better as a safeguard against recurring inflation rather than as a tool to control initial high inflation. According to Cukierman (1994), experience has shown that in high-inflation economies, the taming of the inflation level can only be achieved by the involvement of the entire political establishment. However, once inflation has been conquered, central bank authority can function as a preventive device against future episodes of high inflation.

In sum, many central bank independence studies in the 1990s and early 2000s involving both developing and developed countries showed a negative relationship between inflation and central bank independence to varying degrees, and some studies also confirmed this relationship with the variance of inflation as well as other macroeconomic benefits (a topic which will be studied in the following section). For developing countries, however, there still remains some controversy regarding the basic correlation between central bank independence and inflation (Cukierman, 1994).

2.1.2.2 Interest Rates

For the achievement of price stability (i.e. controlling inflation), effective constraints on monetary policy must be in place. Policymakers are subject to inflationary bias because control over monetary policy allows them to have power relatively quickly over such economic conditions as high employment, financing of the budget deficit and low interest rates. However, in many cases inflation and inflationary expectations are created as a result which persists long after the desired effects of monetary expansion have dissipated (Cukierman, 1994). Therefore inflation and interest rates are tightly

intertwined in the central bank independence framework. Moreover, according to the Fisher Effect, over the long term, inflation rates and (nominal) interest rates should move in tandem, leaving the real interest rate unaffected (Mitchell-Innes et al, 2007).

The theoretical origin of central bank autonomy comes from the “Rules vs. Discretion” debates. This is demonstrated in Eijffinger et al (1996), who use panel data estimation techniques to determine how differences in central bank independence are manifested through different degrees of structural pressure to raise or lower money market rates. They state that a central bank that is less independent will have an incentive to lower the money market rate to stimulate economic growth (and thereby employment). Interestingly, they conclude that a more dependent central bank should have relatively lower money market rates than more independent central banks. However, Fischer (1995) refutes this by stating, “The empirical evidence shows not only that greater independence is associated with lower inflation, but also that the central bank’s rights not to finance government and set interest rates independently increases its effectiveness.”

Regardless of the monetary regime in place at the time, central banks tend to use interest rates as a tool to curb inflation, control economic activity, and influence exchange rates. However, central banks that have adopted such exchange rate regimes as exchange rates crawls use interest rates as a device to defend the peg and protect international reserves in times of crisis (Jácome and Parrado, 2007). Moreover, with the adoption of an operational inflation target, central banks also seek to influence the long-term behavior of interest rates charged by all members of the financial community in the country in question. In turn, central banks expect that interest rates will affect aggregate demand and supply of money, and thus inflation. Therefore, the operational inflation

target has become a key factor in the monetary policy transmission mechanism (Carstens and Jácome, 2005).

2.1.2.3 Exchange Rates

The degree of independence granted to the central bank is a decisive factor in the ability of the government to pursue exchange rate targets (Baines, 2001). Grier (2004) finds that central bank independence reduces the average rate of real exchange rate depreciation and reduces uncertainty about future values of the real exchange rate. Additionally, central bank autonomy is associated with a more stable exchange rate (Kuttner and Posen, 2001). Moreover, a central bank's exchange rate policy is one of the most profound influences on inflation, interest rates and the balance of payments (Lin et al, 1994).

Sintim-Aboagye and Chakroborty examine the possible effect of the inverse relationship between measures of central bank independence and the level and stability of inflation of the long-run relationship between prices and exchange rates. They conclude that more evidence of the cointegration of prices and exchange rates should be found in low independence countries than in high independence countries (Sintim-Aboagye and Chakroborty, 2007). In 1990, seven of the 18 countries in Latin America had a flexible exchange rate, reducing to only five in 1995. All other countries had a pegged or hybrid system. Thus, it seemed that a majority of Latin American banks had given up, at least in part, their ability to conduct monetary policy. Unfortunately, the transition back to exchange rate flexibility has been a painful one. It took place amid financial crises that made initial exchange rate regimes unsustainable and led to swift and massive devaluations in many countries. Thus, a key challenge facing most central banks in Latin

America is to restore markets' confidence in the national currencies (Carstens and Jácome, 2005).

2.1.2.4 Foreign Direct Investment

The growth of both inward and outward FDI flows in Asia has been facilitated by various policy changes at the national and regional levels since the 1990s. For example, the Association of Southeast Asian Nations (ASEAN) and China signed an agreement to establish a free trade area by 2010, and several Asian countries have recently signed free trade agreements with the United States. Additionally, in 2004 China was the largest developing country recipient, receiving \$61 billion in FDI inflows while India received \$7 billion (World Investment Report, 2005).

However, for many years, foreign direct investment played a minor role in Latin America. For most of the 1970s, it amounted to less than 1% of GDP per year. Much of this was due to legislation that made it difficult for foreign enterprises to invest in the respective countries of the region. However, after a series of measures enacted in the 1980s and early 1990s in the region, more foreign direct investment began to flow into countries such as Mexico, Brazil, Venezuela and Chile. In fact, Brazil ranked third worldwide in FDI inflows in 2004 (World Competitiveness Report, 2005). As may be expected, though, the amount of capital inflow was not uniform across the countries of the region (Edwards, 1995).

In general, Latin America has experienced three types of market reforms which have had an impact on the amount of foreign direct investment entering into the countries of the region--microeconomic, macroeconomic, and institutional. Microeconomic type reforms have, to a large degree, decentralized economic decision making in the region by

shifting it from the state to the private sector, so that market forces drive competition. Macroeconomic reforms have mainly sought to reduce inflation and stabilize exchange rates. Institutional reforms have changed the state's role from producer to facilitator, so that the private sector is encouraged and empowered to make investments. While competing for foreign direct investment by offering tax and other incentives can sometimes be effective in attracting investors, improving the quality of a country's institutions appears to have much greater impact (Maniam, 2007).

One of the most important determinants of the distribution of foreign direct investment across countries is the soundness of economic policies. Foreign investors tend to avoid countries with major distortions and controls and are attracted to those countries with consistent and predictable macroeconomic policies. Countries that have policies emphasizing greater openness and competitiveness, as well as a reduction in the size and influence of the government, are seen as more desirable candidates for foreign direct investment (Edwards, 1995).

All else equal, international investors should be more willing to invest in a country with an independent central bank for two reasons. First, it is believed that a central bank with authority should be able to keep a country's economy on a steady, stable course. Second, a primary motivation for the CBI trend during the 1990s across the globe was that governments were attempting to attract and retain capital. Government incentives to attract foreign investment, and the perception that an independent central bank was one tool that governments could use to this end, led to the rapid transition to independence in many countries. The level of actual central bank independence in developing countries tends to be correlated with the respective government's perceived

need for credit and investment, with there being a positive relationship between central bank independence and private investment. Thus, politicians are more likely to delegate monetary policymaking to the central bank the more they value private investment in the country. International investors also tend to dislike arbitrary power in the hands of a government and value central bank autonomy (Maxfield, 1997).

In sum, central bank independence has been shown to help create a more stable macroeconomic environment. It has also been one of the most important institutional reforms of the last several decades, essentially removing the central bank from the hands of government and removing it from political influence. This would also assist in creating a more predictable and consistent environment for multinational companies on which to base their investment decisions. Therefore, central bank autonomy is possibly an extremely important determinant in the amount of foreign direct investment a country receives.

2.1.2.5 Savings Rates and Capital Markets

The financial system is essential to any economy because it provides for resource allocation. Financial systems allow people to increase savings rates and enable risk sharing (Naceur et al, 2007). Thus, capital markets allow individuals, firms and countries to finance spending in excess of their current incomes. They also allow these entities to lend savings they could not otherwise employ profitably themselves (Khamfula, 2005). Additionally, savings rates have shown to be an important determinant of stock market development (which could be considered the “next step” in the development of a nation’s capital market). Furthermore, measures of the size and/or liquidity of the banking sector is shown to be highly correlated with growth of GDP per capita and seem to have a

causal impact on economic growth (Beck et al, 2000). Banks tend to dominate the financial systems of many countries, even developed ones (Naceur et al, 2007).

Financial markets also greatly value stability and effective regulation. Thus, ever more sophisticated financial markets increasingly necessitate central bank independence. It seems that the most effective manner for achieving central bank independence is to focus first on financial market development. In fact, it is difficult for the central bank to conduct any of its basic functions if a country's financial markets are not robust (Maxfield, 1997). Additionally, governments play an important role in building effective financial systems (Demirgüç-Kunt, 2006).

The transition from a state controlled economy to a market economy (which has been the case in a number of the sample countries of Asia and Latin America) necessitates the development of an appropriate financial infrastructure. Privatized firms will need more funds for modernization; and new firms will, in most cases, need more funds than the owners can supply by themselves. If stock markets are to function properly, appropriate accounting, legal and regulatory frameworks must be established. Among the prerequisites of a well-functioning equity market are an efficient clearing and settlement system, a stable currency, and adequate custodial facilities--in sum, a sound banking system (Mendelson and Peake, 1993).

Wider capital markets also lead to higher growth, as the initial level of a country's financial development is a very reliable predictor of future growth rates. To the extent that central bank independence and financial development are positively related, this provides another channel through which central bank independence and growth interact. Thus, broad capital markets may enhance growth both directly and indirectly by

leading political authorities to delegate more authority to the central bank (Cukierman et al, 1993).

2.1.2.6 Employment Levels

In addition to inflation considerations, policymakers also believe that an independent central bank can assist in fine-tuning employment levels. Alan Greenspan, chairman of the U.S. Federal Reserve in the 1990s, stated “the independence of central banks is an element in keeping inflation down, and just as importantly, the lower the rates of inflation, the higher the growth rate in productivity”. One of the most common arguments against central bank independence is the claim that it is not social-welfare optimizing by leading to economic policy that is less employment-promoting. However, the cost of macroeconomic instability brought about by subordinating the central bank to the government likely greatly outweighs any potential employment loss (Maxfield, 1997). Additionally, delegation of authority to a conservative central bank reduces inflationary tendencies without having any negative impact on employment, and is therefore welfare improving (Cukierman and Lippi, 1999).

2.1.2.7 Exports

Most central banks hold foreign exchange reserves to help lessen the effects of external shocks and help restore the external competitiveness of its exports caused by a financial crisis. Central banks can also play a more active role in promoting exports without jeopardizing price stability through sound fiscal policy (Lin et al, 1994). To the extent that there is a stable foreign exchange rate for the nation’s currency, exports will increase. In fact, macroeconomic stability should precede the liberalization of trade in developing countries (Edwards, 1995).

2.1.2.8 Economic Growth

As stated above, although there is still no causal link established, in a number of empirical studies examining both developed and developing countries, there seems to be a correlation between central bank independence and low inflation. The belief that inflation should be maintained at a low level has much to do with the recent proliferation of central bank independence. Experience from recent history has shown that a country cannot achieve macroeconomic goals such as growth, low unemployment, and overall economic well-being in the presence of high inflation. To the extent that inflation is tamed, and as a result, more efficient resource allocation and investment take place, one could expect central bank independence to lead to higher growth (Cukierman et al, 1993). However, to date there appears to be very little established correlation between central bank independence and long-term growth or variability of growth (Cukierman, 1994).

Central bank independence can potentially assist a country in achieving a certain level of financial stability that lays the groundwork for future economic growth (Ribeiro, 2002). One of the seminal works on the relationship between financial market development and economic growth was by Goldsmith (1969). Utilizing data from the years 1860 to 1963, he measured the financial development of a country by dividing the value of financial intermediary assets by GNP. He found that over the course of several decades, there is a rough parallelism between economic growth and financial development. Additionally, there is some evidence of more rapid economic growth being accompanied by an above-average rate of financial development (Levine, 1997).

In general terms, greater financial efficiency reduces the disincentives to risk taking and the accumulation of human capital, thereby increasing the rate of

technological progress and the long-run economic growth rate. Therefore, *ceteris paribus*, a country with higher financial efficiency will have a higher long-run economic growth rate (Trew, 2006). Gurley and Shaw (1955) were the first to study the relationship between financial markets and real activity. They pinpointed that one of the reasons a country was more developed than another was because of its financial markets, with the argument being that in an economy with a more advanced financial system, organizations and investors can borrow money and thereby fund projects easier, thus leading to growth (Caporale et al, 2004).

There were at least two earlier studies that attempted to link central bank independence and not only inflation levels, but also macroeconomic growth variables such as GDP per capita growth. In Cukierman et al (1993), the authors investigate the effects of central bank independence on growth, the variability of growth, private investment and productivity growth. Although the authors found a clear relationship between central bank independence and inflation only in developed countries (but not developing ones), they conclude that there is a relatively robust positive correlation between central bank independence and economic growth (and that a low degree of central bank independence has a negative impact on growth). This growth occurs to a large degree through an increase in private investment with greater independence (and vice versa) and the finding that in economies with more dependent central banks, governments tend to interfere more in the capital markets, resulting in greater resource allocation inefficiencies. In short, the greatest advantage of having a stable macroeconomic environment is stability and a reduction of uncertainty, which promotes investment.

In the same vein, Alesina and Summers (1993) predict that curbing high inflation would tend to positively affect the overall performance of an economy due to the elimination of price and other distortions. They seek to determine if a correlation can be found between central bank independence and macroeconomic variables such as growth, unemployment, and real interest rates. The authors postulate that it is likely to have an impact on these types of variables due to the autonomy from political pressures and the resulting predictability of behavior and economic stability. They conclude that while central bank independence advances price stability and reduces the variability of interest rates, it does not impact macroeconomic growth to any significant degree.

More recently, Hicks (2004) concludes that some scholars have found that central bank independence does not adversely affect macroeconomic variables such as employment or growth, and may actually have a positive effect on investment in developing countries. Some studies using central bank governor turnover as a proxy for operational central bank independence did seem to produce the expected relationship between independence and improved economic performance, along with lower inflation. However, De Haan and Kooi (2000) conclude that central bank independence is related to inflation but not economic growth.

2.1.3 Legal vs. Operational Independence

The conclusion of many empirical studies from the 1990s and early 2000s was that inflation is negatively related to central bank independence in both developed and developing countries. However, for central bank independence to be effective, it must not only be independent in the legal (“de jure”) sense, but also in the operational (“de facto”) sense as well. There is perhaps not a great distinction in the two in most developed

countries, but in many developing countries the difference can be quite large. As long as central bank independence is only “legal” (i.e., formalized), and not operational in scope, the expected benefits will not occur. In fact, it may actually be counterproductive (Cukierman, 1994).

Operational independence implies that central banks are empowered, among other things, to control their own interest rates, and thusly, those of commercial banks in the respective country. The differences between “de jure” and “de facto” independence is great in countries where institutions and rule of law is weak (Carstens and Jácome, 2005). Therefore, one explanation for the weak empirical relationship between central bank independence and inflation is that the legal measure of independence fails to capture operational independence, particularly in developing countries where the rule of law may be weaker (Maxfield, 1997).

In reality, some central banks are more independent than what is implied by law (such as the Bank of Brazil), while others are less independent than what legal independence would imply (such as Venezuela). Therefore, legal provisions may not be consistent with “de facto” independence (Jácome and Vazquez, 2005). For example, although legal indices are important, other aspects of central bank independence such as informal arrangements, tradition and culture of monetary stability are also influential in the setting and carrying out of monetary policy. Therefore, legal independence is not the same as actual independence in many cases for at least two reasons. First, central bank laws are incomplete. Second, even if scope of authority and procedures are described explicitly in the law, actual practice may be differing (Eijffinger et al, 1996).

Moreover, although the degree of legal independence is normally dictated by the central bank laws in a given country, the level of conservativeness can generally be chosen by the central bank. Thus, a lack of legal independence can be compensated for by selecting more conservative central bank governors and other directors. A conservative central bank governor will generally place more weight on inflation stabilization (i.e. price stability) than society. In other words, a conservative central bank governor leads to a lower inflationary bias, which is optimal for society (Eijffinger and Hoeberichts, 1998).

The implementation of independent central banks constitutes an institutional reform that could add creditability to macroeconomic policy. This measure would at least partially isolate the monetary and exchange rate policy from partisan short-term political battles. However, as the Mexican crisis of 1994-1995 clearly indicates, the formal independence of a nation's central bank is not enough to avoid monetary crisis. It is imperative that appropriate monetary policies also be implemented and that it also has political legitimacy. One implication of this is that a central bank should have not only legal independence but also operational independence. For example, as the Venezuelan experience of April 1994 illustrates, when the majority of the central bank board members resigned under political pressure from the executive branch, where there are extreme differences of opinion among the members of the central bank and the executive branch, purely legal independence of the central bank may not be sufficient (Edwards, 1995). However, it seems that the likelihood of independence as well as the degree of independence rises as the differences in political parties and ideologies rise (as was the case in Chile in the 1980s). The rationale for this is that there is an incentive for both

political parties to have an independent, neutral central bank that will pursue policies that are roughly in the middle between the desires of the two parties (Alesina, 1988).

In Arnone et al (2006), the authors examine an updated index of central bank autonomy and apply the index to a set of developed, developing and emerging market countries. The results indicate an increase in central bank autonomy between 1992 and 2003, especially for developing countries. In most cases, this increase is achieved over three stages, from the initial stage in which the foundation for autonomy is laid to the second stage in which operational autonomy develops to the final stage in which further autonomy in terms of policy formulation and leadership is achieved.

Lastly, a large number of central banks around the world have been granted “instrument independence” but not “goal independence.” For example, the Bank of England and the Reserve Bank of New Zealand are given full independence in the setting of monetary policy, but the government sets inflation targets (Sirivedhin and Hataiseree, 2000). However, to reap the full benefits of CBI, it may be better to give the central bank full independence to set its own inflation targets (Muscatelli, 1998). Thus, given the distinct differences between legal and operational independence, laws granting independence may be necessary, but they are not sufficient for true central bank independence to exist. Similarly, formal central bank independence is a necessary, but not sufficient condition for ensuring monetary policy credibility (Sirivedhin and Hataiseree, 2000).

2.1.3.1 Rule of Law and Political Constraints

According to the seminal paper by La Porta et al (1998), common law countries generally have the strongest rule of law, German and Scandinavian civil law countries

have an intermediate level of rule of law strength, and French civil law countries (deriving from the old Roman law) have the weakest rule of law. The quality of law enforcement itself is the highest in Scandinavian and German civil law countries and lowest in French civil law countries, with common law countries being in the middle. These forms of law were developed over several hundred years in the country of origin (i.e. Germany, France, and England) and then disseminated across the world. In the La Porta et al study, the authors examined such variables as corruption, efficiency of the judicial system, accounting standards, risk of expropriation and creditor rights. The countries under their study are many of the same ones that are examined here and can be classified into the following rule of law origins:

Table 2.1.1- Rule of Law Origins: Sample Countries

English Common Law	French Civil Law	Germanic Law	Other
England	Argentina	Japan	China
India	Brazil	Korea	
Malaysia	Chile		
New Zealand	Indonesia		
Thailand	Mexico		
United States	Paraguay		
	Peru		
	Philippines		
	Portugal		
	Spain		
	Uruguay		
	Venezuela		

Source: La Porta et al (1998)

In a similar vein, Acemoglu et al (2008) examine constraints (i.e. “checks and balances”) on countries and determine that distortionary policies are more likely to be adopted when politicians are unaccountable to the citizens of the particular country. As such, even though reforms may be adopted in a country with weak political constraints, the underlying problems are not altered or improved by these reforms. They find that policy reforms have the greatest effect in intermediate constraint countries. In countries

with a high level of constraints, policy reform such as the passing of legal central bank independence may not have much effect since there was already a high regard for the rule of law. Their tabulation includes Constraints of the Executive, Rule of Law and Control of Corruption. A partial categorization as it relates to the sample countries examined here is the following (Thailand is not listed in this study):

Table 2.1.2- Constraint Level: Sample Countries

High	Medium High	Medium	Medium Low	Low
New Zealand	Japan	Argentina	China	Indonesia
England	United States	Brazil	Peru	Paraguay
		Chile	Venezuela	
		Korea		
		India		
		Malaysia		
		Mexico		
		Philippines		
		Portugal		
		Spain		
		Uruguay		

Source: Acemoglu et al (2008)

There is some overlap between the two preceding tables and categorizations. It seems that English speaking countries (that at one time were under some form of British control and/or influence) tend to adhere to the strongest type of law as it relates to the items of interest in this study (common law) and have a high level of constraints on politicians. These types of countries (i.e. the United States, New Zealand and England) may not benefit as much from CBI given this rationalization. Latin America countries tend to operate under the weakest type of law as it related to this study (French Civil Law) and have a medium to low level of constraints on politicians. Asian countries tend to vary widely, although for the most part they tend to operate under either English or Germanic Law and have a medium level of constraints on politicians. Therefore, it is

likely that, given this rationale, Latin American and Asian countries could benefit from CBI, especially if it is operational (vs. purely legal) in scope.

Although for econometric reasons these two studies are not operationalized as dummy variables of any sort, hypotheses will be formed based on these concepts and countries will be examined based on their respective categorizations.

2.1.3.2 Inflation Targeting

Inflation Targeting (IT) refers to a policy framework marked by public announcements indicating the inflation target to be achieved over the next several periods. Low, stable inflation is described as the primary objective of monetary policy (“Inflation Targeting Framework”). IT has become relatively popular since the early 1990s. Some of the benefits of Inflation Targeting are greater transparency, credibility and accountability in conducting monetary policy (Lin and Ye, 2007). Additional benefits include a solution for the time inconsistency problem as well as inflation variability (Ball and Sheridan, 2003). An independent central bank appears to be a strong pre-requisite to successful IT implementation (Agbeja, 2007).

New Zealand was the first to implement Inflation Targeting in 1990 and now approximately 23 countries worldwide (both developed and developing) have implemented “full-fledged” Inflation Targeting, in which an explicit inflation target is the primary objective for a nation’s central bank (Ghazanfar and Sevcik, 2008). However, many countries (such as Indonesia, Japan, and even the United States) have implemented a form of IT, in which inflation target is a secondary objective (along with targets for economic growth, unemployment, etc.).

Although some earlier studies corroborated the effectiveness of IT, more recent studies have placed the framework's value into question. For example, Lin and Ye (2009) state, "once controlling for self-selectivity, inflation targeting has no significant effects on either inflation or inflation variability in developed targeting countries". However, the authors do find that for developing countries, there is a benefit through the "credibility gain", although this benefit can vary greatly based on country specific factors such as the length of time since policy adoption, the government and central bank's commitment to reform and fiscal discipline, etc.

2.1.4 Measuring Central Bank Independence

As has been previously stated, there are many nuances of central bank autonomy and independence. Therefore, reliable procedures for measuring the degree of sovereignty must be found in order to gauge the impact of independence or dependence of a respective country's central bank. Cukierman et al (1993) note the difficulty in quantifying central bank independence, and how legal vs. operational independence can vary, especially in developing countries. In their study, among other things, the authors use the turnover rate of central bank governors as a proxy for actual independence in developing countries, while legal independence is considered an adequate proxy for independence in developed countries. The proxy of central bank governor turnover has since been used in many studies, such as Carstens and Jácome (2005), who found that the average rate in Latin America is .4 (meaning that the average central bank governor remains in office around 2.5 years). This high rate of turnover suggests that Latin American central banks still remain vulnerable to political pressures. A central bank is

considered more politically independent the longer the governor's term in office (Jácome and Vazquez, 2005).

Several other measures of central bank independence have been proposed in the literature, such as the (1991) Grilli, Masciandaro and Tabellini (GMT) Index, which was one of the first formulated indices for central bank independence (which measures political and economic autonomy), and the original (1992) Cukierman Index, which is based on 16 detailed criteria of central bank independence. However, some more recent authors (such as Crowe and Meade, 2008) use the Cukierman, Webb and Neyapti (i.e. the Modified Cukierman Index) measure described in part above because it is very comprehensive and includes developing as well as developed countries. The index has four components relating to appointment procedures (including a proxy for central bank governor turnover), resolution of conflict, the use of explicit policy targets, and rules limiting lending to the government and is an updated version of the original Cukierman Index.

2.2 Central Bank Autonomy around the World

As previously noted, the phenomenon of central bank independence has taken hold around the world, in both developed and developing countries operating under a variety of political regimes. The following section will examine in more detail the circumstances surrounding the individual regions and countries that have undergone central bank independence.

2.2.1 Asia and the Pacific Rim

In the aftermath of the Asian Currency Crisis, the idea of central bank independence once again rose to the forefront of economic thought. It is believed by

many that the lack of independence of this region's central banks was one of the leading causes of the crisis (Sirivedhin and Hataiseree, 2000). Thus, central bank independence began to take hold in Asia and the Pacific Rim during the 1990s and early 2000s. South Korea and Japan, among other countries, made moves toward greater central bank independence during this time period, although not without some setbacks. In sum, it seems that the push for central bank independence has been less in Asia than in Latin America and other parts of the world. However, this may be related to the historically lesser felt need to compete for international capital and creditworthiness (Maxfield, 1997).

2.2.1.1 Thailand

Throughout its history (and varying in degree) the Bank of Thailand has enjoyed a situation of mutual consultation type independence (in lieu of formal independence), in which it has had virtual veto power over decisions made by government fiscal authorities. Thailand itself has traversed through periods of time in which the perceived need for international creditworthiness has varied. For example, the uncertain political landscape of the 1970s with a fledgling democracy attempting to emerge caused a temporary decrease in the need felt for central bank independence and international credibility. Conversely, in the late 1970s and early 1980s the need for central bank autonomy began to come to the forefront once more as the Thai government sensed the need to join the international financial community (Maxfield, 1997).

Although Thailand's macroeconomic management has greatly improved, there is still a lack of legal independence for Thailand's central bank. The legal position of the Bank of Thailand is demonstrated by the turnover of the central bank governor. No

governor has served to retirement age in the history of the Bank of Thailand (McCauley, 2007). Although the Bank of Thailand does possess a certain amount of operational independence, the benefits of a legally independent central bank with clear accountability should be taken into consideration. Although much has been accomplished, without legal guarantee these accomplishments are dependent upon individuals' convictions (Waiquamdee, 2007). Furthermore, the Bank of Thailand's Act of 1942 was developed when central bank independence was considered more of a hindrance than an asset and could most likely benefit from revising (McCauley, 2007).

In sum, although operational independence has tended to rise and fall over the last several decades depending on the government's need for international funds and credibility, the degree of legal independence has remained essentially unchanged since the 1950s. However, as stated before, in many cases legal independence is not as meaningful in developing countries as it would be in developed countries, as the rule of law is uncertain in many cases (Cukierman et al, 1993).

2.2.1.2 South Korea

During the 1970s and 1980s there was little incentive for the South Korean government to grant autonomy to its central bank. It was able to borrow at relatively low interest rates on the international market, economic growth was strong and capital controls minimized the likelihood of massive capital flight, so the government felt little need to increase international credibility. Thus, throughout this time period autonomy in South Korea's central bank remained low. In fact, in 1988 a statute that would have granted more autonomy to the central bank was rejected due to the domestic pressure against granting it more independence. Additionally, a 1993 reform draft passed by

South Korea drew widespread public criticism for its omission of key elements (Maxfield, 1997).

Although the Bank of Korea has undergone reforms, the new structure remains under the influence of the government in terms of such things as inflation targeting and ministry appointments. In 1997, the Korean President established the Presidential Commission for Financial Reform, which had as its only recommendation the redesign of the Bank of Korea. The recommendation was motivated by four general reasons. The first was the fact that although inflation had come down since the 1980s, central bank independence was seen as the next step in its reduction. Second, the granting of central bank independence to the Bank of Korea would bring it in line with other modern central banks around the world. Third, central bank independence would hopefully lead to a more independent private banking system. Lastly, central bank independence was seen as a large part of overall financial and supervisory framework redesign (Cargill, 2001).

When the 1997 Southeast Asian currency crisis struck, the timetable for implementing central bank independence was moved up (partly because it was a condition for IMF assistance) and the Bank of Korea was made independent on April 1, 1998. In fact, the redesigned Bank of Korea was given a good deal of credit for lessening the impact of the crisis and speeding up recovery time. However, Cargill (2001) argues that the Bank of Korea Act of 1997 was not well thought out, was done under pressure from the IMF and was implemented during a time of great economic and financial crisis. Due to the policy outcomes of the Bank of Korea since that time, he suggests that formal independence is neither necessary nor sufficient for price stability, although formal independence is more desirable than formal dependence. Thus, despite the newly granted

autonomy to the Bank of Korea, it was still relatively dependent compared to other central banks in industrial countries (Cargill, 2001).

2.2.1.3 Malaysia

The Central Bank Act of Malaysia Act of 1958 was developed during peacetime and was revised in 1994. The Central Bank of Malaysia has low inflation and stable exchange rates as its two primary monetary policy goals. Since the inception of the country's central bank in 1959, double digit inflation rates have been rare. In fact, during the 1997-1998 Asian Crisis, the decline in value of the country's currency caused a rise in inflation rates above the long-term average, but was relatively mild. The average tenure of the central bank governor in Malaysia is quite long compared to some of its regional neighbors, such as Thailand (6.6 years vs. 3.2 years). The monetary policies of the Central Bank of Malaysia seek to stabilize inflation, while operating without an explicit inflation targeting network. Within the bank the aspect of broad macroeconomic stability is emphasized (including a priority on stabilization of the exchange rate) and a developmental role in financial markets is stressed. In sum, the central bank boasts considerable operational and legal independence (McCauley, 2007).

2.2.1.4 India

The Reserve Bank of India (RBI), India's central bank, was established in 1935 and nationalized in 1949. The primary duties of RBI upon inception were to regulate the issuance of banknotes, to maintain reserves with the objective of maintaining price stability, and to optimize the country's credit and currency system. It also is considered the "lender of last resort" to other banks, controls the credit operations of banks throughout the country, and has the power to inspect the accounts of any commercial

bank (“Reserve Bank of India”). The Reserve Bank of India Act was last amended in 1997 (Arnone et al, 2006). However, it is not considered fully independent.

2.2.1.5 China

The People’s Bank of China was established in 1948 initially as a consolidation of several large private banks and in 1983 was tasked to act as the central bank. However, it was not until 1995 that legal central bank status was conferred. In 2003, China’s Congress amended several laws to strengthen its role in setting and implementing monetary policy and safeguarding the financial stability of the economy (“History of the People’s Bank of China”). According to the PBC’s website, it “carries out business operations independently according to laws and free from intervention by local governments, government departments at various levels, public organizations or any individuals” (“Institutional Arrangement”). However, other sources consider the Chinese Central Bank’s independence to be limited in the sense that it requires the authorization of the government when matters such as modifying interest rates are concerned. As has been stated throughout this study, many times the government and central bank of a country have contrasting objectives. The Chinese government recently has been concerned with the possibility that higher interest rates will further reduce the profitability of state owned enterprises, leading to more bankruptcies and unemployment. Besides interest rates, it also lacks autonomy regarding matters such as money supply targets and foreign exchange rates (Kwan, 2004). Therefore, it would seem that the People’s Bank of China is still lacking in the area of operational independence.

2.2.2 Latin America

Latin America has been in the midst of a very complex historical transition. Countries in this region are overcoming deeply rooted authoritarian legacies in route to more liberal, competitive and participatory democracies (Smith et al, 1994). In 1950, Latin America was the most developed region outside of the industrialized world, with an average GDP per capita more than two and a half times that of East Asia, and about one-fourth that of the U.S. However, three factors can account for much of Latin America's lagging economic performance during the second half of last century: its problems with macroeconomic instability, its low level of integration with the global economy and the poor quality of its public institutions (Elson, 2006). Beginning in the mid-1960s the majority of the Latin American governments began to encourage inward-looking growth and underestimate the importance of macroeconomic stability, as opposed to East Asian economies which implemented and encouraged policies that promoted exports and preserved macroeconomic stability (Edwards, 1995).

Recurrent banking crises and lack of fiscal discipline have traditionally been, and still are, the primary risks for the success of monetary policy in Latin America. Many Latin American countries experienced hyperinflation and economic recession during the 1980s and first part of the 1990s due in part to expansionary monetary policies. In 1990 average inflation in Latin America reached 500%, with three of the countries in the region (Argentina, Peru and Brazil) posting quadruple-digit inflation rates. Until the 1990s, central banks lacked independence from policymakers and thusly the independence to formulate and execute monetary and exchange rate policy. Therefore, for many years, price stability was at most a secondary objective (Carstens and Jácome, 2005).

As a result of these inflation numbers and the economic havoc that was brought about as a result, various governments in the region made central bank reform a key component of the new economic agenda. With monetary policy independently managed, governments' chances of potential expansionary policies associated with a country's political and business cycles were drastically reduced (Carstens and Jácome, 2005). Since the late 1980s and early 1990s, many Latin American countries have implemented major financial reforms, of which granting independence to their central banks is a part, aimed at increasing the degree of financial intermediation and raising the efficiency of investment (Edwards, 1995). Inflation began to improve when many countries in the region began to reform their constitutions and included a chapter devoted explicitly to central bank restructuring and independence (Gutiérrez, 2003).

As has been stated throughout, dependent central banks are commonly at the whim or mercy of specialized agencies in charge of the country's overall financial activities. Independent central banks, on the other hand, have much more freedom to carry out monetary policy (Edwards, 1995). For example, Chile was one of the first countries in the region to grant autonomy to its central bank. According to its web site, the Central Bank of Chile is "an autonomous entity of technical nature, created in accordance with constitutional provisions, has full legal capacity, possesses its own assets and has an indefinite duration" (www.bcentral.cl/eng/index.htm).

The goals of financial deregulation and central bank autonomy in most Latin American countries have been to stabilize prices and remove government from directly participating in the allocation of credit and setting of interest rates. At the time the reforms were implemented, the feeling among Latin American leaders was that direct

government involvement had failed badly and that the Latin American countries needed to move toward a very different model. The debt crisis of 1982 and the failure of early policies in the region to deal with it also played an important role in reshaping policy views in Latin America (Edwards, 1995). Moreover, the external shocks of the debt crises of the early 1980s and again in the mid- to late 1990s led to the implementation of policies with strong common elements--trade liberalization and the promotion of foreign investment, deregulation of markets, fiscal reforms and macroeconomic equilibrium (Smith et al, 1994).

Thus, it became increasingly apparent that relying heavily on the state to run the economy did not generate the expected results. Instead of protecting the public from major external shocks, the state greatly weakened the ability of the respective country's economy to react to foreign disturbances. By the end of the 1980s, many political leaders of the region were looking to market based policies to turn their respective economies around. Part of the process of fiscal and macroeconomic reform has been the granting of independence to many central banks in the region. Unfortunately, however, accountability of central banks in some Latin American countries has improved but transparency is still weak (Edwards, 1995).

Although common motives have been a need for economic retooling for export-led growth and a need for international creditworthiness as a result of a history of financial mismanagement, countries in the region have also had their own unique rationales for seeking CBI. For example, in Chile, independence was granted in order to limit the consequences of poor judgment of future administrations. In Argentina, a conditional IMF loan agreement prompted central bank independence. In Venezuela,

although the central bank was granted a great deal of legal monetary policy authority, the country's vast oil wealth decreased the importance that many politicians placed on independence; as a result, operational CBI has been severely stymied. In Brazil, more inclusive objectives have been assigned to the semi-independent central bank, such as the advancement of social progress. Lastly, for Mexico, international creditworthiness was a major consideration in granting autonomy to the central bank due to this country's desire to continue to attract foreign capital (Maxfield, 1997).

Although the macroeconomic results have not fully met the expectations that were raised at the time the reforms were instituted, there has indeed been an improvement over the previous decade. This is especially true when the tremendous consequences of persistent hyperinflation, macroeconomic instability and the regional financial shocks that took place during the 1990s are taken into consideration. The decline of inflation rates in Latin America was astonishing, from an average of 500% at the beginning of the 1990s to single digits by the early 2000s (Carstens and Jácome, 2005).

In sum, the history of Latin America abounds with incidents where fiscal balance was lost and inflation took over due to political pressures. Some countries in the region have tried to deal with these problems by implementing reforms that lessen the extent to which the central bank is influenced by short-term politics, arguably the most important of which has been making their central banks independent from the executive branch, a policy which was widely adopted during the 1990s. However, it is important to realize that although autonomous central banks, with the responsibility of achieving monetary stability, can have a great deal of influence in reducing macroeconomic imbalances, independent central banks are in no way substitutes for prudent and consistent monetary

and fiscal policy (Edwards, 1995). Additionally, the tremendous reduction of inflation in these countries was due to an entire group of reforms, although undoubtedly central bank independence played an important role (Jácome and Vazquez, 2005).

2.2.2.1. Brazil

The Central Bank of Brazil was created in 1964 and is considered an autonomous federal institution. However, from that time until the mid-1980s, monetary authority was shared between the Central Bank, the Bank of Brazil (which acted in a sort of government bank capacity) and the National Treasury. In 1985, financial reorganization of the government was undertaken and continued on until 1988, at which time the functions of monetary authority were completely transferred to the Central Bank from other monetary authorities in the country (such as the Bank of Brazil). The 1988 Constitution establishes the Central Bank's responsibilities, such as the issuance of money, the nomination of Central Bank governors, and the prohibition of loans to the National Treasury ("Banco Central History").

However, despite worsening macroeconomic conditions in the country during the 1980s, including the massive rise of external government debt as well as several episodes of hyperinflation, the issue of legal central bank independence remained a low priority for the once again civilian government (which had been under military rule since the mid-1960s). This was most likely due to the easy money available during that time period through different sources (such as private bankers) in the international markets as well as a massive growth in exports. Unfortunately, although talks pursuing renewed interest in central bank autonomy rose again in 1993-94, they were quickly buried among scandals of political corruption (Carstens and Jácome, 2005). Therefore, at this point it seems that

there is a tacit agreement of some degree of operational independence between the Central Bank of Brazil and the government (with responsibilities established by the Constitution), but no formal central bank independence has been declared in the Constitution or in any other legal statute.

2.2.2.2 Mexico

The concept of central bank autonomy in Mexico has evolved through three distinct stages. During the first stage, known as Statutory Autonomy (1925-1938), rules were established which governed the ceilings on growth of key operative variables. In the second stage, known as Charismatic Autonomy (1955-1970), sound public finances were emphasized and defended by important political figures. During the most recent stage, known as Institutional Autonomy (1994 to present), the central bank's autonomy has been based on a constitutional mandate that determines the bank's functions, provides for a self-governing directive body, and administrative and budgetary independence (Turrent y Diaz, 2007). When Mexico's Central Bank was granted legal (vs. simply some degree of operational) independence in 1994, the treasury secretary explained that the changes were designed to ensure that the bank would gain control over the country's anti-inflation policies, which were tied directly to the central bank's lending practices ("Presidency Introduces Legislation Seeking Autonomy for Central Bank").

The 1994 statute essentially made three changes. The stabilization of the peso's purchasing power was stipulated as the central bank's primary goal. The statute also prohibited the extension of central bank credit to the government and provided for appointment terms and conditions of the central bank governor and directors as those "that would be conducive to exercising autonomy." However, the peso crisis of 1994-

1995 demonstrated how little impact legal independence can sometimes have. However, the example of Mexico strongly suggests how legal independence can arise from operational independence (although sometimes with a considerable lag). It also helps to lend credence to the intuitively sound thought that markets shape ideology more than ideology shapes markets (Grier, 2004).

2.2.3 OECD

OECD countries such as England, Spain, Portugal, Italy, France and Belgium have all moved toward central bank independence since the early to mid-1990s. The creation of the European Central Bank has been at the forefront of economic thought regarding the financial health of a respective country's central bank and how independence assists in this aspect (Stella and Lonnberg, 2008). Western European countries such as Spain and Portugal also granted full autonomy to their respective central banks, with Spain passing the Bank of Spain statute in 1994, and Portugal passing new laws in 1990 and again in 1992. The Bank of Spain statute emphasized the job tenure of central bank governors and the board of directors, while legislation set by the Portuguese authorities granted greater freedom to the Bank of Portugal in setting monetary and exchange rate policies as well as more influence in the regulation of financial markets and institutions (Maxfield, 1997).

New Zealand has been one of the extraordinary success stories over the last twenty years in controlling inflation through the increase of central bank independence, primarily through the resulting successful inflation targeting that took place. The Reserve Bank of New Zealand Act of 1989 granted this country's central bank independence from the federal government and established an explicit inflation target, becoming the first of

many countries to do so. In fact, many scholars have credited the success of this country's central bank endeavor a key factor in other countries' decision to also pursue CBI (Carlstrom and Fuerst, 2006).

The Bank of England undoubtedly has the longest history of any central bank in this study. It was initially founded in 1694; however, it was nationalized in 1946 and only granted independence in 1997. It is committed to promoting and maintaining monetary and financial stability. However, as established by the most recent legislation (the Bank of England Act of 1998), the Banking Supervision functions responsible for financial stability effectively was divided among three entities--the Bank of England, the Financial Services Authority and the Treasury. Although a Memorandum of Understanding was established, there is no formal legislation setting out the respective responsibilities regarding financial stability ("About the Bank").

2.2.4. The United States

The Central Bank of the United States, referred to as the Federal Reserve, was created in December 1913 and has four major areas of responsibility--the conducting of monetary policy in pursuit of full employment and price stability, supervising and regulating banking institutions, maintaining the stability of the financial system, and providing key financial services to the U.S. government and financial institutions. It is composed of a central governmental agency (the Board of Governors) and twelve regional Reserve Banks throughout the nation. It is considered to be an independent entity within the government and derives its authority from the U.S. Congress. Its decisions do not have to be ratified by the President of the U.S. or anyone else in the executive or legislative branch of government. However, it is subject to governmental

oversight by Congress and has, therefore, been described as “independent within the government.” However, it is also self-financed and is, therefore, not subject to the congressional budgetary procedure (“The Federal Reserve Board Frequently Asked Questions--Federal Reserve System”).

Of the Federal Reserve’s tools for influencing monetary policy, the most important is considered to be open market operations, in which it buys and sells U.S. Treasury and other federal agency securities in the open market. This helps the Federal Reserve to influence one of the key interest rates in the economy, which helps to influence other interest rates, foreign exchange rates, the money supply, inflation and ultimately the entire spectrum of macroeconomic variables. However, it is interesting to note that the Federal Reserve does not attempt to control the interest rates that commercial or other banks in the country set, allowing this to be done by the forces of supply and demand (“The Federal Reserve Board Frequently Asked Questions--Monetary Policy”). The Federal Reserve Act was last amended in December 2000 (Arnone et al, 2006).

2.3 Central Bank Autonomy and Financial Crises

Throughout the last few decades, several waves of financial crises have swept throughout both the developed and the developing world, leaving a trail of financial damage after each. In general terms, a financial crisis is a situation in which financial institutions or assets lose a large part of their value. Some examples of financial crises are stock market crashes, currency crises (a sudden large devaluation of a nation’s currency), and sovereign defaults (the failure to pay back debt) (Kindleberger and Aliber, 2005).

Although financial crises have been studied quite prodigiously over the last

decade or so, there are still many things that remain unknown about the phenomena. Various explanations of crises can be offered. First, trade linkages can transmit a crisis, as currency depreciation in one country makes its exports cheaper on the world market, thus placing pressure on its competitors to also devalue their currencies. Second, financial interdependence can also contribute to crisis transmission. Third, a crisis in one country can worsen market participants' perceptions of the economies with similar characteristics as the one in question--in other words, investors "lump together" all markets with similar characteristics (Pesanti and Tille, 2000). In fact, the current global financial crisis (of late 2007 to the present) has reinforced the idea that even central banks in developed countries behave in a very similar way when faced with the same types of economic shocks (Posen, 2007). Table A3 in the Appendix lists the financial crises under consideration in this study.

There are several common factors regarding the antecedents to the four major emerging market financial crises of the 1990s. In nearly all cases, serious banking and financial sector weaknesses played an important role. Fixed exchange rates without the necessary monetary policy commitments were present as antecedents to crisis in all the cases. Additionally, overly inflationary monetary policies, large fiscal deficits, or even large current account deficits, were present in several cases. Lastly, national balance sheet weaknesses, including large short-term liabilities either of government or the private sector, were important elements in each of the crises (Corsetti et al, 1998).

The "domino effect" is also a powerful force when examining currency crises and can be described in the following way: "Yet it is also true that small events at times have large consequences, that there are such things as chain reactions and cumulative forces.

And economic collapse often has the character of a cumulative process. Let it go beyond a certain point, and it will tend for a time to gain strength from its own development as its effects spread and return to intensify the process of collapse” (Friedman et al, 1963). In other words, there is a “contagion” or “spillover” effect many times from the country of origin to other countries, many of whom perhaps do not share the same economic fundamentals as the country of origin. For example, as a result of the Asian crisis, other disparate countries such as Russia, Chile, Mexico and Brazil also experienced a degree of financial crisis (Tannuri-Pianto, 2006).

This “contagion” effect of the 1990s financial crises should not be underestimated. There are many explanations and models of contagion such as common shocks that simultaneously hurt commodity exporting countries, competitive devaluations among countries, asset market correlations through financial linkages, market illiquidity, investor irrationality (i.e. the “herding” effect), and “reputational externalities” (a crisis in one country affecting investors’ expectations and perceptions about common structural conditions and vulnerabilities in other countries and likely policy responses to such vulnerabilities) (Summers, 2000). In sum, contagion can influence investors’ sentiments and generate self-fulfilling crises if foreign investors do not discriminate among different macroeconomic fundamentals across countries and can create a “stampede” effect across countries (Garay, 2003).

There has been shown to be a relationship, at least in the long-term, between the pursuit of price stability and financial stability. Financial crises are frequently linked to inflation and interest rate instability. Sharp fluctuations in interest rates induce large changes in the asset values of banks and make it more likely that financial crises will

occur (Cukierman, 1994). Hattori (2002) concurs in that a country's central bank has two means of defending their currency when its value comes under attack--they can either use foreign exchange reserves or they can raise domestic interest rates. Furthermore, a currency crisis can cause hardship for a weak banking sector if the government defends its currency and increases interest rates (Rojas-Suarez and Weisbond, 1995).

Also, a speculative attack on the currency can precipitate a banking crisis if deposit money is used to speculate on the currency and banks are thereby depleted of funds (Miller, 1996). A weak banking sector may itself cause a currency crisis if rational speculators anticipate that policymakers will not endure the costs of defending their currency and that an internal drain (i.e. a bank run) can cause an external drain (i.e. a speculative attack on a currency) (Calvo, 1995). Interestingly, emerging markets with more rigid exchange rate regimes were less likely to experience currency crisis during the last two decades (Peltonen, 2006).

All told, an independent central bank should have more flexibility to pursue the monetary policy it feels is most advantageous for the country at a given point in time without regard for political considerations. Therefore, if a country is on the verge of crisis (or is already suffering from a crisis) or is threatened by a possible contagion effect, an autonomous central bank would likely be more able to defend the country's economy and steer the country through to calmer economic waters.

2.3.1 Asia and the Pacific Rim

The Asian Currency Crisis of the late 1990s unfolded against the backdrop of several decades of spectacular economic performance in the region. Until the currency crisis broke out in the summer of 1997, East Asia was the most dynamically growing part

of the world and was often called the “growth center” of the world economy. Its economic miracle was made possible by such factors as rapid capital accumulation, productivity growth and an increased supply of labor. This remarkable economic development in the latter part of this era was aided by stable macroeconomic environments in the respective countries and outward-oriented policy regimes. However, the experience of the Asian Currency Crisis shows that even the supposedly most dynamic economies can come under serious economic attack and experience severe capital outflows once confidence is lost. Before the crisis, the Asian financial and banking sectors suffered from lax supervision, corrupt lending practices, and weak regulations. All these things caused weakness in the sector as a whole. (Masahiro, 1998). The financial services sector in most of these newly industrialized economies had also been developing rapidly and without sufficient regulation, oversight, and government controls (Nanto, 1998).

The devaluation of the Thai baht on July 2, 1997, generated waves of turbulence in currency and equity markets that surpassed the contagion effects in the midst of the 1994-1995 devaluation of the Mexican peso. The crisis first spread through East Asia in the form of a string of devaluations and stock market collapses. As the problems intensified, the currencies of other Asian countries, including Hong Kong and South Korea, came under speculative pressure. As a result of a possible “domino” effect, countries outside the region such as Argentina, Brazil, and Russia suffered sharp declines in their equity markets and periodic bouts of speculation against their currencies (Kaminsky and Reinhart, 1999).

The series of country specific financial crises that plagued the Asian region from the middle of 1997 forced countries such as Thailand, Indonesia, and South Korea to have to turn to the International Monetary Fund and other financial institutions for assistance in getting through their mounting debt problems. In general terms, blame has been laid on, among others, the greed of international speculators, the ill-timed Chinese currency devaluation in 1994, the aggressive push by the United States to open Asian markets to U.S. goods and services, and the Japanese failure to address its own financial problems. However, the immediate cause can be put down to short-term foreign over-borrowing by Asian banks, corporations, and individuals (Bridges, 1999).

In a broader sense, however, there were a number of country-specific structural defects that contributed to the severity of the crisis, such as the use of political rather than commercial criteria in the allocation of loans (especially in Thailand), the over-expansion of manufacturing capacity and reliance on very thin profit margins (South Korea), poor supervision of financial institutions and systems (several countries), and a penchant for mega-projects with few cost controls (Malaysia and Indonesia) (Bridges, 1999). Another view is that weaknesses in Asian financial systems were at the root of the crisis. These weaknesses were caused largely by the lack of incentives for effective risk management created by implicit or explicit government guarantees against failure. The weaknesses of the financial sector were masked by rapid growth and accentuated by large capital inflows, which were partly encouraged by pegged exchange rates (“What Caused East Asia’s Financial Crisis?”, 1998).

Although the crisis came to light in July 1997, signs of trouble had appeared a year or so earlier. Doubts over what had become known as the “East Asian Miracle”

began growing around 1996, when the region's export growth showed signs of slowing down, and economic growth rates declined as a result. However, at first glance, the general consensus was that the Asian economy was taking a pause and that it would resume its strong growth in a few years. Specifically, the first negative news was talk of an interest rate increase in Japan and to a lesser extent in the U.S. These countries had been the two main sources of the world's inexpensive, yet reliable, money. When government officials in Japan and the chairman of the U.S. Federal Reserve began talking about the need to raise interest rates, the traders concluded that it was time to liquidate their investments in Asia, creating a "run" on the respective nations' currencies (Ushikoshi, 1998).

In the early 1990s, when capital began to flow to emerging market economies, countries in East Asia were enjoying a substantial amount of Foreign Direct Investment with short-term capital inflows representing a relatively low share of the total. However, by 1996, persistent bouts of sterilized intervention in several East Asian nations kept short-term interest rates high relative to international levels and acted as a magnet for short-term capital inflows. The rising volume of short-term flows was largely intermediated by the poorly regulated and ill-supervised domestic banking sectors. Thus, it appears that the combination of volatile international capital and weaknesses in the financial sector may be at the heart of the 1997 crisis in East Asia and may explain its severity (Kaminsky et al, 1999).

The relevance of fundamental economic imbalances is illustrated by the different experiences that the region's countries had during the crisis. Taiwan, Singapore, and Hong Kong were relatively less hard hit by the turmoil. Hong Kong, although losing

some of the value of its equity market, was able to maintain its currency peg. Taiwan and Singapore decided to let their currencies float instead of losing reserves trying to defend them. However, depreciation rates were modest in comparison to that of many regional currencies. Most importantly, they did not experience the widespread financial panic and fund reversals that their neighbors did (Pesanti and Tille, 2000).

As the Asian financial crisis progressed, the affected governments (except for Hong Kong) eased the rigidity of their exchange rate systems. Exchange rates are now allowed to move in wider bands, in a crawling band, or to float. This flexibility reduces the potential for large and sudden changes in these exchange values, as the rates respond continuously and in smaller increments to market forces (Nanto, 1998). It is quite clear that whatever exchange rate regime a country chooses to follow, that sound economic fundamentals and a strong banking sector are crucial for long-term success (Nakamae et al, 2005).

In sum, the rapidity with which the Asian economies grew and liberalized their financial markets meant that the development of the financial systems in some countries did not keep pace with the development of the financial markets. To varying degrees, lax lending standards, weak supervisory regimes, inadequate capitalization, and excessive inter-connected lending were at the heart of the crisis (Nanto, 1998).

2.3.1.1. Thailand

The regional currency crisis began with a sharp fall of the Thai baht. Beginning in May 1997, the baht came under strong selling pressure, which monetary authorities responded to with market intervention. However, on July 2, as concern grew that further defense of the baht would harm the domestic economy, Thailand shifted from a currency

basket to a managed float, which effectively devalued the currency. From July to August, the baht's fall impacted other Southeast Asian currencies, such as the Indonesian rupiah, Philippine peso, and Malaysian dollar. The decline in East Asian stock markets, which had been soft since that July, accelerated in August (Ushikoshi, 1998).

The Thai baht experienced downward pressure in July 1997 when some foreign investors recognized its potential weakness. Thailand had been receiving a massive amount of foreign (portfolio) investment and was susceptible to withdrawal. The Bank of Thailand (the Thai Central Bank) tried to defend the baht in the foreign exchange market but ran out of foreign reserves and was forced to surrender its attempt. The IMF intervened in exchange for financial reform. Since the Asian Crisis of 1997-1998, the Bank of Thailand has regained confidence to a certain degree by utilizing inflation targeting and a managed float exchange rate system (Waiquamdee, 2007).

2.3.1.2. South Korea

The dynamic in South Korea was due to the country's credit downgrade as a result of the high percentage of problem loans and financial institutions that had difficulty acquiring dollars began selling won to buy dollar-denominated assets. The Bank of Korea, despite being low in foreign reserves, decided against market intervention. As a result, the won's depreciation began to accelerate, the capital exodus continued, and the stock market plunged (Ushikoshi, 1998). However, it was the near collapse of the Korean economy during this time that caused independence of the Bank of Korea to come to the forefront of thought in that country (Cargill, 2001).

2.3.2 Latin America

Banking crises have been a common and recurring event in Latin America with catastrophic macroeconomic effects. Since 1990, banking crises have been the main cause of inflation reversals and economic contraction in many countries of the region (Carstens and Jácome, 2005). Additionally, throughout the last few decades, several waves of currency crises have swept through Latin America, provoking very serious financial damage after each crisis. These currency crises have been of two general sorts--those that originated in the region and spread to neighboring countries and those that originated outside the region but through a “domino effect” eventually affected Latin America. Historically, financial crises in Latin America have been more frequent and severe than in other regions (Kaminsky and Reinhart, 1999). Moreover, several Latin American countries have experienced simultaneous currency and banking crises, such as Chile (1982), Argentina (1982 and 1995), Venezuela (1994) and Mexico (1994-95). Causation for these dual crises may run in either direction, as the two can many times be complementary (Miller, 1998).

2.3.2.1 Brazil

In Brazil, the Real Plan was initiated in 1994 and had several successful years of controlling the hyper-inflation that had plagued the economy for the previous decade. The currency, the Real, was placed within a “mini-band” against the U.S. dollar that allowed very small daily changes in the currency’s value. However, the dollar began to strengthen in the mid-1990s and the 1997-1998 Asian crisis heightened fears among international investors regarding emerging markets. Nevertheless, Brazil was able to protect its currency fairly well from the effects of the Asian crisis because its Central Bank had a large amount of U.S. dollar reserves and was able to support its currency in

the foreign exchange markets. However, interest rates were raised to extreme levels to keep foreign investment in the country. Because Brazil is the region's largest and most powerful market, the fear was that if its currency were devalued, the entire region would be eventually forced to devalue their respective currencies. Nevertheless, the Central Bank of Brazil's international reserves eventually declined by about \$9 billion with the value of the Real having been under pressure and losing several percent of value in late 1998 (Kaminsky et al, 2003).

The final blow came in January 1999 when a state governor declared a moratorium on debt payments to the Central Bank. Within a matter of only a few weeks, investors had removed several billion dollars from the country and the Central Bank had no choice but to eventually abandon the band and let the currency float (Langley and Bolling, 1999). On January 13, 1999, the Brazilian Real was devalued and several weeks later left to float freely, eventually losing 70 percent of its value (Kaminsky et al, 2003).

Additionally, during the height of the Asian boom in the 1990s, South Korean banks had extended credit to several governments, with Brazil being one of them. Brazilian banks also had invested in Russian short-term treasury securities. When the Asian crisis hit, South Korea was affected and the consequences spread from that country to Brazil and other countries of Latin America, again demonstrating how financial linkages can lead to contagion during times of financial crisis (Khalid and Rajaguru, 2005).

2.3.2.2 Argentina

The Argentine peso differs from the other currencies in question because the Argentine government in the early 1990s adopted a currency board to set the value of the

country's currency on a one-to-one parity with the U.S. dollar and as a result caused its own market to be very illiquid (Fukuhama, 2003). Mexico's currency crisis of 1994-1995 was the first serious test of Argentina's currency board and its commitment to maintain its parity with the U.S. dollar (Goldberg and Veitch, 2002). In late 2001 it was decided that the parity could no longer be sustained, and the currency was allowed to float. It immediately devalued, bottoming out at approximately 3.5 pesos per dollar soon afterward. Although it has strengthened somewhat since then, it has never approached the value of the parity era.

2.3.2.3. Mexico

In the 1970s and 1980s, Mexico faced several severe monetary crises. In the mid-1970s, the presidential administration in power at that time decided to abandon the stabilization process of the previous decade. In the wake of severely skewed distribution of income as well as a growing current account deficit, the administration forced the Bank of Mexico to print more money to cover the growing public sector deficit. This increase in the money supply had a telling impact on the consumer price index. Inflation jumped from only 3.7% in 1971 to 22.5% by 1975, along with increases in the debt service ratio and capital flight (Ramirez, 1989).

A few years later, in August 1982, Mexico defaulted on its external bank debt. By the end of that year, the peso had depreciated 100%. Prior to Mexico's default, several Latin American countries had already experienced currency crises, banking crises or both. When Mexico defaulted, the highly leveraged foreign banks pulled back from emerging markets in general and Latin America in particular (Kaminsky et al, 2003). The then President of Mexico, Lopez Portillo, was accused of postponing devaluation

(perhaps making it worse than it would have been had it been more timely), to which he replied, “when a president devalues, he devalues himself.” Despite efforts to control the crisis, capital flight continued to a great degree and the administration decided to nationalize the banking system on September 10, 1982, which placed the banking sector in the hands of the government (Ramirez, 1989).

The Mexican peso came under attack again in December 1994. After several attempts at defending it by the Mexican Central Bank, the currency was left to float freely in January 1995. Speculative attacks on other Latin American countries immediately ensued. The most severely affected countries in the region were Argentina, Brazil, Peru and Venezuela. Interestingly, Chile was a notable exception (Glick and Rose, 1998). The Mexican Crisis of 1994-1995 spread across emerging markets not only due to weaknesses in economic fundamentals, but also due to a high degree of financial interdependence among the affected economies (Fratzcher, 2003).

Moreover, similar to the Asian crisis, the Mexican currency crisis was preceded by a loss in profitability of the banking system and by an increase in non-performing loans. Commercial banks’ net worth was affected negatively by the peso depreciation and sharply increased the value of foreign liabilities of commercial banks. Also, many banks, for a variety of reasons, were not hedged against exchange rate risk and immediately became dangerously overleveraged (Buch and Heinrich, 1999).

2.3.3 The Current International Financial Crisis

The current financial crisis of 2008-2009, although originating in whole or in part in the United States, has spread quickly across national boundaries and could now be considered international in scope. It is generally believed that the origins of this crisis can

be traced back to risky mortgage lending practices in the U.S. during which time credit was easily available during the first half of the 2000s (Roubini, 2009). By 2007, however, real estate prices in many other industrialized countries, such as England, New Zealand, Spain, and Australia, as well as developing countries such as China and India, were already being bid up and could be considered “bubbles”. In fact, it is considered by some to be the largest asset bubble in history (“The Global Housing Boom”).

The National Bureau of Economic Research stated in December 2008 that the U.S. had technically been in a recession since December 2007. It is considered the worst recession since the Great Depression of the 1930s (Lightman, 2009). Despite stimulus packages announced by both developed and developing countries (i.e. U.S., China, Germany, India, Russia and Saudi Arabia, among others), trade volumes, production and employment have all shrunk rapidly. A global contraction of .5 to 1% (annualized) in the first quarter of 2009 is anticipated, which is the first global contraction in 60 years (“Advanced Economies to Contract Sharply in ‘09”).

According to the IMF, the root of the market failure was “optimism bred by a long period of high growth, low real interest rates and volatility, and policy failures in financial regulations, macroeconomic policies, and global architecture”. The roots of the crisis can actually be traced back to the beginning of the decade, during which time low interest rates and high world growth caused investors to be willing to assume greater and greater risk, while at the same time being overly optimistic about the risks ahead (“Initial Lessons of the Crisis”). These record low real interest rates tended to reflect monetary policy of the recent years. Central banks found that they could maintain stable and low inflation rates with low interest rates, due in part to the strong demand from Asia for

developed country government backed assets (“Lessons of the Global Crisis for Macroeconomic Policy”).

Central banks are much to blame for this crisis. They should have adopted a broader “macro-prudential” view, taking into account in their decisions asset price movements, credit booms, leverage and the build up of systemic risk (“Initial Lessons of the Crisis”). Some central banks had geared monetary policy almost exclusively toward the stabilization of inflation, while others gave more explicit weight to aggregate activity. However, almost none made sufficient assessments of risk from asset price increases or leverage (“Lessons of the Global Crisis for Macroeconomic Policy”).

However, although central banks are to blame, there are many others who share the responsibility for the global crisis. There existed “fragmented surveillance in silos of expertise” among nations and international economic entities, including the Bank for International Settlements and the IMF (“Initial Lessons of the Crisis”).

Chapter 3-Hypotheses

Given the related theory and evidence provided in Chapter 2, the following hypotheses are put forth. For simplicity, they are written in the form of alternative hypotheses, with the null being that of no correlation:

H1- Central Bank Independence is negatively related to Inflation.

H2- Central Bank Independence is negatively correlated to Money Supply growth.

H3- Central Bank Independence is negatively related to Interest Rates.

H4- There is a positive correlation between Central Bank Independence and the strength of the Exchange Rate.

H5- Central Bank Independence, with the resulting macroeconomic stability that it has engendered, is positively related to Direct Investment.

H6- Central Bank Independence, with the resulting macroeconomic stability that it has brought about, is positively related to the Savings level.

H7- There is an inverse correlation between Central Bank Independence and Unemployment.

H8- Central Bank Independence, with the resulting macroeconomic stability that it has engendered, is positively related to Exports.

H9- There is a positive correlation between Central Bank Independence and Economic Growth.

H10- There is stronger evidence of CBI effectiveness among countries with lower levels of Corruption.

H11- There is stronger evidence of CBI effectiveness among countries operating under English Common Law than other types of legal systems.

H12- There is stronger evidence of CBI effectiveness among countries operating under inflation targeting regimes.

H13- There is stronger evidence of CBI effectiveness among countries operating under a medium level of constraints than either a high or low level of constraints.

H14- There is stronger evidence of CBI effectiveness among countries that are more economically developed.

H15- There is stronger evidence of CBI effectiveness among countries that have a higher level of Tax Revenue amassed by the government.

H16- There is stronger evidence of CBI effectiveness among countries that have a higher level of Government Consumption.

H17- There is stronger evidence of CBI effectiveness among countries operating under a higher level of Operational Independence.

H18- There is a beneficial “Signaling Effect” present with the announcement of Central Bank Independence.

H19- Among countries that experienced a financial crisis both before and after having an independent central bank, post-CBI time frames fared better than pre-CBI time frames.

H20- Among those countries that experienced a financial crisis, those countries that had an independent central bank at the time of crisis fared better than those that had a dependent central bank.

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Chapter 4- Methodology and Data

The methodology will use a variety of econometric techniques to test for volatility, correlation and causation with results organized by each individual hypothesis. Macroeconomic data from Q1:1960 to Q1:2009 will be taken entirely from the International Monetary Fund (considered a highly regarded, neutral, global financial institution) in order to maintain consistency of measurements, data collection processes and to avoid biases of individual country central banks or national data collection agencies.

4.1 Variables

The following macroeconomic variables will be used as dependent variables to measure the impact of central bank independence and outside phenomena on a country's growth and stability and are measured on a quarterly basis. For econometric reasons, the first iteration (i.e. "first difference") of each variable will be used.

1. Inflation Rates (as measured through changes in the Consumer Price Index)
2. Money (M1) Supply
3. Interest (Deposit) Rates
4. Exchange Rates (National Currency per US\$)
5. Direct Investment in the Economy (% Change)
6. Time Deposits (% Change)
7. Unemployment Rates (%)
8. Exports (% Change)
9. Gross Domestic Product Volume Index

The following independent variables will be used to examine the impact of the nuances (political and economic) of central bank independence and related phenomena on the previously mentioned dependent variables. Dummy variables are used frequently to classify data into mutually exclusive categories (Gujarati, 2003) and will be used in the following equations to represent, among other things, the effects of country development, foreign exchange rate regimes, the occurrence of currency crises, the degree of international inflation, and the occurrence of legal CBI.

1. Central Bank Independence (Legal Date). This is a dummy variable designed to distinguish between years in which no central bank independence was in place and years in which legal central bank independence was in existence. Although for most countries it takes the value of either 0 (representing no legal independence) or 1 (representing full legal independence), it also takes the value of .5 for countries (such as Brazil) that after a certain date were operating under a semi-independent/informal accord type arrangement
Source: Central Bank websites.

2. Modified Cukierman Index. This is a composite CBI index measuring various facets of independence. It was originally developed by Cukierman et al and has been recently updated. It is comprised of 18 criteria of political and economic independence with a total score ranging from zero to one. Regarding political independence, it focuses on the appointment and dismissal of the central bank's governor. The specific criteria are the following:

- a.) Term of office of Governor (more than one presidential period, the period does/does not coincide with presidential elections, or not specified)

- b.) Channel through which the governor is appointed (via executive and legislative, only executive, etc.)
 - c.) Appointment and term of office for the rest of the board.
 - d.) Dismissal of board members
 - e.) Governor allowed to hold another government office or not
 - f.) Fundamental objective of central bank (price stability, price stability that does/does not conflict with other economic goals, etc.)
 - g.) Channel through which monetary policy is formulated
 - h.) Government directives and resolution of conflict
 - i.) Central bank involvement in debt approval
 - j.) Limitations and advances on central bank lending
 - k.) Lending to government (not allowed, restricted, fully allowed, etc.)
 - l.) Channel through which financing conditions to the government are decided
 - m.) Beneficiaries of central bank financing
 - n.) Interest rates in advances of lending (at market rates, below or above)
 - o.) Lender of Last Resort
 - p.) Financial autonomy
 - q.) Accountability of central banks (i.e., who do they report to and how often?)
 - r.) Central bank transparency (i.e., who do they disclose to and how often?)
- (Jácome and Vázquez, 2005).

The following table presents original and update index scores for the countries examined in this study:

Table 4.1.1--Modified Cukierman Index: Original and Updated Country Scores

Country	Original Score	Updated Score
Argentina	0.48	0.77
Brazil	0.20	0.30
Chile	0.26	0.73
China	na	na
England	0.29	0.37
India	na	na
Indonesia	0.27	0.27
Japan	0.18	0.35
Korea	0.32	0.45
Malaysia	0.40	0.40
Mexico	0.34	0.56
New Zealand	0.22	0.34
Paraguay	na	na
Peru	0.43	0.83
Philippines	0.43	0.67
Portugal	0.56	0.83
Spain	0.09	0.92
Thailand	0.24	0.24
United States	na	na
Uruguay	0.24	0.51
Venezuela	0.31	0.68

(Source: Jácome and Vázquez, 2005)

3. Grilli Masciandaro Tabellini (GMT) Index. This is a composite index measuring political and economic independence in various areas. It was originally developed by Grilli, Masciandaro and Tabellini in the early 1990s and has been updated since by several authors. According to the original creators of this index, credibility is the primary asset of having an independent central bank (Hicks, 2004). It is based on 15 criteria, each with a binary score of zero to one. The criteria are then simply added together. The specific criteria are the following:

- a.) Appointment of the governor (with or without government involvement).
- b.) Length of time of appointment (greater than or less than five years).
- c.) Appointment of the board of directors (with or without government involvement).
- d.) Length of time of board appointment (greater than or less than five years).

- e.) No mandatory participation of government representatives on the board.
 - f.) No government approval required in formulating monetary policy.
 - g.) Requirements in the charter for monetary stability to be a primary objective.
 - h.) Legal protections for the central bank in the event of conflict with the government.
 - i.) No automatic procedure for the government to obtain credit directly.
 - j.) Direct credit is extended to the government at market interest rates.
 - k.) Credit is extended on a temporary basis.
 - l.) Credit is extended for a limited amount.
 - m.) The central bank does not participate in the primary market for public debt.
 - n.) The central bank is responsible for setting the discount rate.
 - o.) The central bank has no responsibility for overseeing the banking sector (or at least shares this with other institutions).
- (Arnone et al, 2006).

As above, the following table presents original and update index scores for the countries examined in this study:

Table 4.1.2- GMT Index: Original and Updated Country Scores

Country	Original Score	Updated Score
Argentina	6	13
Brazil	7	10
Chile	14	14
China	na	na
England	6	11
India	6	7
Indonesia	na	na
Japan	6	7
Korea	na	na
Malaysia	na	na
Mexico	6	13
New Zealand	4	7
Paraguay	7	11
Peru	10	13
Philippines	9	10
Portugal	3	13
Spain	5	14
Thailand	na	na
United States	12	12.5
Uruguay	7	10
Venezuela	7	12

(Source: Arnone et al, 2006).

4. CB Governor Turnover. This has been used in the CBI literature as a proxy for independence, with the idea being that if there is a high ratio of turnover (as measured on a yearly basis), this is an indicator that the central bank is actually under the firm control of the executive branch and that the central bank governor will be dismissed if he or she does not comply with the government's demands. Therefore higher turnover is usually seen as a sign of a lesser degree of independence. See Carstens and Jácome, 2005, among others. Source: Central Bank websites and author's own calculations.

5. Development Dummy Variable (from 0 to 1 in .1 increments). This measures the economic development of the respective sample country in terms of Gross Domestic Product, with each .1 increment representing \$3,000 of GDP in constant dollars Source: The World Economy: Historical Statistics, OECD Development Center, 2003.

6. Exchange Rate Regime Dummy Variable. This is designed to capture the effects of a particular type of exchange rate system on a country's economy. It takes a value of 0 for pure fixed and 1 for pure floating in .1 increments. Sources: Crowe and Meade (2008), Musa et al (2000), Baig (2002), Bubula and Otker-Robe (2003), and IMF (2003).

7. Financial Crisis Dummy Variable. This is designed to capture the effects of currency crises on a country. It takes a value of 0 for years in which no crisis was experienced and 1 otherwise. Sources: Carstens and Jácome (2005) and author's own calculations.

8. International Inflation Dummy Variable. This is designed to control for the effects of international inflation on a country, with the idea being that many countries (including developed ones) are susceptible to outside economic forces. It takes a value from 0 to 1 in .1 increments. Sources: Jácome and Vázquez (2005) and IMF Database.

9. Inflation Targeting Regime Dummy. This is devised to capture the effects of Inflation Targeting as either a primary or secondary objective for a nation's central bank. It will take the value of 1 if IT is considered a primary objective and .5 if it is considered a secondary objective, with the value of 0 given for the years prior to IT implementation. Additionally, there will be other distinctions made, such as if a central bank (or executive branch) is monitoring inflation or has a very informal inflation goal, but has not made a formal commitment to the Inflation Targeting framework. Also, for those that have formally committed to IT, a distinction will be made between a changing yearly inflation target and a fixed target (i.e. the same goal over the course of many years), with the latter taking a higher value. Sources: Stone (2003), Agbeja (2007), central bank websites.

10. Government Consumption as a Percentage of GDP. This will be used as a proxy measure of government spending or government expenditure in order to measure the

effects of fiscal policy on the dependent variables already described. Government consumption is considered to be government purchases of goods and services for current use (excluding military spending) and is considered to be a major portion of the gross domestic product Source: The Heritage Foundation.

11. Total Tax Revenue as a Percentage of GDP. This will also be a proxy measure of fiscal policy, representing the respective government's policy toward and ability to collect taxes relative to the size of the economy (although not a problem in developed economies, in many economies the government hampered in its ability to collect tax revenues though lack of enforcement and proper tax codes). Source: Revenue Statistics, 1965-2007 (2008).

4.2 Nonstationarity, Autocorrelation, and Hysteresis

In order to obtain an accurate regression, the time series under measurement must be stationary (its mean, variance and autocovariance remain the same regardless of what point in time they are measured). Nonstationarity in a time series can be defined as having a time-varying mean and/or variance and can lead to a spurious (nonsense) regression (Gujarati, 2003). Therefore, before engaging in regression analysis of any sort the time series will be differenced in order to achieve stationarity. Another method to verify that the regressions are not spurious (through non-stationary time series) is having the Durbin-Watson test result (see below) greater than the Adjusted R^2 (Brooks, 2002).

Autocorrelation occurs when values obtained are dependent upon previous values. Technically, it may be defined as "correlation between members of a series of observations ordered in time (as in time series data) or space (as in cross-sectional data)" (Kendall and Buckland, 1971, p.8). The classical regression model assumes that there is

no relationship between the disturbance term in one observation and that of any other observation. Autocorrelation can occur due to inertia (or “sluggishness”) in economic time series. The most well-known test for detecting autocorrelation is the Durbin-Watson test. Using the DW d statistic, if d is found to be 2 in application, no autocorrelation is assumed. However, the closer this number is to 0, the more evidence of positive autocorrelation. In a similar vein, the closer the number is to 4, the more evidence of negative autocorrelation (Gujarati, 2003).

Hysteresis refers to path dependence and the permanent effects of a temporary stimulus (i.e. “persistence”). Many aspects of economic systems are hysteretic (Cross et al, 2009). The persistence of first order difference equations with unit roots is also considered as such (Gocke, 2002). Although it is difficult to detect, some indication of persistence is evident if the sum of the coefficients on the lagged squared error and lagged conditional variance on the regression results are very close to unity (1.00) (Brooks, 2002).

4.3 Regression

Regression analysis is one of the most useful econometric techniques in practice. According to Gujarati (2003), “Regression analysis is concerned with the study of the dependence of one variable, the dependent variable, on one or more other variables, the explanatory variables, with a view to estimating and/or predicting the mean or average value of the former in terms of the known or fixed (in repeated sampling) values of the latter” (p. 18). The basic individual regression in this study will take the following form:

$$y = \alpha + \beta_1 \text{CBI}_1 + \beta_2 \text{CBI}_2 + \beta_3 \text{CBI}_3 + \dots + \varepsilon \quad (1)$$

The specific regressions given the above dependent and independent variables will be the following:

$$\begin{aligned} \textbf{Inflation Rate} = & \alpha - \beta_1 \text{Development Dummy} - \beta_2 \text{Modified Cukierman Index} - \beta_3 \text{ GMT} \\ & \text{Index} - \beta_4 \text{ Foreign Exchange Rate Regime Dummy} + \beta_5 \text{Currency Crisis Dummy} + \\ & \beta_6 \text{International Inflation Dummy} - \beta_7 \text{Legal Central Bank Independence Dummy} - \\ & \beta_8 \text{Operational Legal Independence Dummy} + \beta_9 \text{Central Bank Governor Turnover} - \text{IT} \\ & \text{Dummy} + \text{Cons} - \text{TaxRev} + \varepsilon \end{aligned} \quad (2)$$

$$\begin{aligned} \textbf{Money Supply} = & \alpha - \beta_1 \text{Development Dummy} - \beta_2 \text{Modified Cukierman Index} - \beta_3 \text{ GMT} \\ & \text{Index} - \beta_4 \text{ Foreign Exchange Rate Regime Dummy} + \beta_5 \text{Currency Crisis Dummy} + \\ & \beta_6 \text{International Inflation Dummy} - \beta_7 \text{Legal Central Bank Independence Dummy} - \\ & \beta_8 \text{Operational Legal Independence Dummy} + \beta_9 \text{Central Bank Governor Turnover} - \text{IT} \\ & \text{Dummy} + \text{Cons} - \text{TaxRev} + \varepsilon \end{aligned} \quad (3)$$

$$\begin{aligned} \textbf{Deposit Rate} = & \alpha - \beta_1 \text{Development Dummy} - \beta_2 \text{Modified Cukierman Index} - \beta_3 \text{ GMT} \\ & \text{Index} - \beta_4 \text{ Foreign Exchange Rate Regime Dummy} + \beta_5 \text{Currency Crisis Dummy} + \\ & \beta_6 \text{International Inflation Dummy} - \beta_7 \text{Legal Central Bank Independence Dummy} - \\ & \beta_8 \text{Operational Legal Independence Dummy} + \beta_9 \text{Central Bank Governor Turnover} - \text{IT} \\ & \text{Dummy} + \text{Cons} - \text{TaxRev} + \varepsilon \end{aligned} \quad (4)$$

$$\begin{aligned} \textbf{Foreign Exchange Rate (National Currency per \$US)} = & \alpha - \beta_1 \text{Development Dummy} - \\ & \beta_2 \text{Modified Cukierman Index} - \beta_3 \text{GMT Index} + \beta_4 \text{Foreign Exchange Rate Regime} \\ & \text{Dummy} + \beta_5 \text{Currency Crisis Dummy} + \beta_6 \text{International Inflation Dummy} - \beta_7 \text{Legal} \\ & \text{Central Bank Independence Dummy} - \beta_8 \text{Operational Legal Independence Dummy} + \\ & \beta_9 \text{Central Bank Governor Turnover} - \text{IT Dummy} - \text{Cons} + \text{TaxRev} + \varepsilon \end{aligned} \quad (5)$$

$$\begin{aligned} \textbf{Direct Investment in Economy \% Change} = & \alpha + \beta_1 \text{Development Dummy} + \beta_2 \text{Modified} \\ & \text{Cukierman Index} + \beta_3 \text{ GMT Index} + \beta_4 \text{ Foreign Exchange Rate Regime Dummy} - \\ & \beta_5 \text{Currency Crisis Dummy} - \beta_6 \text{International Inflation Dummy} + \beta_7 \text{Legal Central Bank} \\ & \text{Independence Dummy} + \beta_8 \text{Operational Legal Independence Dummy} - \beta_9 \text{Central Bank} \\ & \text{Governor Turnover} + \text{IT Dummy} - \text{Cons} + \text{TaxRev} + \varepsilon \end{aligned} \quad (6)$$

$$\text{Time Deposits \% Change} = \alpha + \beta_1 \text{Development Dummy} + \beta_2 \text{Modified Cukierman Index} + \beta_3 \text{GMT Index} + \beta_4 \text{Foreign Exchange Rate Regime Dummy} - \beta_5 \text{Currency Crisis Dummy} - \beta_6 \text{International Inflation Dummy} + \beta_7 \text{Legal Central Bank Independence Dummy} + \beta_8 \text{Operational Legal Independence Dummy} - \beta_9 \text{Central Bank Governor Turnover} + \text{IT Dummy} + \text{Cons} - \text{TaxRev} + \varepsilon$$

(7)

$$\text{Unemployment \%} = \alpha - \beta_1 \text{Development Dummy} - \beta_2 \text{Modified Cukierman Index} - \beta_3 \text{GMT Index} - \beta_4 \text{Foreign Exchange Rate Regime Dummy} + \beta_5 \text{Currency Crisis Dummy} + \beta_6 \text{International Inflation Dummy} - \beta_7 \text{Legal Central Bank Independence Dummy} - \beta_8 \text{Operational Legal Independence Dummy} + \beta_9 \text{Central Bank Governor Turnover} - \text{IT Dummy} + \text{Cons} - \text{TaxRev} + \varepsilon$$

(8)

$$\text{Exports \% Change} = \alpha + \beta_1 \text{Development Dummy} + \beta_2 \text{Modified Cukierman Index} + \beta_3 \text{GMT Index} + \beta_4 \text{Foreign Exchange Rate Regime Dummy} - \beta_5 \text{Currency Crisis Dummy} - \beta_6 \text{International Inflation Dummy} + \beta_7 \text{Legal Central Bank Independence Dummy} + \beta_8 \text{Operational Legal Independence Dummy} - \beta_9 \text{Central Bank Governor Turnover} + \text{IT Dummy} - \text{Cons} + \text{TaxRev} + \varepsilon$$

(9)

$$\text{GDP Volume Index} = \alpha + \beta_1 \text{Development Dummy} + \beta_2 \text{Modified Cukierman Index} + \beta_3 \text{GMT Index} + \beta_4 \text{Foreign Exchange Rate Regime Dummy} - \beta_5 \text{Currency Crisis Dummy} - \beta_6 \text{International Inflation Dummy} + \beta_7 \text{Legal Central Bank Independence Dummy} + \beta_8 \text{Operational Legal Independence Dummy} - \beta_9 \text{Central Bank Governor Turnover} + \text{IT Dummy} + \text{Cons} - \text{TaxRev} + \varepsilon$$

(10)

This study will first use a non-linear form of multivariate regression to test for correlation and relationships among the variables in question. Regressions will be done in a time series fashion over the time period under consideration (quarterly data for almost 50 years, to equal more than 190 observations for most time series). The “goodness of fit” ratio, or Adjusted R^2 will be used to gauge how well each dependent variable is explained by the independent variables. The Durbin-Watson Test will also be used for autocorrelation. A comparison of the Adjusted R^2 and Durbin-Watson test will be used as verification that the regression is not spurious (as mentioned above, if the Durbin-Watson results are quantitatively higher than the Adjusted R^2 , this is a sign that the regression is

not spurious, or non-sensical). Only results that are at least at the 10% statistical significance level will be listed throughout the results section and all will be listed with coefficients.

4.4 Volatility Clustering

The assumption under the Classical Linear Regression Model is that the variance of the errors is constant (homoscedastic). However, most time series are assumed to be heteroscedastic (the variance of errors is not constant). In other words, a current level of volatility tends to be positively correlated with its level in the previous periods. This “volatility clustering” or “volatility pooling” describes the tendency of large changes of time series values (of either sign) to follow large changes and small changes (of either sign) to follow small changes (Brooks, 2002). According to Frances (1998), “Given that news can lead to various economic interpretations, and also given that specific economic events like an oil crisis can last for some time, we often observe that large positive and large negative observations in financial time series tend to appear in clusters” (p. 155).

Given this phenomenon of volatility clustering, a model that does not assume constant variances and that describes how the error variances evolve would be of value. This is because when the time series data is no longer in a linear form, traditional Ordinary Least Squares cannot be used with any degree of accuracy. Therefore, Generalized Autoregressive Conditional Heteroskedasticity (GARCH) techniques are frequently used to examine time series which may exhibit wide swings in volatility, such as the ones presented in this study (i.e. volatility which occurs in “bursts, or “clusters”). It is a newer, more parsimonious and more generally accepted form of the Autoregressive Conditional Heteroscedastic (ARCH) form of models (Brooks, 2002).

Under the traditional ARCH model, the “autocorrelation in volatility” is modeled by allowing the conditional variance of the error term, σ_t^2 , to depend on the immediately previous value of the squared error. However, the ARCH model does not consider the conditional mean (which describes how the dependent variable varies over time). Researchers realized this problem and developed the GARCH model to bypass this shortcoming. The model thus considers not only the previous value of the squared error, but also that of the conditional mean/variance (Brooks, 2002). In other words, with these techniques, the error variance is assumed to be related to squared error terms several periods in the past as well as its conditional variance in the previous time period (Gujarati, 2003). Additionally, in the Eviews software package, to be used in this study, the Marquardt algorithm incorporates a “correction effect” which in effect pushes the coefficient estimates more quickly to their optimal values (Brooks, 2002). A GARCH model, then, takes volatility clustering into consideration and takes the following form:

$$\sigma_t^2 = \alpha_0 + \alpha_1 u_{t-1}^2 + \beta \sigma_{t-1}^2 \quad (11)$$

Therefore, another of the contributions of this study is that instead of traditional regression analysis using Ordinary Least Squares under the Classical Linear Regression Model (either via time series, cross sectional, or panel data techniques), this study will implement GARCH techniques in order to take into consideration the effects of common occurrences of volatility clustering. In this study, the significant variables (as measured by z-statistics), along with their coefficients, will be listed. As mentioned previously, an additional distinction that can be made by using GARCH models is that of noting the “persistence of shock” phenomenon and will be noted throughout the results sections.

4.5 Causation

Although regression analysis examines the dependence of one variable on other variables, it does not necessarily imply causation (Gujarati, 2003). According to Kendall and Stuart (1961), “A statistical relationship, however suggestive, can never establish causal connection” (p. 279). Therefore, once correlation is established, causality will be established using Granger Causality techniques. Although Granger Causality does not prove absolute causality, it based on the idea that “time does not run backward. That is, if event A happens before event B, then it is possible that A is causing B. However, it is not possible that B is causing A. In other words, events in the past can cause events to happen today. Future events cannot” (Koop, 2000, p. 175).

Therefore, Granger Causality, although powerful, could be considered more as precedence or predictive causality than true causality. In other words, it only measures whether one thing happens before another thing and helps predict it. However, as Diebold (2001) states, “The statement ‘ y_i causes y_j ’ is just shorthand for the more precise, but long-winded statement, ‘ y_i contains useful information for predicting y_j (in the linear least squares sense) over and above the past histories of the other variables in the system’. To save space, we simply say that y_i causes y_j ” (p. 254).

The Granger Causality test can only be applied to pairwise (two at a time) variables, so also may produce somewhat misleading results when examining a system of variables. One may assume that the Granger Causality exists if the null of “Variable X does not Granger Cause Variable Y” is rejected through the F-test (significance at 1%, 5%, or 10%). In this study, only statistically significant relationships will be listed, either unidirectionally (A causes B) or bidirectionally (A and B cause each other). Overall,

Granger Causality is a useful technique but should be interpreted with caution. It should also be noted that, as often happens, the absence of correlation (here done through regression) does not necessarily mean a lack of causation, as many times major reforms such as those of CBI take time to have any effect (Carlstrom and Fuerst, 2006).

Chapter 5- Results and Interpretations

Results will be organized by hypothesis. To help the reader, operational and legal CBI variables are italicized. In later hypotheses, the variable by which results are organized is also placed in bold font.

5.1 Inflation

H1- Central Bank Independence is negatively related to Inflation.

Regression results will be viewed for Inflation using GARCH. Again, only statistically significant (at 10% or less) variables are listed.

Table 5.1.1a-- Inflation GARCH Results: Overall and Regional

Region	Adj. R ²	DW	Significant Variables (Coefficients)	ARCH Coefficient (z-stat)	GARCH Coefficient (z-stat)
Overall Sample	.03	.78	<i>MCI</i> (.10), <i>GMT</i> (-.01), <i>CBGov</i> (-.10), <i>LCBI</i> (.60), IID (.07), DD (-.42), FXD (.33), IT (.25), Cons (.01)	1.04 (15.60)	.49 (31.61)
Asia	.11	.96	<i>MCI</i> (-.32), <i>GMT</i> (.01), <i>CBGov</i> (-.08), IID (.57), DD (-1.42), FCD (1.64), FXD (.80), Cons (-.01), IT (.53)	1.08 (14.96)	.46 (29.89)
Latin America	.07	.74	<i>MCI</i> (-.21), <i>GMT</i> (.00), <i>CBGov</i> (-.36), <i>LCBI</i> (.74), IID (.04), DD (.29), FCD (-.01), FXD (.36), IT (.51), Cons (.00), TaxRev (.00)	1.35 (3.38)	.45 (7.19)
Developed West	-.12	1.21	<i>GMT</i> (.03), <i>CBGov</i> (-.13), <i>LCBI</i> (-.23), IID (.26), DD (.46), FXD (.07), IT (.47), TaxRev (-.01)	.65 (15.76)	.57 (30.89)

Several items are worth noting here. For the Overall Sample as well as for all regional samples, central bank governor turnover can be slightly beneficial (indicated by a negative relationship between Inflation and CBBov), indicating that overall (and no doubt in moderation), that “new blood” can be advantageous. The coefficients are small; however, indicating that the link between turnover and inflation is not extremely strong (although statistically significant). Additionally, on the Overall level, both MCI and LCBI are positively correlated with inflation. It is perhaps surprising to note that Inflation

Targeting has not been effective overall. Government Consumption and Tax Revenue, while statistically significant, are not strongly correlated with inflation. The Foreign Exchange Rate Regime (FXD) variable is positively correlated with Inflation in all subsamples, reinforcing the notion that a floating exchange rate is positively associated with inflation levels, regardless of country-specific differences.

For Asia, the picture presented is unclear regarding inflation. It seems that operationally, there is a negative relationship between CBI and Inflation. As the region has developed, it appears that inflation rates have decrease. However, financial crises have taken their toll on the region, with a strong positive link between the two.

It is difficult to make any broad generalizations regarding Latin America, as this region unfortunately experienced extreme financial turmoil such as hyperinflation during much of the time period under question. Operational CBI appears to have been effective at stemming Inflation (although legal CBI has not). Like Asia, the region seems to be influenced by international inflation levels and a floating exchange rate tends to be correlated with greater inflation levels. Also encouraging is the tendency for country development to be negatively related to inflation.

One would expect for the developed countries in the overall sample to adhere better to what theory predicts. To some degree this is the case, with the LCBI negatively correlated with inflation. However, operationally, there appears to be little or no statistical relationship. Many of these countries' central banks were already fairly operationally independent, such as the Bank of England, which was only granted legal independence in 1997. However, it has been considered to be highly operationally

independent for decades (as has the United States' Federal Reserve). Perhaps the marginal benefit of more independence is not substantial.

Now, the pre- vs. post-CBI periods will be examined to pinpoint any gains achieved from the legal and/or operational rise in central bank autonomy.

Table 5.1.1b-- Inflation GARCH Results: Overall and Regional (Pre-CBI)

Region	Adj. R ²	DW	Significant Variables (Coefficients)	ARCH Coefficient (z-stat)	GARCH Coefficient (z-stat)
Overall Sample	.00	.45	<i>MCI</i> (1.02), <i>GMT</i> (-.03), <i>CBGov</i> (-.02), <i>IID</i> (.33), <i>DD</i> (-.44), <i>FCD</i> (-.13), <i>FXD</i> (-.45), <i>TaxRev</i> (.02)	2.32(3.70)	.27(9.20)
Asia	.09	.82	<i>MCI</i> (16.10), <i>GMT</i> (-.22), <i>CBGov</i> (.07), <i>IID</i> (.02), <i>FCD</i> (-.56), <i>FXD</i> (.57), <i>Cons</i> (.02), <i>TaxRev</i> (-.07)	2.02 (3.83)	.33(7.71)
Latin America	.44	.44	<i>MCI</i> (15.92), <i>GMT</i> (-.57), <i>CBGov</i> (.00), <i>IID</i> (.00), <i>DD</i> (.01), <i>FXD</i> (-9.05), <i>Cons</i> (.00), <i>TaxRev</i> (.52)	4.45(1.74)	.14(2.89)
Developed West	.18	.82	<i>MCI</i> (-.20), <i>GMT</i> (-.01), <i>CBGov</i> (-.09), <i>IID</i> (.69), <i>FCD</i> (.26), <i>FXD</i> (.45), <i>TaxRev</i> (-.01)	.57(9.30)	.61(21.96)

Table 5.1.1c-- Inflation GARCH Results: Overall and Regional (Post-CBI)

Region	Adj. R ²	DW	Significant Variables (Coefficients)	ARCH Coefficient (z-stat)	GARCH Coefficient (z-stat)
Overall Sample	.09	.72	<i>MCI</i> (-4.08), <i>GMT</i> (.18), <i>CBGov</i> (2.09), <i>IID</i> (-.97), <i>DD</i> (1.47), <i>FCD</i> (.28), <i>FXD</i> (2.76), <i>TaxRev</i> (-.15)	1.45(3.64)	.09(1.63)
Asia	-.23	1.35	<i>Cons</i> (.11), <i>IT</i> (.47)	.71(2.97)	.44(2.74)
Latin America	-.01	.68	<i>MCI</i> (-2.17), <i>GMT</i> (.21), <i>CBGov</i> (2.02), <i>IID</i> (-.37), <i>FXD</i> (1.52), <i>Cons</i> (.05), <i>IT</i> (.63), <i>TaxRev</i> (-.17)	1.74(4.16)	.08(2.09)
Developed West	-.01	2.15	<i>DD</i> (1.07), <i>FCD</i> (-.39), <i>Cons</i> (.06)	.37(3.67)*	.61(11.93)

* = evidence of hysteresis

In the pre-CBI period, there is little to indicate that operational independence has any power to lower inflation rates—there is a strong positive relationship between the MCI and Inflation in the Overall Sample as well as in Asia and Latin America. After legal CBI takes hold, one can observe the more expected negative relationship both

Overall and in Latin America. Again, it would seem that Inflation Targeting has not been as helpful in lowering inflation rates as hoped for.

Next, Granger Causality will be used to determine the effects of the independent variables on Inflation.

Table 5.1.2a-- Inflation Causality: Overall and Regional

Region	Unidirectional	Bidirectional
Overall Sample	None	MCI, GMT, LCBI, FCD, IID, IT, FXD
Asia	LCBI, FCD	FCD, IT, FXD, Cons
Latin America	GMT	LCBI, IID, IT, FXD
Developed West	None	CBGov, FCD, IID, DD, FXD, Cons

As the above table indicates, most of the relationships are bidirectional (indicating that the variables Granger Cause each other. Only in Asia and Latin America does one find unidirectional relationship regarding either operational or legal CBI and Inflation. Also worthy of note is that financial crises have been very influential in causing (either unidirectionally or bidirectionally) inflation rates.

Conclusion- The effectiveness of operational independence in lowering Inflation increased post-CBI. However, legal CBI does not appear to be quite as effective (except in the Developed West region).

5.2 Money Supply

H2- Central Bank Independence is negatively correlated to Money Supply growth.

Next, the effects of inflation will be examined from the supply side in terms of M1 Money Supply (currency circulation plus checking accounts). As was noted earlier, one of the key benefits of an independent central bank is that it does not have to acquiesce to the government's demands to print money. Therefore, the supply of money could perhaps grant some insight into the effectiveness of an autonomous central bank.

Table 5.2.1a-- Money Supply GARCH Results: Overall and Regional

Region	Adj. R ²	DW	Significant Variables (Coefficients)	ARCH Coefficient (z-stat)	GARCH Coefficient (z-stat)
Overall Sample	-.01	1.86	<i>MCI</i> (-.61), <i>GMT</i> (-.03), <i>CBGov</i> (1.38), <i>LCBI</i> (5.68), <i>IID</i> (1.93), <i>DD</i> (1.66), <i>FCD</i> (.19), <i>FXD</i> (.47), <i>Cons</i> (-.15), <i>IT</i> (-9.43), <i>TaxRev</i> (.04)	3.24(95.82)	.32(83.14)
Asia	-.01	2.08	<i>MCI</i> (-2.75), <i>GMT</i> (.08), <i>CBGov</i> (.00), <i>IID</i> (-.16), <i>DD</i> (.91), <i>FXD</i> (-.01), <i>Cons</i> (-.01), <i>TaxRev</i> (.05)	4.53(3.72)	.27(7.02)
Latin America	-.02	1.01	<i>MCI</i> (1.81), <i>GMT</i> (-.11), <i>CBGov</i> (.78), <i>LCBI</i> (9.65), <i>IID</i> (2.73), <i>DD</i> (-4.56), <i>FXD</i> (-.32), <i>Cons</i> (.03), <i>TaxRev</i> (.01)	4.94(3.39)	.24(3.72)
Developed West	-.04	2.16	<i>MCI</i> (-3.25), <i>GMT</i> (.27), <i>CBGov</i> (-3.09), <i>LCBI</i> (-3.49)	1.36(3.59)	.49(11.22)

There are several key relationships present. Operational independence (as expressed through the MCI and in some cases the GMT) is strongly negative, as theory would predict. This could be viewed as an indication that central bank autonomy is indeed working in some cases to “tighten the reins” on monetary policy. Another striking relationship is the extremely strong positive one between central bank governor turnover and money supply. Perhaps extremely high turnover leads to an increase in the money supply (i.e. through a “weak hand” at the helm of the country’s respective central bank). Many countries that have experienced periods of high or even hyperinflation have also simultaneously gone through periods of high CB governor turnover. Although an increase in the money supply does not by itself increase inflation, it does create upward pressure.

Unfortunately, Legal CBI seems to be strongly positively correlated with M1 except in the Developed West. Factors which may lead to this outcome (such as economic development and respect for the rule of law) will be examined in later results.

Now pre- vs. post-CBI results will be examined.

Table 5.2.1b-- Money Supply GARCH Results: Overall and Regional (Pre-CBI)

Region	Adj. R ²	DW	Significant Variables (Coefficients)	ARCH Coefficient (z-stat)	GARCH Coefficient (z-stat)
Overall Sample	-.04	.80	<i>MCI</i> (9.20), <i>GMT</i> (-.28), <i>CBGov</i> (.00), <i>IID</i> (.03), <i>DD</i> (-.99), <i>FCD</i> (2.50), <i>FXD</i> (-4.82), <i>Cons</i> (.00), <i>TaxRev</i> (.27)	8.40(1.64)	.09(3.92)
Asia	-.03	2.42	<i>IID</i> (-.01), <i>DD</i> (-3.99), <i>FCD</i> (-.23), <i>Cons</i> (.00), <i>TaxRev</i> (.10)	1.70(4.11)	.22(21.36)
Latin America	-.04	.79	<i>IID</i> (2.07), <i>DD</i> (-8.33), <i>Cons</i> (-.20)	5.16(.78)	.11(.82)
Developed West	-.12	2.03	<i>MCI</i> (1.24), <i>GMT</i> (.45), <i>CBGov</i> (-5.93), <i>IID</i> (-1.41), <i>DD</i> (-5.79), <i>FXD</i> (.66), <i>Cons</i> (-.07), <i>TaxRev</i> (.13)	1.72(4.80)	.44(7.80)

Table 5.2.1c-- Money Supply GARCH Results: Overall and Regional (Post-CBI)

Region	Adj. R ²	DW	Significant Variables (Coefficients)	ARCH Coefficient (z-stat)	GARCH Coefficient (z-stat)
Overall Sample	.09	1.05	<i>MCI</i> (20.12), <i>GMT</i> (-2.62), <i>CBGov</i> (24.88), <i>IID</i> (18.38), <i>DD</i> (-30.87), <i>FXD</i> (8.26), <i>Cons</i> (-.60), <i>TaxRev</i> (1.15)	.80(4.55)	.52(10.16)
Asia	-.21	2.29	<i>MCI</i> (19.49), <i>IID</i> (-19.61), <i>DD</i> (-16.61), <i>FCD</i> (2.61), <i>FXD</i> (-9.00), <i>TaxRev</i> (.61)	2.28(4.31)	.00(-.02)
Latin America	.06	1.05	<i>CBGov</i> (26.64), <i>FXD</i> (11.11), <i>Cons</i> (-1.19)	.75(3.75)	.50(9.56)
Developed West	.00	2.26	<i>GMT</i> (.22), <i>CBGov</i> (-6.49), <i>IID</i> (.59), <i>DD</i> (-.37), <i>FCD</i> (.82), <i>FXD</i> (1.71), <i>Cons</i> (.04), <i>TaxRev</i> (-.07)	3.04(4.59)	.23(6.27)

It is difficult to make any new distinctions regarding M1 in the pre- vs. post-CBI periods. It is apparent, however, that financial crises are, by in large, related to an increase in M1.

Next, causality as it relates to Money Supply will be examined.

Table 5.2.2a—Money Supply Causality: Overall and Regional

Region	Unidirectional	Bidirectional
Overall Sample	CBGov, IID	None
Asia	None	None
Latin America	CBGov, FXD	IID
Developed West	LCBI	None

It is apparent that there is a causal link between central bank governor turnover and Money Supply, both overall and in Latin America. Latin America has, especially

during times of crisis, experienced extremely high central bank governor turnover. Perhaps this high turnover is indeed a sign of underlying dependence on (and being answerable to) the executive branch of government, which at times throughout the last several decades has used the central bank for political purposes.

Although many times the relationship of the causal link (either positive or negative) is unknown, it is worthy to note that in some cases (in M1 supply and other measures) that there is bidirectional causality. In other words, each variable influences the other. For example, in Latin America, International Inflation has a bidirectional relationship with Money Supply.

Conclusion- Perhaps Central Bank Independence, when working properly, is able to stem the creation of money (as the literature suggests), thereby indirectly stemming inflation growth due to an independent, non-politically motivated or influenced central bank.

5. 3 Interest Rates

Hypothesis 3 will now be examined.

H3- Central Bank Independence is negatively related to Interest Rates.

The following regression results are obtained.

Table 5.3.1a--Deposit Rate GARCH Results: Overall and Regional

Region	Adj. R ²	DW	Significant Variables (Coefficients)	ARCH Coefficient (z-stat)	GARCH Coefficient (z-stat)
Overall Sample	-.04	2.79	<i>MCI</i> (369.27), <i>CBGov</i> (845.08), IID (259.55), DD (882.57), FXD (176.36), Cons (-35.71), TaxRev (-11.91)	4.37(3.14)	.00(-.98)
Asia	-.01	1.33	<i>MCI</i> (-.26), <i>GMT</i> (.02), <i>CBGov</i> (.23), IID (.16), DD (.42), FCD (.08), FXD (-.26), Cons (-.02)	2.22(5.85)	.48(13.59)
Latin America	-.01	2.36	TaxRev (-.72)	1.51(.79)	.00(-.46)
Developed West	-.02	2.11	<i>MCI</i> (-.16), IID (.80), DD (.21), FCD (-.60), FXD (-.34), Cons (-.01), IT (.16)	1.68(6.36)	.35(10.10)

Overall, there is a strikingly positive relationship between both the MCI and CBGov with deposit rates, which may derive in the vast part from Latin America's extended periods of hyperinflation and accompanying high interest rates (Asia and the Developed West each have extremely small coefficients, indicating a statistically significant, but relatively weak, relationship). Again, this seems to corroborate the belief that high central bank turnover is a sign of a weak and/or highly dependent central bank.

There is no indication that legal CBI is statistically related to deposit rates in any of the samples.

Now, the pre- vs. post-CBI effects will be measured.

Table 5.3.1b--Deposit Rate GARCH Results: Overall and Regional (Pre-CBI)

Region	Adj. R ²	DW	Significant Variables (Coefficients)	ARCH Coefficient (z-stat)	GARCH Coefficient (z-stat)
Overall	-.15	2.63	<i>CBGov</i> (1950.67), IID (2305.78), Cons (-177.21)	1.13(2.18)	-.01(-.87)
Asia	.00	1.43	<i>MCI</i> (8.92), <i>GMT</i> (-.21), <i>CBGov</i> (-.04), DD (.07), FCD (-.63), Cons (.01), TaxRev (-.02)	1.49(3.20)	.64(9.76)
Latin America	-.13	2.71	Cons (-506.71)	1.12(1.22)	-.01(-.31)
Developed West	-.02	2.16	<i>MCI</i> (-.27), <i>CBGov</i> (.03), IID (.65), DD (-.25), FCD (-.73), FXD (-.12), Cons (-.03), TaxRev (.02)	1.83(6.97)	.32(10.30)

Table 5.3.1c--Deposit Rate GARCH Results: Overall and Regional (Post-CBI)

Region	Adj. R ²	DW	Significant Variables (Coefficients)	ARCH Coefficient (z-stat)	GARCH Coefficient (z-stat)
Overall	-.03	2.36	None	.54(3.42)	-.01(-.04)
Asia	-.25	1.64	<i>MCI</i> (-.23), DD (.26), FXD (-.63)	.62(2.88)	.69(4.50)
Latin America	-.03	2.38	None	1.39(1.92)	.00(-.05)
Developed West	.08	1.05	<i>GMT</i> (-.05), IID (.55), DD (1.56), FCD (-.68), FXD (-.87), IT (.91)	.59(4.21)	.49(4.36)

In the Pre-CBI period, one sees the extraordinary positive coefficients between the indicators of operational CBI and interest rates (signs that during this time CBI was not effective). In the Post-CBI time frame, there is more indication that operational CBI is related to interest rates (as shown by the negative relationship in Asia and the Developed West).

Next, Granger Causality will be used to determine to what extent these independent variables influence the dependent variables under question.

Table 5.3.2a--Deposit Rates Causality: Overall and Regional

Region	Unidirectional	Bidirectional
Overall Sample	None	None
Asia	FCD	LCBI, FXD
Latin America	Cons	IID
Developed West	None	None

As can be seen above, overall and regionally, there is no causality of CBI on deposit rates (except in Asia on the bidirectional level). However, the Financial Crisis variable appears to have a statistically significant causal relationship in Asia as does the Foreign Exchange Rate Regime variable (on a bidirectional basis).

Conclusion- There is some weak evidence of operational CBI being negatively related to interest rates in Asia and the Developed West. However, no legal CBI relationship is present.

5.4 Foreign Exchange Rates

H4- There is a positive correlation between Central Bank Independence and the strength of the Exchange Rate.

The following are GARCH results for foreign exchange rates.

Table 5.4.1a--FX Rate GARCH Results: Overall and Regional

Region	Adj. R ²	DW	Significant Variables (Coefficients)	ARCH Coefficient (z-stat)	GARCH Coefficient (z-stat)
Overall Sample	-.03	1.88	<i>MCI</i> (-.05), <i>GMT</i> (-.04), <i>CBGov</i> (.20), <i>LCBI</i> (.42), <i>IID</i> (-.68), <i>DD</i> (-.28), <i>FCD</i> (.07), <i>FXD</i> (.40), <i>IT</i> (-.37), <i>TaxRev</i> (.01)	.70(131.11)	.85(4981.35)
Asia	.00	1.93	<i>MCI</i> (-13.58), <i>GMT</i> (-1.78), <i>CBGov</i> (.30), <i>LCBI</i> (-8.26), <i>IID</i> (.59), <i>DD</i> (-64.31), <i>FCD</i> (37.54), <i>FXD</i> (17.30), <i>Cons</i> (-.54), <i>IT</i> (17.01), <i>TaxRev</i> (1.72)	3.75(5.64)	.07(3.18)
Latin America	-.01	1.52	<i>MCI</i> (-3.34), <i>GMT</i> (.07), <i>CBGov</i> (-.01), <i>LCBI</i> (.08), <i>IID</i> (.01), <i>DD</i> (3.15), <i>FCD</i> (-.02), <i>FXD</i> (.61), <i>IT</i> (.20), <i>Cons</i> (.00), <i>TaxRev</i> (-.04)	2.00(54.82)	.57(342.22)
Developed West	-.01	1.85	<i>MCI</i> (-.36), <i>GMT</i> (-.01), <i>CBGov</i> (.87), <i>LCBI</i> (-.03), <i>IID</i> (-.06), <i>DD</i> (.07), <i>FCD</i> (.05), <i>FXD</i> (-.05), <i>Cons</i> (.01), <i>IT</i> (.13), <i>TaxRev</i> (-.01)	4.20(2.82)	.27(5.44)

Overall, it would seem that legal central bank independence has little or no relationship with foreign exchange rates (except in Asia, where it appears to have been very effective). However, in the Overall as well as several regional samples, operational independence (as expressed by the MCI and/or GMT index) is typically associated with a stronger exchange rate (fewer national currency units per \$US). Additionally, as theory would predict, there is a positive relationship between CBBov and exchange rates, indicating that high turnover is related to a weaker exchange rate.

Additionally, as the Development Dummy would suggest, there has been a relationship between country development and a stronger exchange rate. Financial crises have also had an effect in weakening exchange rates, especially in Asia. Lastly, a floating exchange rate seems to be positively correlated with the number of currency units per US\$ (i.e. a weaker exchange rate).

Now the pre- vs. post-CBI results will be examined.

Table 5.4.1b--FX Rate GARCH Results: Overall and Regional (Pre-CBI)

Region	Adj. R ²	DW	Significant Variables (Coefficients)	ARCH Coefficient (z-stat)	GARCH Coefficient (z-stat)
Overall Sample	.03	1.09	<i>MCI</i> (-.89), <i>GMT</i> (.10), <i>CBGov</i> (.24), <i>DD</i> (-1.20), <i>FCD</i> (-.01), Cons (.02), TaxRev (-.03)	2.24(2.34)	.37(8.44)
Asia	-.04	1.19	<i>MCI</i> (-99.81), <i>GMT</i> (2.93), <i>CBGov</i> (-.02), <i>DD</i> (-2.66), <i>FCD</i> (-.16), Cons (.01), TaxRev (.03)	3.67(3.29)	.23(4.12)
Latin America	.02	.53	<i>GMT</i> (.08), <i>CBGov</i> (-.53), <i>IID</i> (.26), <i>DD</i> (-.24), <i>FCD</i> (.20), Cons (.03), TaxRev (-.04)	1.01(1.25)	.56(3.56)
Developed West	-.02	1.31	<i>MCI</i> (-.47), <i>GMT</i> (.01), <i>CBGov</i> (.72), <i>IID</i> (.17), <i>DD</i> (-.71), <i>FXD</i> (.11), Cons (.01), TaxRev (-.01)	2.70(2.20)	.44(3.22)

Table 5.4.1c--FX Rate GARCH Results: Overall and Regional (Post-CBI)

Region	Adj. R ²	DW	Significant Variables (Coefficients)	ARCH Coefficient (z-stat)	GARCH Coefficient (z-stat)
Overall Sample	-.03	1.57	<i>MCI</i> (-.30), <i>GMT</i> (.08), <i>CBGov</i> (-1.28), <i>IID</i> (.10), <i>DD</i> (.61), <i>FCD</i> (-.02), <i>FXD</i> (-.11), Cons (.01), <i>IT</i> (.12), TaxRev (-.01)	5.04(3.71)	.09(3.92)
Asia	-.09	1.85	<i>CBGov</i> (-21.23), <i>IID</i> (1.96), <i>FCD</i> (-.42), <i>FXD</i> (-.48), Cons (.52)	.60(4.47)	.52(12.17)
Latin America	-.05	1.53	<i>MCI</i> (-.20), <i>GMT</i> (.06), <i>CBGov</i> (-.56), <i>DD</i> (-.48), <i>FCD</i> (-.13), <i>FXD</i> (.07), Cons (.00), <i>IT</i> (-.32), TaxRev (-.01)	2.44(4.17)	.39(10.20)
Developed West	.01	2.05	<i>MCI</i> (-15.93), <i>GMT</i> (-.73), <i>CBGov</i> (-29.46), <i>IID</i> (-15.93), <i>FXD</i> (-7.77), TaxRev (1.17)	-.01(-1.46)	.54(1.03)

It would seem that in both time periods, operational CBI has been related to lower (i.e. stronger) exchange rates. Next, foreign exchange rates will be examined through a causality perspective as in the previous hypotheses.

Table 5.4.2--FX Rates Causality: Overall and Regional

Region	Unidirectional	Bidirectional
Overall Sample	None	FCD
Asia	None	FCD
Latin America	None	MCI, FCD, DD, IT
Developed West	MCI, LCBI	IID

It seems that CBI, either operationally or legally, can have a causal impact on exchange rates. However, perhaps the most noteworthy item regarding causality is the occurrence of the financial crisis dummy, which left its mark on all regions except the Developed West (also to note that the relationship is bidirectional in all cases where it is present, indicating financial crises affecting exchange rates and vice versa).

Conclusion- There is evidence of a robust relationship between operational CBI and a stronger exchange rate. However, legal CBI, while showing that it can have influence (as seen in Asia), does not seem to be as dominant a force.

5.5 Direct Investment

H5- Central Bank Independence, with the resulting macroeconomic stability that it has engendered, is positively related to Direct Investment received.

After achieving the necessary stationarity, GARCH regression results are obtained. First, overall and regional statistics are presented, followed by individual country results.

Table 5.5.1a—Direct Investment GARCH Results: Overall and Regional

Region	Adj. R ²	DW	Significant Variables (Coefficients)	ARCH Coefficient (z-stat)	GARCH Coefficient (z-stat)
Overall Sample	-.04	1.92	<i>MCI</i> (396.30), <i>GMT</i> (25.37), <i>LCBI</i> (-249.79), <i>IID</i> (398.64), <i>DD</i> (-2160.63), <i>FCD</i> (261.65), <i>FXD</i> (-329.06), <i>Cons</i> (27.54),	3.33(3.32)	.00(-.03)
Asia	.00	2.00	<i>DD</i> (-2836.35)	3.62(1.52)	.00(-.02)
Latin America	-.01	2.01	<i>MCI</i> (17.16), <i>GMT</i> (.02), <i>CBGov</i> (-6.72), <i>IID</i> (.69), <i>DD</i> (-15.50), <i>FCD</i> (-.14), <i>FXD</i> (-2.05), <i>Cons</i> (.06), <i>IT</i> (-5.39), <i>TaxRev</i> (.12)	19.37(4.44)	.11(2.60)
Developed West	-.01	2.00	<i>LCBI</i> (-54.90), <i>FCD</i> (-45.47), <i>TaxRev</i> (-1.44)	14.41(1.20)	.00(-.68)

As can be observed, there is some evidence on the Overall as well as in Latin America that operational CBI is related to an increase in Direct Investment. However, legal CBI is strongly negatively correlated with Direct Investment. In terms of some of

the secondary variables, Government Consumption shows to be positively correlated with Direct Investment, as does Tax Revenue in Latin America.

Now, pre- vs. post CBI periods will be examined.

Table 5.5.1b—Direct Investment GARCH Results: Overall and Regional (Pre-CBI)

Region	Adj. R ²	DW	Significant Variables (Coefficients)	ARCH Coefficient (z-stat)	GARCH Coefficient (z-stat)
Overall Sample	-.01	1.93	<i>MCI</i> (79.12), <i>GMT</i> (-1.08), <i>CBGov</i> (.65), <i>IID</i> (4.98), <i>DD</i> (-8.90), <i>FCD</i> (-.54), <i>FXD</i> (-2.99), <i>Cons</i> (.03), <i>TaxRev</i> (-.30)	3.08(4.09)	.26(2.40)
Asia	-.03	1.92	<i>CBGov</i> (-.63), <i>IID</i> (-25.12), <i>DD</i> (63.19), <i>FCD</i> (1.32), <i>FXD</i> (12.48), <i>Cons</i> (-.70), <i>TaxRev</i> (.45)	5.62(1.89)	-4.63(-23.04)
Latin America	-.10	2.15	<i>MCI</i> (311.76), <i>GMT</i> (-13.75), <i>CBGov</i> (9.05), <i>IID</i> (-8.02), <i>DD</i> (63.46), <i>FCD</i> (-2.24), <i>FXD</i> (-206.10), <i>Cons</i> (1.01), <i>TaxRev</i> (11.58)	7.91(2.15)	.03(1.72)
Developed West	-.01	1.96	<i>MCI</i> (13.91), <i>GMT</i> (-1.53), <i>CBGov</i> (10.70), <i>IID</i> (-30.15), <i>DD</i> (56.23), <i>FXD</i> (14.24), <i>Cons</i> (2.12), <i>TaxRev</i> (-1.63)	3.50(1.44)	.36(3.39)

Table 5.5.1c—Direct Investment GARCH Results: Overall and Regional (Post-CBI)

Region	Adj. R ²	DW	Significant Variables (Coefficients)	ARCH Coefficient (z-stat)	GARCH Coefficient (z-stat)
Overall Sample	-.05	1.95	None	2.84(.90)	.00(-.02)
Asia	-.02	2.28	None	4.06(.71)	-.01(-.07)
Latin America	.00	2.23	<i>MCI</i> (128.18), <i>GMT</i> (-8.12), <i>CBGov</i> (-88.10), <i>IID</i> (56.85), <i>FCD</i> (-32.84), <i>FXD</i> (-122.81), <i>IT</i> (44.76), <i>TaxRev</i> (5.92)	5.26(2.28)	.00(-9.13)
Developed West	-.03	2.05	None	-.01(-.75)	.46(10.35)

Again, there is evidence that operational independence is positively linked with the amount of Direct Investment received in a country. If the rationale for this is examined, it seems logical that given that legal and/or operational central bank independence could be considered a form of positive signal to international investors

regarding the overall business climate and that these variables could be correlated with the inflow of Direct Investment.

Next, direct investment will be examined from a causality viewpoint.

Table 5.5.2a—Direct Investment Causality: Overall and Regional

Region	Unidirectional	Bidirectional
Overall Sample	DD	None
Asia	DD	None
Latin America	MCI, LCBI, IT	DD
Developed West	LCBI, DD, IT	None

From the above tables, it is worth noting that the operational (MCI) and legal (LCBI) variables have a causal relationship with Direct Investment in both Latin America and the Developed West (thus, this may indicate that any type of country may benefit from CBI, regardless of level of development).

Conclusion- There is strong evidence of operational CBI being positively correlated with Direct Investment. However, evidence for legal CBI in this regard is much weaker.

5.6 Savings

H6- Central Bank Independence, with the resulting macroeconomic stability that it has brought about, is positively related to the Savings level.

Below are regression results for Time Deposits (% change). First, as in the preceding hypotheses, overall and regional results for the entire time frame will be followed by a pre- vs. post CBI analysis.

Table 5.6.1a—Time Deposits GARCH Results: Overall and Regional

Region	Adj. R ²	DW	Significant Variables (Coefficients)	ARCH Coefficient (z-stat)	GARCH Coefficient (z-stat)
Overall Sample	.00	2.00	<i>MCI</i> (-11.81), <i>GMT</i> (-.09), <i>CBGov</i> (10.59), <i>LCBI</i> (4.46), <i>IID</i> (13.75), <i>DD</i> (-20.95), <i>FCD</i> (1.06), <i>FXD</i> (-3.83), <i>Cons</i> (-.27), <i>IT</i> (3.52), <i>TaxRev</i> (.54)	5.24(.95)	.38(11.00)
Asia	-.01	2.11	<i>MCI</i> (-11.26), <i>GMT</i> (.37), <i>CBGov</i> (-.21), <i>LCBI</i> (3.32), <i>IID</i> (-.06), <i>DD</i> (-30.52), <i>FXD</i> (-5.41), <i>Cons</i> (.05), <i>TaxRev</i> (.96)	3.08(3.19)	.32(5.28)
Latin America	-.01	1.98	<i>CBGov</i> (4.48), <i>LCBI</i> (4.69), <i>IID</i> (15.91), <i>DD</i> (-30.16), <i>FXD</i> (-2.83), <i>IT</i> (2.31)	8.29(.54)	.37(8.20)
Developed West	-.04	1.86	<i>MCI</i> (-4.11), <i>CBGov</i> (-2.12), <i>LCBI</i> (-2.82), <i>IID</i> , (-1.03), <i>DD</i> (-3.81), <i>FXD</i> (1.43), <i>IT</i> (-2.18), <i>TaxRev</i> (.39)	1.23(4.43)	.47(11.23)

It appears that evidence is mixed for this variable for the entire time frame. There is evidence that operational independence may actually be negatively correlated with Time Deposits, while legal CBI may have a positive relationship with this variable. Also worth noting is that there is a very consistent (for Overall as well as each regional sample) negative relationship between the Development Dummy and Time Deposits. In other words, it would seem that as a country develops economically, its population tends to save less. There is also a fairly consistently positive relationship between Direct Investment and Tax Revenue collected (which would be consistent with fiscal policy theory, as an increase in tax revenue should lead to a depreciation of the nation's currency, making its assets "cheaper" for international investors).

Now, as before, pre- vs. post-CBI results will be examined.

Table 5.6.1b—Time Deposits GARCH Results: Overall and Regional (Pre-CBI)

Region	Adj. R ²	DW	Significant Variables (Coefficients)	ARCH Coefficient (z-stat)	GARCH Coefficient (z-stat)
Overall Sample	-.02	.97	<i>MCI</i> (28.57), <i>GMT</i> (-.64), <i>CBGov</i> (-.12), <i>IID</i> (-.09), <i>DD</i> (.36), <i>FXD</i> (-8.97), <i>Cons</i> (.03), <i>TaxRev</i> (.39)	7.34(4.73)	.05(1.92)
Asia	.08	1.71	<i>DD</i> (-11.45), <i>Cons</i> (.46)	-.01(-13.56)	.55(1.13)
Latin America	-.04	.95	<i>MCI</i> (11.94), <i>GMT</i> (-.07), <i>CBGov</i> (1.93), <i>IID</i> (3.60), <i>DD</i> (-10.12), <i>FCD</i> (1.55), <i>Cons</i> (-.38), <i>TaxRev</i> (.32)	12.68(3.62)	.01(.81)
Developed West	-.11	1.31	<i>MCI</i> (-4.81), <i>GMT</i> (-.32), <i>DD</i> (-8.05), <i>FXD</i> (1.85), <i>Cons</i> (-.29), <i>TaxRev</i> (.58)	1.05(2.42)	.51(4.73)

Table 5.6.1c—Time Deposits GARCH Results: Overall and Regional (Post-CBI)

Region	Adj. R ²	DW	Significant Variables (Coefficients)	ARCH Coefficient (z-stat)	GARCH Coefficient (z-stat)
Overall Sample	.09	.64	<i>MCI</i> (27.47), <i>GMT</i> (-2.42), <i>CBGov</i> (13.75), <i>IID</i> (16.41), <i>DD</i> (-22.71), <i>FCD</i> (-1.41), <i>FXD</i> (2.19), <i>IT</i> (2.89), <i>TaxRev</i> (.60)	1.88(4.38)	.31(6.37)
Asia	.11	2.14	<i>GMT</i> (-6.47), <i>FXD</i> (-5.90), <i>IT</i> (-8.38), <i>TaxRev</i> (1.27)	1.86(.98)	.00(-1.44)
Latin America	.12	.67	<i>MCI</i> (30.31), <i>GMT</i> (-5.24), <i>CBGov</i> (35.17), <i>IID</i> (20.35), <i>DD</i> (23.85), <i>FCD</i> (-5.10), <i>FXD</i> (-.79), <i>Cons</i> (-1.01), <i>TaxRev</i> (1.87)	1.30(4.27)	.34(5.72)
Developed West	.07	1.89	<i>MCI</i> (-1.26), <i>CBGov</i> (2.02), <i>IID</i> (.11), <i>FXD</i> (1.70), <i>Cons</i> (.05), <i>TaxRev</i> (-.02)	1.84(6.85)	.28(10.86)

Again, results seem to be unclear. Perhaps a causality analysis will provide more insight.

Table 5.6.2a—Time Deposits Causality: Overall and Regional

Region	Unidirectional	Bidirectional
Overall Sample	IID	None
Asia	None	None
Latin America	IID	None
Developed West	GMT, DD, IT, Cons, TaxRev	LCBI

As can be seen, there is very little causal evidence to support the claim that central bank independence is associated with Savings.

Conclusion- There seems to be relatively little relationship between Time Savings and CBI. However, it is noteworthy that the secondary/control variables of Development and Tax Revenue both appear consistently (the former with a negative relationship to the variable in question, the latter with a positive relationship).

5.7 Unemployment

H7- There is an inverse correlation between Central Bank Independence and Unemployment.

The following table presents the regression results for Unemployment. Once more, there is significant influence from financial crises, international inflation, the foreign exchange regime in place, and the development level of the country. It would be expected that an increase in central bank independence would engender more economic stability and thus a lower unemployment rate. However, the results seem to be mixed and on occasion contradict one another. As has been the case throughout, the Financial Crisis dummy variable proves to be a force, being positively related to Unemployment in many of the sample countries below (i.e. unemployment increases in the years in which a currency crisis takes place). Inflation Targeting seems to be a beneficial force, being inversely related to Unemployment throughout (perhaps through helping to establish a stable economic environment).

Table 5.7.1a--Unemployment GARCH Results: Overall and Regional

Region	Adj. R ²	DW	Significant Variables (Coefficients)	ARCH Coefficient (z-stat)	GARCH Coefficient (z-stat)
Overall Sample	.00	2.19	<i>MCI</i> (.03), <i>CBGov</i> (-.03), <i>LCBI</i> (-.04), <i>IID</i> (.00), <i>DD</i> (.23), <i>FCD</i> (.31), <i>FXD</i> (.00), <i>Cons</i> (.00), <i>IT</i> (-.21), <i>TaxRev</i> (.00)	1.08(6.29)	.77(43.80)
Asia	-.01	2.59	<i>MCI</i> (.04), <i>GMT</i> (.00), <i>CBGov</i> (.01), <i>LCBI</i> (.00), <i>IID</i> (.01), <i>DD</i> (-.01), <i>FCD</i> (.00), <i>FXD</i> (.01), <i>Cons</i> (.00), <i>IT</i> (-.01), <i>TaxRev</i> (.00)	6.22(21.44)	.56(67.71)
Latin America	-.07	1.91	<i>MCI</i> (.22), <i>GMT</i> (-.01), <i>CBGov</i> (-.10), <i>LCBI</i> (.34), <i>IID</i> (-.02), <i>DD</i> (.12), <i>FCD</i> (-.22), <i>FXD</i> (.06), <i>Cons</i> (-.01), <i>IT</i> (-.35), <i>TaxRev</i> (.00)	1.35(7.21)	.71(64.19)
Developed West	-.01	2.04	<i>MCI</i> (-.01), <i>GMT</i> (.00), <i>CBGov</i> (.01), <i>IID</i> (-.03), <i>DD</i> (.03), <i>FCD</i> (.01), <i>FXD</i> (.12), <i>Cons</i> (.01), <i>IT</i> (-.11), <i>TaxRev</i> (-.01)	1.32(6.36)	.60(30.22)

There is mixed evidence regarding CBI and Unemployment for the full time frame (although for the Overall Sample, one can note a negative relationship between the dependent variable and LCBI). However, from the pre- vs. post-CBI analysis, one can see that only in the Developed West is operational CBI negatively linked to Unemployment (i.e. an increase in operational CBI would be related to a drop in Unemployment).

Table 5.7.1b--Unemployment GARCH Results: Overall and Regional (Pre-CBI)

Region	Adj. R ²	DW	Significant Variables (Coefficients)	ARCH Coefficient (z- stat)	GARCH Coefficient (z-stat)
Overall Sample	-.01	2.01	<i>MCI</i> (.33), <i>GMT</i> (-.01), <i>CBGov</i> (-.05), <i>IID</i> (.03), <i>FCD</i> (.02), <i>FXD</i> (-.02), <i>Cons</i> (.00), <i>TaxRev</i> (.00)	7.33(4.22)	.40(8.44)
Asia	.01	2.23	<i>CBGov</i> (.00), <i>IID</i> (.01), <i>DD</i> (.01), <i>FCD</i> (-.04), <i>FXD</i> (.00), <i>Cons</i> (.00), <i>TaxRev</i> (-.01)	2.96(2.54)	.43(8.31)
Latin America	.02	2.02	<i>CBGov</i> (.02), <i>IID</i> (-.02), <i>FCD</i> (-.02), <i>Cons</i> (-.01)	10.12(4.64)	.28(12.71)
Developed West	-.02	2.05	<i>MCI</i> (-.01), <i>GMT</i> (.00), <i>CBGov</i> (.01), <i>IID</i> (-.05), <i>FXD</i> (.13), <i>Cons</i> (.01), <i>TaxRev</i> (-.01)	1.44(4.96)	.61(33.46)

Table 5.7.1c--Unemployment GARCH Results: Overall and Regional (Post-CBI)

Region	Adj. R ²	DW	Significant Variables (Coefficients)	ARCH Coefficient (z-stat)	GARCH Coefficient (z-stat)
Overall Sample	.01	1.99	MCI (.68), FCD (.22), FXD (-1.14), TaxRev (.02)	.24(3.23)*	.78(15.27)
Asia	-.05	2.05	DD (-1.49)	-.04(-.68)	.93(4.20)
Latin America	.03	2.05	MCI (1.01), IID (.84), FXD (-1.12)	.12(2.19)*	.85(8.63)
Developed West	.04	2.20	MCI (-.46), GMT (-.08), CBGov (-4.27), IID (-.51), FCD (.19), Cons (.09)	.43(3.08)*	.62(8.69)

* = evidence of hysteresis

Next, causality will be examined to determine to what degree the independent variables can be shown to Granger Cause the dependent variable, Unemployment %.

Table 5.7.2a--Unemployment Causality: Overall and Regional

Region	Unidirectional	Bidirectional
Overall Sample	FCD, IT	None
Asia	MCI, IT, FXD	LCBI, FCD
Latin America	None	None
Developed West	FCD	None

Again, it is difficult to ascertain to what extent (if any) central bank independence has been able to lead to a causal effect on unemployment.

Conclusion- Although there is some evidence related to operational CBI and Unemployment, it is only for one region (Developed West). Legal CBI does not appear to play much (if any) of a role.

5.8 Exports

H8- Central Bank Independence, with the resulting macroeconomic stability that it has engendered, is positively related to Exports.

First, regression results for the overall and regional portions will be examined. As can be seen, there is fairly strong evidence of Exports being positively related to both legal and operational independence in the Overall Sample as well as in Asia. There is not much supporting evidence in Latin America or the Developed West, however.

Interestingly, there is a large display of persistence for exports, indicating that hysteresis may be present. In other words, an initial “shock” or large increase or decrease in exports may have very lasting effects. Inflation Targeting also appears positive and statistically positive in several areas, indicating that stable inflation may also be related to an increase in exports.

Table 5.8.1a--Exports GARCH Results: Overall and Regional

Region	Adj. R ²	DW	Significant Variables (Coefficients)	ARCH Coefficient (z-stat)	GARCH Coefficient (z-stat)
Overall Sample	-.01	2.43	<i>GMT</i> (.07), <i>CBGov</i> (-.01), <i>LCBI</i> (1.38), <i>IID</i> (5.56), <i>DD</i> (-.74), <i>Cons</i> (.02), <i>IT</i> (1.99), <i>TaxRev</i> (-.08)	.30(9.05)*	.79(50.24)
Asia	-.03	2.58	<i>MCI</i> (17.27), <i>GMT</i> (.30), <i>CBGov</i> (.05), <i>IID</i> (.76), <i>DD</i> (.53), <i>Cons</i> (.12), <i>TaxRev</i> (-.32)	.55(5.39)	.69(20.53)
Latin America	-.01	2.33	<i>IID</i> (8.01), <i>DD</i> (-1.03), <i>IT</i> (3.93)	.31(3.55)	.79(18.43)
Developed West	-.01	2.90	<i>GMT</i> (-.27), <i>IID</i> (2.64), <i>DD</i> (8.00), <i>FXD</i> (-3.45), <i>Cons</i> (.54), <i>TaxRev</i> (-.27)	.22(5.45)*	.81(30.24)

* = evidence of hysteresis

Now the pre- vs. post-CBI effects will be observed.

Table 5.8.1b--Exports GARCH Results: Overall and Regional (Pre-CBI)

Region	Adj. R ²	DW	Significant Variables (Coefficients)	ARCH Coefficient (z-stat)	GARCH Coefficient (z-stat)
Overall Sample	-.03	2.41	<i>GMT</i> (.78), <i>IID</i> (.52), <i>Cons</i> (.11), <i>TaxRev</i> (-.28)	1.44(7.68)	.44(20.86)
Asia	-.05	2.89	<i>MCI</i> (-74.89), <i>GMT</i> (3.00), <i>CBGov</i> (.23), <i>IID</i> (-.15), <i>DD</i> (.96), <i>Cons</i> (.14), <i>TaxRev</i> (-.31)	3.92(3.95)	.09(2.60)
Latin America	-.01	2.27	<i>IID</i> (7.90), <i>DD</i> (-.94)	.38(2.79)	.79(11.47)
Developed West	-.02	2.89	<i>GMT</i> (-.45), <i>IID</i> (2.72), <i>FCD</i> (4.46), <i>Cons</i> (.40)	.31(3.97)*	.75(15.95)

* = evidence of hysteresis

Table 5.8.1c--Exports GARCH Results: Overall and Regional (Post-CBI)

Region	Adj. R ²	DW	Significant Variables (Coefficients)	ARCH Coefficient (z-stat)	GARCH Coefficient (z-stat)
Overall Sample	-.01	2.29	None	.07(1.96)*	.92(22.34)
Asia	.01	2.35	DD (48.64)	.22(1.35)*	.75(3.23)
Latin America	-.03	2.28	IID (8.51), IT (5.93)	-.09(-1.92)	.36(.75)
Developed West	-.03	3.00	DD (12.57)	.32(3.84)*	.74(13.86)

* = evidence of hysteresis

The pre- vs. post analysis seems to add to the complexity of the relationship, with mixed results. Next, causality will be examined. Interestingly, there is a good bit of causal relationship present (with the purely CBI variables as well as the control variables), as the following tables show.

Table 5.8.2a--Exports Causality: Overall and Regional

Region	Unidirectional	Bidirectional
Overall Sample	MCI, CBGov, LCBI, FCD, IID, DD, IT, FXD, Cons, TaxRev	None
Asia	CBGov, IID, DD, TaxRev	LCBI, IT, Cons
Latin America	MCI, GMT, CBGov, LCBI, IID, DD, FXD, Cons, TaxRev	None
Developed West	LCBI, FCD, IID, DD, IT, FXD, TaxRev	GMT, Cons

Taking the above regional tables together, it is interesting to note that there is some degree of CBI causality present in many of the countries for the overall time period.

It is noteworthy that the variable CBGov (Central Bank Governor Turnover) is a causal variable in many cases, overall, pre- and post-CBI. This could be due to the signaling effect that has been seen in other cases in this study. For example, after a country's central bank is made independent, more attention is focused on it. If there is an increase in turnover, this could perhaps bode negatively for economic stability in general and for exporters specifically. However, results are inconclusive at this point.

Conclusion- There is moderate evidence that both operational and legal CB independence are positively related to exports. Other disciplinary measures such as Inflation Targeting are also positively related, seeming to indicate that exporters (and their customers) benefit from the economic stability brought about by CBI and its supplementary measures.

5.9 Economic Growth

H9- There is a positive correlation between Central Bank Independence and Economic Growth.

Arguably the most important outcome CBI could ultimately provide is to be correlated with economic growth. With that in mind, regression results are now presented for GDP Volume Index.

Table 5.9.1a--GDP GARCH Results: Overall and Regional

Region	Adj. R ²	DW	Significant Variables (Coefficients)	ARCH Coefficient (z-stat)	GARCH Coefficient (z-stat)
Overall Sample	.00	2.92	<i>MCI</i> (.21), <i>GMT</i> (.00), <i>CBGov</i> (-.12), <i>LCBI</i> (.32), <i>IID</i> (.01), <i>DD</i> (.30), <i>FCD</i> (-.37), <i>FXD</i> (.35), <i>Cons</i> (.00), <i>IT</i> (-.31), <i>TaxRev</i> (-.01)	2.60(8.87)	.28(16.78)
Asia	-.01	2.82	<i>MCI</i> (-.45), <i>GMT</i> (.02), <i>LCBI</i> (.83), <i>IID</i> (-.15), <i>DD</i> (-1.15), <i>FCD</i> (-.32), <i>FXD</i> (.31), <i>Cons</i> (.01), <i>IT</i> (-1.03), <i>TaxRev</i> (.01)	1.58(9.30)	.39(30.03)
Latin America	-.01	3.12	<i>MCI</i> (.12), <i>GMT</i> (.01), <i>CBGov</i> (-.08), <i>LCBI</i> (-.07), <i>IID</i> (.09), <i>DD</i> (.15), <i>FCD</i> (-.03), <i>FXD</i> (.11), <i>Cons</i> (-.01), <i>IT</i> (.06), <i>TaxRev</i> (-.01)	8.60(14.99)	.03(6.98)
Developed West	.02	2.20	<i>CBGov</i> (-.01), <i>LCBI</i> (.39), <i>IID</i> (-.08), <i>DD</i> (.31), <i>FCD</i> (-.58), <i>Cons</i> (.01), <i>IT</i> (.25), <i>TaxRev</i> (-.01)	1.36(7.27)	.46(17.02)

There is quite a bit of evidence to support the hypothesis that economic growth is positively related to CBI (both operationally and legally). *LCBI* appears significant and fairly large (coefficient-wise) in the Overall, Asia and Developed West samples.

Operational independence (as expressed by the MCI and GMT) appear positively correlated for the most part as well. Interestingly (and true to theory), there is a negative relationship between CBGov and economic growth, possibly indicating that central bank governor turnover, in excess, is harmful for an economy's growth.

Again, one sees the financial crises and the possible negative implications that these phenomenon have on growth in the Overall Sample as well as in all regions. Also, the inverse relationship with Inflation Targeting perhaps contradicts theory, while the negative relationship with Tax Revenue tends to support fiscal policy theory.

Table 5.9.1b--GDP GARCH Results: Overall and Regional (Pre-CBI)

Region	Adj. R ²	DW	Significant Variables (Coefficients)	ARCH Coefficient (z-stat)	GARCH Coefficient (z-stat)
Overall Sample	.00	2.54	<i>MCI</i> (1.02), <i>GMT</i> (-.03), <i>CBGov</i> (.01), <i>IID</i> (.16), <i>FCD</i> (-.02), <i>FXD</i> (-.54), <i>Cons</i> (-.01), <i>TaxRev</i> (.03)	5.82(2.13)	.07(1.85)
Asia	-.01	2.61	<i>MCI</i> (17.32), <i>GMT</i> (-.40), <i>CBGov</i> (.00), <i>IID</i> (-.01), <i>DD</i> (-.03), <i>FXD</i> (.00), <i>Cons</i> (.00), <i>TaxRev</i> (.00)	.64(7.55)	.50(21.51)
Latin America	.03	2.44	<i>MCI</i> (4.87), <i>GMT</i> (-.16), <i>IID</i> (.14), <i>DD</i> (-.47), <i>FCD</i> (-.01), <i>FXD</i> (-2.67), <i>Cons</i> (.00), <i>TaxRev</i> (.15)	4.10(1.10)	.00(-.22)
Developed West	-.01	2.37	<i>MCI</i> (.10), <i>GMT</i> (.01), <i>CBGov</i> (-.01), <i>ID</i> (-.11), <i>DD</i> (1.05), <i>FXD</i> (.09), <i>Cons</i> (.00), <i>TaxRev</i> (-.02)	1.65(7.21)	.37(18.66)

Table 5.9.1c--GDP GARCH Results: Overall and Regional (Post-CBI)

Region	Adj. R ²	DW	Significant Variables (Coefficients)	ARCH Coefficient (z- stat)	GARCH Coefficient (z-stat)
Overall Sample	-.02	2.95	<i>MCI</i> (1.13), <i>CBGov</i> (.20), <i>IID</i> (-.03), <i>DD</i> (.05), <i>FCD</i> (-.01), <i>FXD</i> (.40), <i>Cons</i> (.00), <i>IT</i> (-.12), <i>TaxRev</i> (-.05)	3.20(3.24)	.17(3.18)
Asia	.02	1.61	<i>CBGov</i> (-2.68), <i>DD</i> (6.70), <i>FCD</i> (-.03), <i>FXD</i> (-.05), <i>TaxRev</i> (-.16)	2.31(3.84)	.13(5.10)
Latin America	-.03	2.97	<i>GMT</i> (.07), <i>CBGov</i> (.60), <i>IID</i> (-.01), <i>DD</i> (1.34), <i>FCD</i> (-.02), <i>FXD</i> (.34), <i>Cons</i> (.01), <i>IT</i> (-.08), <i>TaxRev</i> (-.09)	2.10(4.00)	.23(6.16)
Developed West	.10	1.93	<i>DD</i> (1.73), <i>FCD</i> (-1.11), <i>FXD</i> (-.97), <i>Cons</i> (-.08), <i>IT</i> (2.05)	.33(2.73)*	.72(9.20)

* = evidence of hysteresis

To summarize the above tables, several key inferences can be made from the above tables regarding GDP Growth:

- a.) There is strong evidence of a positive relationship between both legal and operational CBI and economic growth.
- b.) Although Central Bank Governor Turnover (CBGov) proves significant in many cases, it is mixed (with no suggestive pattern). As has been stated before, for some developed countries (for example, England and the United States) with low governor turnover, a marginal increase in this regard may actually prove beneficial (i.e. “new blood”). However, in general, this proxy regarding the effects of CBI on economic growth in particular and all economic variables in general must be used with care.
- c.) Financial crises are negatively correlated with economic growth in all cases.

Now Granger Causality techniques will be used.

Table 5.9.2a--GDP Causality: Overall and Regional

Region	Unidirectional	Bidirectional
Overall Sample	MCI, GMT, FCD, IID, FXD, TaxRev	LCBI, DD, IT
Asia	MCI, FCD, FXD, TaxRev	LCBI, DD, IT
Latin America	MCI, GMT, LCBI, DD, IT, FXD	None
Developed West	FCD, FXD, Cons	GMT, CBBov, LCBI, DD, IT, TaxRev

The evidence above indicates that the control variables of International Inflation, Country Development, Financial Crises, and the Foreign Exchange Rate regime, among other factors, have had a causal impact on GDP as well as the measurements of actual central bank independence.

Conclusion- There is a good deal of evidence that central bank independence, either operationally or legally, is related to economic growth.

For the next set of hypotheses, issues such as corruption, legal systems, inflation targeting, and level of constraints (i.e., “checks and balances”) will be examined with countries divided into groups according to their ranks (High, Medium, Low or High, Medium High, Medium, Medium Low, Low) in these categories. Each hypothesis will have three variables—Inflation, the GDP Volume Index, and Unemployment—which will be measured in turn.

5.10 Corruption Levels

H10- There is stronger evidence of CBI effectiveness among countries with lower levels of Corruption.

a.) Inflation

Table 5.10.1a—Inflation GARCH Results: Corruption Levels

Level	Adj. R ²	DW	Significant Variables (Coefficients)	ARCH Coefficient (z-stat)	GARCH Coefficient (z-stat)
Low	-.15	1.06	<i>MCI</i> (-.14), <i>GMT</i> (-.01), <i>CBGov</i> (.01), <i>IID</i> (.01), <i>DD</i> (.80), <i>FXD</i> (.15), <i>IT</i> (.41), <i>Cons</i> (.00), <i>TaxRev</i> (.00)	.90(18.68)	.48(50.39)
Medium	.03	.55	<i>MCI</i> (.11), <i>GMT</i> (.01), <i>CBGov</i> (.03), <i>LCBI</i> (.67), <i>IID</i> (.58), <i>DD</i> (-1.33), <i>FXD</i> (-.12), <i>IT</i> (.75), <i>Cons</i> (-.01), <i>TaxRev</i> (.01)	1.84(10.83)	.29(17.68)
High	.11	.93	<i>MCI</i> (.34), <i>GMT</i> (.00), <i>CBGov</i> (-.08), <i>LCBI</i> (.64), <i>IID</i> (-.01), <i>DD</i> (.16), <i>FCD</i> (-.17), <i>IT</i> (5.06), <i>Cons</i> (-.02), <i>TaxRev</i> (-.01)	1.47(7.84)	.42(18.23)

As can be seen, the effect of operational independence is moderately strong at low levels of corruption and loses strength as corruption levels increase. Interestingly, the IT variable increases dramatically as corruption levels move from Low to High.

b.) GDP Growth

Table 5.10.1b—GDP GARCH Results: Corruption Levels

Level	Adj. R ²	DW	Significant Variables (Coefficients)	ARCH Coefficient (z-stat)	GARCH Coefficient (z-stat)
Low	.00	2.55	<i>MCI</i> (-.03), <i>GMT</i> (.00), <i>CBGov</i> (-.01), <i>LCBI</i> (.50), <i>IID</i> (.23), <i>FCD</i> (-.55), <i>FXD</i> (.17), <i>IT</i> (.27), <i>Cons</i> (.00), <i>TaxRev</i> (.00)	1.75(9.74)	.36(21.73)
Medium	-.01	3.13	<i>MCI</i> (.48), <i>GMT</i> (.00), <i>CBGov</i> (.04), <i>LCBI</i> (-.04), <i>IID</i> (.11), <i>DD</i> (.53), <i>FCD</i> (-.06), <i>FXD</i> (.17), <i>IT</i> (-.11), <i>Cons</i> (.01), <i>TaxRev</i> (-.02)	3.64(6.26)	.20(7.77)
High	-.02	2.69	<i>MCI</i> (.16), <i>GMT</i> (.00), <i>CBGov</i> (.01), <i>LCBI</i> (-.12), <i>IID</i> (.10), <i>DD</i> (.01), <i>FCD</i> (.05), <i>FXD</i> (-.06), <i>IT</i> (.49), <i>Cons</i> (.00), <i>TaxRev</i> (.00)	13.14(1.66)	.08(2.47)

It appears that both operational and legal CBI are positively related to economic growth (however, the degree of correlation varies somewhat over regions). In other words, operational and/or legal CBI can make a difference in economic growth regardless of corruption levels (with legal CBI perhaps playing a larger role in low corruption countries and operational CBI playing a larger role in highly corrupt countries).

c.) Unemployment

Table 5.10.1c—Unemployment GARCH Results: Corruption Levels

Level	Adj. R ²	DW	Significant Variables (Coefficients)	ARCH Coefficient (z-stat)	GARCH Coefficient (z-stat)
Low	-.07	1.92	<i>MCI</i> (-.45), <i>GMT</i> (-.03), <i>CBGov</i> (.29), <i>IID</i> (.07), <i>DD</i> (.37), <i>FCD</i> (.14), <i>FXD</i> (-.08), <i>IT</i> (.11), <i>Cons</i> (.02), <i>TaxRev</i> (-.01)	.65(6.55)	.71(22.67)
Medium	-.04	1.92	<i>MCI</i> (-.10), <i>GMT</i> (.00), <i>CBGov</i> (-.03), <i>IID</i> (-.02), <i>DD</i> (.37), <i>FXD</i> (.04), <i>Cons</i> (.00), <i>TaxRev</i> (.00)	2.22(3.63)	.58(24.30)
High	-.01	2.34	<i>MCI</i> (.31), <i>GMT</i> (.01), <i>CBGov</i> (-.01), <i>LCBI</i> (.12), <i>IID</i> (.10), <i>DD</i> (-.53), <i>FXD</i> (.08), <i>IT</i> (-.34), <i>Cons</i> (.00), <i>TaxRev</i> (-.01)	1.66(3.55)	.54(8.97)

Operational independence is strongly positively correlated with Unemployment in Low corruption level countries, but gradually changes direction as corruption levels increase.

Conclusion- For Inflation, a low level of corruption is linked to CBI effectiveness (as expressed by operational and/or legal independence). For GDP growth, low and medium levels of corruption are associated with CBI effectiveness. A low level of corruption is most strongly linked with central bank effectiveness regarding Unemployment.

H11- There is stronger evidence of CBI effectiveness among countries operating under English Common Law than other types of legal systems.

a.) Inflation

Table 5.11.1a—Inflation GARCH Results: Legal System

Legal System	Adj. R ²	DW	Significant Variables (Coefficients)	ARCH Coefficient (z-stat)	GARCH Coefficient (z-stat)
English Common	-.02	1.17	<i>MCI</i> (1.16), <i>GMT</i> (.03), <i>IID</i> (.51), <i>DD</i> (.37), <i>FXD</i> (.18), <i>IT</i> (.21), <i>TaxRev</i> (-.02)	.64(13.06)	.59(29.46)
French Civil	.02	.74	<i>MCI</i> (.01), <i>GMT</i> (-.01), <i>CBGov</i> (-.14), <i>LCBI</i> (.73), <i>IID</i> (.01), <i>DD</i> (.16), <i>FCD</i> (-.01), <i>FXD</i> (.12), <i>IT</i> (.41), <i>Cons</i> (.00), <i>TaxRev</i> (.00)	1.74(87.91)	.31(71.10)
Germanic	.05	1.31	<i>MCI</i> (-2.51), <i>IID</i> (1.74), <i>IT</i> (1.03), <i>TaxRev</i> (.03)	1.09(6.49)	.43(9.65)

Regarding Inflation, it appears that the only legal system with an inverse relationship with CBI (either operational or legal) is Germanic Law (practiced by some Asian countries in the sample).

b.) GDP Growth

Table 5.11.1b—GDP GARCH Results: Legal System

Legal System	Adj. R ²	DW	Significant Variables (Coefficients)	ARCH Coefficient (z-stat)	GARCH Coefficient (z-stat)
English Common	.00	2.03	<i>MCI</i> (-.17), <i>GMT</i> (-.01), <i>CBGov</i> (-.01), <i>LCBI</i> (.54), <i>IID</i> (-.27), <i>DD</i> (.02), <i>FCD</i> (-1.07), <i>FXD</i> (.32), <i>IT</i> (.17), <i>Cons</i> (.01), <i>TaxRev</i> (.00)	3.47(5.22)	.26(8.16)
French Civil	-.01	3.10	<i>MCI</i> (.18), <i>GMT</i> (.01), <i>CBGov</i> (-.09), <i>LCBI</i> (-.04), <i>IID</i> (.07), <i>DD</i> (.01), <i>FCD</i> (-.08), <i>FXD</i> (.05), <i>IT</i> (.06), <i>Cons</i> (-.01), <i>TaxRev</i> (.00)	7.10(3.92)	.10(3.31)
Germanic	-.03	3.20	<i>IID</i> (-1.43), <i>DD</i> (2.66), <i>FCD</i> (-1.42)	.39(8.74)*	.66(23.96)

* = evidence of hysteresis

In the area of economic growth, English Common Law is more strongly related to legal CBI, while French Civil Law is more strongly associated with operational independence.

c.) Unemployment

Table 5.11.1c—Unemployment GARCH Results: Legal System

Legal System	Adj. R ²	DW	Significant Variables (Coefficients)	ARCH Coefficient (z-stat)	GARCH Coefficient (z-stat)
English Common	-.03	2.18	<i>MCI</i> (-.35), <i>GMT</i> (-.04), <i>CBGov</i> (.17), <i>DD</i> (.68), <i>FCD</i> (.06), <i>FXD</i> (.05), <i>IT</i> (-.08), <i>Cons</i> (.02), <i>TaxRev</i> (-.01)	.54(3.30)	.70(8.67)
French Civil	-.01	2.15	<i>MCI</i> (-.05), <i>GMT</i> (.00), <i>DD</i> (.23), <i>FCD</i> (.37), <i>FXD</i> (.00), <i>IT</i> (-.40), <i>Cons</i> (.00), <i>TaxRev</i> (.00)	.78(7.29)	.77(106.75)
Germanic	-.02	1.94	<i>MCI</i> (.61), <i>GMT</i> (.03), <i>CBGov</i> (.00), <i>LCBI</i> (-.12), <i>IID</i> (.00), <i>DD</i> (.01), <i>FXD</i> (.01), <i>Cons</i> (.00), <i>TaxRev</i> (-.01)	1.62(3.81)	.50(10.06)

There seems to be a mixed pattern regarding Unemployment and CBI here. However, the desired inverse relationship seems to be strongest under English Common Law.

Conclusion- Germanic Law is most strongly associated with CBI effectiveness in the area of Inflation. However, English Common Law (followed by French Civil Law) seems to be most strongly associated with CBI effectiveness regarding Economic Growth. For Unemployment, results are mixed (however, strongest effect under English Common Law).

H12- There is stronger evidence of CBI effectiveness among countries operating under inflation targeting regimes.

a.) Inflation

Table 5.12.1a—Inflation GARCH Results: Inflation Targeting

Level	Adj. R ²	DW	Significant Variables (Coefficients)	ARCH Coefficient (z-stat)	GARCH Coefficient (z-stat)
High	-.02	.62	<i>MCI</i> (-.10), <i>GMT</i> (-.01), <i>CBGov</i> (.01), <i>LCBI</i> (.70), <i>IID</i> (-.03), <i>DD</i> (.84), <i>FCD</i> (-.01), <i>FXD</i> (-.02), IT (-.06) , <i>Cons</i> (.02), <i>TaxRev</i> (-.01)	1.06(12.94)	.44(21.39)
Medium	.11	.97	<i>MCI</i> (-.21), <i>GMT</i> (-.01), <i>LCBI</i> (.18), <i>IID</i> (.17), <i>DD</i> (-.51), <i>FCD</i> (1.52), <i>FXD</i> (.68), IT (.67) , <i>Cons</i> (.01)	1.20(12.29)	.44(27.74)
Low	.09	.87	<i>MCI</i> (-.31), <i>GMT</i> (-.01), <i>CBGov</i> (-.01), <i>LCBI</i> (.70), <i>IID</i> (.03), <i>DD</i> (.18), <i>FCD</i> (-.02), <i>FXD</i> (.20), IT (3.45) , <i>Cons</i> (-.01), <i>TaxRev</i> (.00)	1.78(6.00)	.38(17.60)

Results here are mixed. There is actually a stronger inverse relationship between MCI/GMT and Inflation in Low Inflation Targeting countries (although the Legal CBI variable is still strongly positive). However, the IT variable becomes gradually more positive as the level goes from High to Low (as would be expected).

b.) GDP Growth

Table 5.12.1b—GDP GARCH Results: Inflation Targeting

Level	Adj. R ²	DW	Significant Variables (Coefficients)	ARCH Coefficient (z-stat)	GARCH Coefficient (z-stat)
High	.00	3.12	<i>MCI</i> (.57), <i>GMT</i> (-.02), <i>CBGov</i> (-.07), <i>LCBI</i> (-.06), <i>IID</i> (.08), <i>DD</i> (1.03), <i>FXD</i> (.14), IT (.37) , <i>Cons</i> (.01), <i>TaxRev</i> (-.02)	1.40(10.02)	.39(25.62)
Medium	.00	3.13	<i>MCI</i> (.10), <i>GMT</i> (.03), <i>CBGov</i> (-.25), <i>LCBI</i> (.33), <i>IID</i> (-.13), <i>FCD</i> (-.68), <i>FXD</i> (.23), IT (.24) , <i>Cons</i> (-.01), <i>TaxRev</i> (.00)	1.70(4.96)	.38(7.58)
Low	-.01	2.29	<i>MCI</i> (.09), <i>GMT</i> (.00), <i>CBGov</i> (.02), <i>LCBI</i> (-.01), <i>IID</i> (.13), <i>DD</i> (.06), <i>FCD</i> (-.01), <i>FXD</i> (-.08), IT (.14) , <i>Cons</i> (-.01), <i>TaxRev</i> (.00)	18.86(29.78)	.00(15.07)

It seems that the High and Medium level countries perform relatively equally regarding GDP and Legal and/or Operational CBI. However, also interesting to note is the gradual decline in the positive IT coefficient as the level moves from High to Low.

Also of note is the positively correlated foreign exchange rate regime variable (FXD) in both High and Medium IT regimes, but not at the Low level.

c.) Unemployment

Table 5.12.1c—Unemployment GARCH Results: Inflation Targeting

Level	Adj. R ²	DW	Significant Variables (Coefficients)	ARCH Coefficient (z-stat)	GARCH Coefficient (z-stat)
High	.00	2.06	<i>GMT</i> (.00), <i>CBGov</i> (-.04), <i>LCBI</i> (.31), <i>IID</i> (.01), <i>DD</i> (.09), <i>FCD</i> (.12), <i>FXD</i> (.02), <i>IT</i> (-.33) , <i>Cons</i> (.01), <i>TaxRev</i> (-.01)	1.44(3.59)	.67(23.81)
Medium	-.02	2.37	<i>MCI</i> (-.18), <i>GMT</i> (.01), <i>CBGov</i> (-.01), <i>LCBI</i> (-.02), <i>IID</i> (.18), <i>DD</i> (-.05), <i>FCD</i> (-.02), <i>FXD</i> (.07), <i>IT</i> (.03) , <i>Cons</i> (-.01), <i>TaxRev</i> (.00)	1.62(4.13)	.64(10.02)
Low	.01	2.04	<i>MCI</i> (.19), <i>GMT</i> (.01), <i>CBGov</i> (.03), <i>LCBI</i> (-.02), <i>IID</i> (.00), <i>DD</i> (-.37), <i>FCD</i> (.23), <i>FXD</i> (-.02), <i>IT</i> (-.25) , <i>Cons</i> (.00), <i>TaxRev</i> (.00)	1.48(4.19)	.71(28.64)

Regarding Unemployment, there seems to be no readily observable pattern regarding the impact of CBI effectiveness as categorized by Inflation Targeting regimes. However, it seems that countries with a Medium level of IT have had better CBI effectiveness in this regard. It is also noteworthy that for countries with a Medium and Low level of Inflation Targeting, there is a slight (Medium) to strong (Low) inverse relationship between the level of country development and the level of Unemployment.

Conclusion- Surprisingly, there seems to be no clear link between CBI effectiveness and high IT countries in the area of Inflation (although there is a negative relationship between the IT variable and inflation). However, there is strong evidence of CBI effectiveness related to GDP growth in both high and medium IT countries. For Unemployment, results are mixed.

H13- There is stronger evidence of CBI effectiveness among countries operating under a medium level of constraints than either a high or low level of constraints.

a.) Inflation

Table 5.13.1a—Inflation GARCH Results: Constraint Level

Level	Adj. R ²	DW	Significant Variables (Coefficients)	ARCH Coefficient (z-stat)	GARCH Coefficient (z-stat)
High	.02	1.17	<i>GMT</i> (.15), <i>CBGov</i> (-.07), <i>LCBI</i> (.68), <i>IID</i> (.39), <i>DD</i> (-.85), <i>FCD</i> (-.08), <i>FXD</i> (.61), <i>IT</i> (-.84), <i>Cons</i> (.09), <i>TaxRev</i> (-.06)	.89(8.00)	.48(17.47)
Medium High	.17	1.61	<i>MCI</i> (.98), <i>GMT</i> (.14), <i>CBGov</i> (-.30), <i>LCBI</i> (-.90), <i>IID</i> (.66), <i>FXD</i> (.15), <i>IT</i> (.44), <i>TaxRev</i> (-.05)	.54(6.41)	.62(17.27)
Medium	.01	.60	<i>MCI</i> (-.17), <i>GMT</i> (-.01), <i>LCBI</i> (.69), <i>IID</i> (.08), <i>DD</i> (.70), <i>FXD</i> (.07), <i>IT</i> (.39), <i>Cons</i> (.00), <i>TaxRev</i> (-.01)	1.58(12.42)	.32(40.63)
Medium Low	.05	1.27	<i>MCI</i> (-1.90), <i>GMT</i> (.06), <i>CBGov</i> (-.02), <i>LCBI</i> (.08), <i>IID</i> (-.03), <i>DD</i> (-3.40), <i>FCD</i> (-.36), <i>FXD</i> (.01), <i>IT</i> (2.07), <i>Cons</i> (.01), <i>TaxRev</i> (.08)	5.27(13.07)	.12(7.74)
Low	.30	1.21	<i>MCI</i> (2.01), <i>CBGov</i> (-.01), <i>IID</i> (.14), <i>DD</i> (3.02), <i>FCD</i> (1.17), <i>FXD</i> (1.11), <i>IT</i> (5.29), <i>TaxRev</i> (-.06)	1.56(8.85)	.36(15.46)

As can be seen, there is some evidence of operational and/or legal CBI effectiveness in the Medium levels (Medium High, Medium, Medium Low) of constraints. Also noteworthy are the *IT* variable (which seems to decrease in effectiveness the lower the constraint level) and the *Tax Revenue* variable (which is inversely related to Inflation in four of the five subsamples).

b.) GDP Growth

Table 5.13.1b—GDP GARCH Results: Constraint Level

Level	Adj. R ²	DW	Significant Variables (Coefficients)	ARCH Coefficient (z-stat)	GARCH Coefficient (z-stat)
High	.02	2.16	<i>GMT</i> (.07), <i>LCBI</i> (.62), <i>IID</i> (.06), <i>DD</i> (-2.41), <i>FCD</i> (-.58), <i>FXD</i> (.44), <i>Cons</i> (.06)	1.51(6.02)	.42(12.75)
Medium High	-.03	2.55	<i>LCBI</i> (-.35), <i>IID</i> (-.68), <i>DD</i> (.69), <i>FCD</i> (-1.33), <i>FXD</i> (.26)	.47(6.71)*	.58(14.51)
Medium	.00	2.94	<i>MCI</i> (.07), <i>GMT</i> (.01), <i>CBGov</i> (.02), <i>LCBI</i> (.67), <i>IID</i> (.12), <i>DD</i> (.22), <i>FCD</i> (-.02), <i>FXD</i> (.08), <i>IT</i> (-.29), <i>Cons</i> (-.01), <i>TaxRev</i> (.00)	3.51(5.31)	.23(10.14)
Medium Low	-.01	3.26	<i>MCI</i> (-.43), <i>GMT</i> (.04), <i>CBGov</i> (-.02), <i>LCBI</i> (-.18), <i>IID</i> (.01), <i>DD</i> (-.04), <i>FCD</i> (-.01), <i>FXD</i> (.23), <i>IT</i> (.36), <i>Cons</i> (.00)	7.52(1.85)	.01(1.36)
Low	-.01	2.04	<i>MCI</i> (4.69), <i>GMT</i> (.09), <i>CBGov</i> (.00), <i>LCBI</i> (-.41), <i>IID</i> (-.05), <i>DD</i> (.15), <i>FXD</i> (.02), <i>IT</i> (.27), <i>Cons</i> (.00), <i>TaxRev</i> (-.07)	6.42(1.54)	.12(5.72)

* = evidence of hysteresis

CBI effectiveness occurs at any constraint level as regards to the GDP Volume Index at the operational and/or legal level (with the legal benefits coming at the higher constraint levels).

c.) Unemployment

Table 5.13.1c—Unemployment GARCH Results: Constraint Level

Level	Adj. R ²	DW	Significant Variables (Coefficients)	ARCH Coefficient (z-stat)	GARCH Coefficient (z-stat)
High	-.01	2.16	<i>GMT</i> (-.04), <i>CBGov</i> (.00), <i>IID</i> (-.03), <i>FCD</i> (.01), <i>FXD</i> (.18), <i>Cons</i> (.00)	3.46(4.52)	.17(9.77)
Medium High	-.02	2.39	<i>CBGov</i> (.00), <i>IID</i> (.02), <i>DD</i> (.00), <i>FCD</i> (.18), <i>Cons</i> (.00)	.33(2.12)	.83(12.36)
Medium	-.01	2.20	<i>GMT</i> (.00), <i>CBGov</i> (-.14), <i>LCBI</i> (-.21), <i>IID</i> (-.07), <i>DD</i> (.39), <i>FCD</i> (.40), <i>Cons</i> (.01), <i>TaxRev</i> (-.01)	.57(10.70)	.82(107.88)
Medium Low	-.11	2.00	<i>MCI</i> (2.06), <i>GMT</i> (-.08), <i>CBGov</i> (-.01), <i>LCBI</i> (.23), <i>IID</i> (.03), <i>DD</i> (.91), <i>FXD</i> (.02), <i>IT</i> (-1.45), <i>Cons</i> (.00), <i>TaxRev</i> (-.02)	2.39(13.27)	.57(44.49)
Low	-.01	2.02	<i>MCI</i> (-1.37), <i>GMT</i> (-.03), <i>CBGov</i> (.00), <i>IID</i> (-.01), <i>FCD</i> (.00), <i>FXD</i> (.02), <i>Cons</i> (.00), <i>TaxRev</i> (.02)	5.98(1.22)	.27(2.55)

Only the Medium constraint level boasts Legal CBI effectiveness in the area of Unemployment (although signs of Operational CBI effectiveness occur at other levels).

Conclusion- Regarding Inflation, there seems to be stronger evidence of CBI effectiveness in the mid-range constraint levels (Medium High, Medium, Medium Low) operationally and/or legally. In terms of GDP growth, there is evidence of legal and/or operational effectiveness at any constraint level. Legal CBI effectiveness occurs in the Medium constraint level in the area of Unemployment (although operational independence is seen at the low level of constraints).

H14- *There is stronger evidence of CBI effectiveness among countries that are more economically developed.*

Table 5.14.1a—Inflation GARCH Results: Economic Development

Level	Adj. R ²	DW	Significant Variables (Coefficients)	ARCH Coefficient (z-stat)	GARCH Coefficient (z-stat)
High	-.02	1.15	<i>MCI</i> (.64), <i>GMT</i> (.08), <i>CBGov</i> (-.07), <i>LCBI</i> (-.64), <i>IID</i> (.67), DD (.18), <i>FXD</i> (.13), <i>Cons</i> (.02), <i>TaxRev</i> (-.04)	.56(10.69)	.61(33.66)
Medium High	-.12	1.55	<i>MCI</i> (.08), <i>CBGov</i> (-.01), <i>IT</i> (1.52), <i>TaxRev</i> (.00)	1.51(13.12)	.29(24.77)
Medium	.12	.47	<i>MCI</i> (-.32), <i>GMT</i> (.00), <i>CBGov</i> (-.18), <i>LCBI</i> (1.34), <i>IID</i> (.15), DD (.60), <i>FCD</i> (-.02), <i>FXD</i> (.22), <i>IT</i> (-.74), <i>Cons</i> (-.02), <i>TaxRev</i> (.00)	1.49(7.54)	.33(12.04)
Medium Low	-.02	.58	<i>MCI</i> (.96), <i>CBGov</i> (-.21), <i>LCBI</i> (.37), <i>IID</i> (.45), DD (-1.86), <i>FCD</i> (.05), <i>FXD</i> (-.10), <i>IT</i> (2.16), <i>Cons</i> (-.03), <i>TaxRev</i> (.02)	1.30(4.84)	.44(6.50)
Low	.18	1.00	<i>MCI</i> (.10), <i>CBGov</i> (-.02), <i>LCBI</i> (.71), <i>IID</i> (.05), DD (-.97), <i>FCD</i> (1.29), <i>FXD</i> (1.20), <i>IT</i> (4.56), <i>Cons</i> (-.02)	1.51(8.78)	.41(17.06)

There is no clear evidence here, although it should be noted that the only level which has a negative LCBI coefficient is the High development level (although, as would not be expected, its operational CBI variables both have positive coefficients). Also of note is the International Inflation variable, which is positively related to Inflation in four

of the five subsamples (perhaps indicating that international inflation levels affect an individual country's inflation level, regardless of level of economic development).

b.) GDP Growth

Table 5.14.1b—GDP GARCH Results: Economic Development

Level	Adj. R ²	DW	Significant Variables (Coefficients)	ARCH Coefficient (z-stat)	GARCH Coefficient (z-stat)
High	-.02	2.46	FCD (-1.04), FXD (.29), IT (.38)	1.09(8.63)	.49(24.85)
Medium High	-.02	3.27	<i>GMT</i> (-.01), <i>CBGov</i> (-.05), <i>LCBI</i> (.03), IID (.00), DD (.00), FCD (.00), FXD (.01), IT (.00), TaxRev (.00)	1.15(3.74)	.55(6.16)
Medium	-.02	2.98	<i>MCI</i> (-.01), <i>GMT</i> (.01), <i>CBGov</i> (-.01), <i>LCBI</i> (.05), IID (.16), DD (.27), FXD (-.02), IT (-.05), Cons (-.01), TaxRev (.00)	6.15(5.55)	.03(2.39)
Medium Low	-.01	2.91	<i>MCI</i> (.86), <i>CBGov</i> (.10), IID (.14), DD (.38), IT (.05), Cons (-.02), TaxRev (.00)	2.65(2.14)	.28(3.37)
Low	-.01	2.63	<i>LCBI</i> (-.20), FCD (.20), IT (.55)	23.84(.61)	.07(1.42)

Legal CBI has a slightly positive coefficient in the Medium High and Medium levels (although it is negative in the Low level). However, Inflation Targeting has larger positive coefficient in several levels at all ends of the economic spectrum. Also noteworthy is the Development Dummy variable, which has a gradually increasing coefficient as the level of development declines. This seems to indicate that as a country develops and “closes the gap” between itself and other countries, its rate of growth seems to begin to decline (i.e. the “convergence theory”).

c.) Unemployment

Table 5.14.1c—Unemployment GARCH Results: Economic Development

Level	Adj. R ²	DW	Significant Variables (Coefficients)	ARCH Coefficient (z-stat)	GARCH Coefficient (z-stat)
High	-.01	2.28	<i>CBGov</i> (.01), <i>IID</i> (-.01), <i>FXD</i> (.08), <i>Cons</i> (.00)	1.22(8.23)	.56(28.35)
Medium High	.02	1.97	<i>MCI</i> (-.04), <i>GMT</i> (.01), <i>CBGov</i> (-.06), <i>IID</i> (-.04), DD (-.25), <i>FCD</i> (.44), <i>FXD</i> (.04), <i>Cons</i> (.02), <i>TaxRev</i> (-.01)	1.78(3.94)	.58(10.58)
Medium	-.11	1.80	<i>MCI</i> (.45), <i>CBGov</i> (.08), <i>LCBI</i> (.49), <i>IID</i> (-.05), DD (-.15), <i>FCD</i> (.03), <i>FXD</i> (-.08), <i>IT</i> (-.56), <i>Cons</i> (-.02), <i>TaxRev</i> (.01)	1.58(3.25)	.57(6.22)
Medium Low	-.15	1.78	<i>MCI</i> (-.05), <i>GMT</i> (-.01), <i>CBGov</i> (.01), <i>LCBI</i> (-.17), <i>IID</i> (-.04), DD (.75), <i>FXD</i> (.08), <i>IT</i> (.34), <i>Cons</i> (.00), <i>TaxRev</i> (.00)	1.27(3.08)	.63(35.46)
Low	-.01	2.54	<i>MCI</i> (.16), <i>GMT</i> (.00), <i>CBGov</i> (.02), <i>LCBI</i> (-.01), <i>IID</i> (.01), DD (.21), <i>FXD</i> (.02), <i>IT</i> (-.08), <i>Cons</i> (.00), <i>TaxRev</i> (.00)	3.44(5.15)	.62(26.56)

Regarding Unemployment, there is evidence that operational CBI is inversely correlated at most any level of economic development. However, it seems to hold the most promise at the Medium Low level of economic development. Also noteworthy is the *CBGov* variable, which is positively related to Unemployment in four of the five subsamples, possibly indicating a harmful effect from excessive central bank governor turnover.

Conclusion- Although operational independence is linked to Inflation in countries with a medium level of development, only highly developed countries show evidence of legal CBI effectiveness. In the area of GDP growth, there is evidence of both legal and operational effectiveness in the mid-level (MH, Med, and ML) developed countries. Regarding Unemployment, it appears that countries in the Medium Low category show the most evidence of CBI effectiveness.

H15- *There is stronger evidence of CBI effectiveness among countries that have a higher level of Tax Revenue amassed by the government.*

Table 5.15.1a—Inflation GARCH Results: Tax Revenue

Level	Adj. R ²	DW	Significant Variables (Coefficients)	ARCH Coefficient (z-stat)	GARCH Coefficient (z-stat)
High	-.05	1.40	<i>MCI</i> (.84), <i>GMT</i> (.04), <i>CBGov</i> (-.05), <i>LCBI</i> (-.22), <i>IID</i> (.41), <i>DD</i> (.45), <i>FXD</i> (.14), <i>IT</i> (.21), <i>Cons</i> (.01), TaxRev (-.03)	.76(11.80)	.51(26.19)
Medium High	-.09	.56	<i>MCI</i> (.19), <i>LCBI</i> (.77), <i>DD</i> (.33), <i>FCD</i> (-.02), <i>FXD</i> (.03), <i>IT</i> (-.61), <i>Cons</i> (.00), TaxRev (-.01)	1.07(11.87)	.42(52.46)
Medium	-.35	.47	<i>MCI</i> (1.24), <i>GMT</i> (.02), <i>CBGov</i> (.03), <i>IID</i> (-.02), <i>DD</i> (.19), <i>FCD</i> (-.01), <i>FXD</i> (.28), <i>IT</i> (.73), <i>Cons</i> (.01), TaxRev (-.05)	1.71(5.09)	.29(9.75)
Medium Low	.14	.92	<i>MCI</i> (1.99), <i>GMT</i> (-.07), <i>LCBI</i> (.82), <i>IID</i> (.05), <i>DD</i> (5.18), <i>FCD</i> (-.01), <i>FXD</i> (.99), <i>IT</i> (-2.13), <i>Cons</i> (.00), TaxRev (-.06)	1.03(9.14)	.52(22.63)
Low	.09	.65	<i>MCI</i> (.78), <i>GMT</i> (-.06), <i>CBGov</i> (-.02), <i>LCBI</i> (1.73), <i>IID</i> (.09), <i>DD</i> (-2.28), <i>FCD</i> (-.05), <i>FXD</i> (.05), <i>Cons</i> (-.02), TaxRev (.09)	1.90(6.91)	.37(17.18)

There are no clear results regarding Inflation and CBI effectiveness when examined through Tax Revenue levels. However, it does lend support to fiscal policy theory (i.e. a decrease in taxes is part of expansionary fiscal policy, leading to more output and, in many cases, higher inflation levels) that there is an inverse relationship between TaxRev and Inflation and the four highest Tax Revenue levels. Perhaps at the Low level an increase in Tax Revenue would be actually be beneficial for economic output, as these would most likely be represented by the poorest nations that are in most dire need of basic infrastructure, healthcare, education, etc.

b.) GDP Growth

Table 5.15.1b—GDP GARCH Results: Tax Revenue

Level	Adj. R ²	DW	Significant Variables (Coefficients)	ARCH Coefficient (z-stat)	GARCH Coefficient (z-stat)
High	.02	1.96	<i>MCI</i> (.63), <i>GMT</i> (.03), <i>LCBI</i> (.15), IID (.08), DD (.68), FCD (-1.05), IT (.22), TaxRev (-.02)	1.49(6.87)	.43(19.88)
Medium High	-.03	2.30	<i>MCI</i> (-.04), <i>GMT</i> (-.01), <i>CBGov</i> (-.14), <i>LCBI</i> (.04), IID (.06), DD (.50), FXD (-.08), IT (-.05), Cons (-.01), TaxRev (.01)	2.13(8.52)	.31(10.69)
Medium	-.01	3.33	<i>MCI</i> (1.12), <i>GMT</i> (-.08), <i>CBGov</i> (.13), IID (.31), DD (.16), FCD (-.07), FXD (.13), IT (.21)	1.24(6.13)	.31(7.76)
Medium Low	-.01	2.91	<i>MCI</i> (.11), <i>GMT</i> (.01), <i>CBGov</i> (.01), <i>LCBI</i> (1.64), IID (.01), DD (-.38), FCD (-.04), FXD (.00), IT (-1.60), TaxRev (.00)	7.24(3.34)	.17(4.18)
Low	-.01	2.75	<i>MCI</i> (.49), <i>GMT</i> (-.03), <i>CBGov</i> (-.21), <i>LCBI</i> (.42), IID (.08), DD (-1.24), FCD (-.03), FXD (.25), IT (-.33), Cons (-.01), TaxRev (.05)	6.59(127.79)	.04(9.81)

There is CBI effectiveness as it pertains to GDP growth at most any level of Tax Revenue (which provides hope to countries at the lower echelons of accumulated revenue). Additionally, it is noteworthy how the pattern of Tax Revenue tends to support expansionary/contractionary fiscal policy theory. In other words, in theory a decrease in taxes would lead to an increase in economic output. However, one can see the coefficient changing in scope gradually as the Tax Revenue levels declines. It could be at lower levels of Tax Revenue collected (again, many times comprised of the poorer nations), any increase in Tax Revenue is actually beneficial to satisfy the basic needs of the country. Also of note is the Financial Crisis Dummy variable, which is again consistently statistically negatively related to GDP growth.

c.) Unemployment

Table 5.15.1c—Unemployment GARCH Results: Tax Revenue

Level	Adj. R ²	DW	Significant Variables (Coefficients)	ARCH Coefficient (z-stat)	GARCH Coefficient (z-stat)
High	-.02	2.04	<i>MCI</i> (-.37), <i>CBGov</i> (.01), <i>IID</i> (-.04), <i>DD</i> (.02), <i>FCD</i> (.01), <i>FXD</i> (.17), <i>Cons</i> (.00)	1.22(6.62)	.59(25.71)
Medium High	.01	1.91	<i>MCI</i> (.19), <i>GMT</i> (.02), <i>CBGov</i> (.00), <i>DD</i> (-.04), <i>FXD</i> (.06), <i>IT</i> (-.34), <i>Cons</i> (.00), Tax (-.01)	.37(5.49)	.86(63.19)
Medium	-.16	1.89	<i>MCI</i> (-.13), <i>GMT</i> (-.01), <i>LCBI</i> (.58), <i>IID</i> (-.01), <i>DD</i> (.08), <i>FCD</i> (-.20), <i>FXD</i> (.01), <i>Cons</i> (-.01), TaxRev (.01)	1.53(4.52)	.68(47.89)
Medium Low	-.05	2.43	<i>MCI</i> (.26), <i>GMT</i> (.00), <i>CBGov</i> (-.01), <i>LCBI</i> (-.07), <i>IID</i> (.01), <i>DD</i> (-1.26), <i>FCD</i> (.20), <i>FXD</i> (.30), <i>IT</i> (-.15), <i>Cons</i> (-.01), TaxRev (.00)	3.35(16.21)	.32(47.33)
Low	-.02	1.97	<i>MCI</i> (-.05), <i>GMT</i> (-.01), <i>CBGov</i> (-.01), <i>LCBI</i> (-.01), <i>IID</i> (.00), <i>DD</i> (.43), <i>FXD</i> (.02), <i>IT</i> (-.02), <i>Cons</i> (.00)	1.67(2.93)	.63(31.42)

Operational CBI effectiveness can be most readily observed at the High Level of Tax Revenue. Government Consumption also appears negatively related to Unemployment in several instances (perhaps indicating that this government spending can help to prevent and/or lessen the degree of unemployment).

Conclusion- Regarding Inflation, there seems to be no clear link. In terms of GDP growth, it would appear that evidence of CBI effectiveness can be seen at any level of tax revenue. In the area of Unemployment, strong evidence of operational independence can be witnessed in the High level of tax revenue category.

H16- *There is stronger evidence of CBI effectiveness among countries that have a higher level of Government Consumption.*

Table 5.16.1a—Inflation GARCH Results: Government Consumption

Level	Adj. R ²	DW	Significant Variables (Coefficients)	ARCH Coefficient (z-stat)	GARCH Coefficient (z-stat)
High	-.19	.93	<i>MCI</i> (-.08), <i>GMT</i> (.06), <i>LCBI</i> (-.58), <i>IID</i> (.23), <i>DD</i> (.45), <i>FXD</i> (.15), <i>IT</i> (.54), Cons (.01) , <i>TaxRev</i> (-.02)	.76(13.37)	.52(31.43)
Medium High	-.07	.69	<i>MCI</i> (.33), <i>GMT</i> (-.07), <i>CBGov</i> (.00), <i>LCBI</i> (-.06), <i>IID</i> (.00), <i>DD</i> (.00), <i>FCD</i> (.00), <i>FXD</i> (-.01), <i>IT</i> (1.07), Cons (.00) , <i>TaxRev</i> (.02)	.92(9.63)	.52(59.98)
Medium	.25	.75	<i>MCI</i> (.19), <i>GMT</i> (-.01), <i>CBGov</i> (-.01), <i>LCBI</i> (2.82), <i>IID</i> 9-.02), <i>DD</i> (1.72), <i>FCD</i> (.59), <i>FXD</i> (.14), <i>IT</i> (-2.65), Cons (-.02)	1.76(5.95)	.35(14.08)
Medium Low	-.01	1.36	<i>CBGov</i> (-.03), <i>LCBI</i> (.15), <i>IID</i> (.03), <i>DD</i> (-1.83), <i>FCD</i> (-.13), <i>FXD</i> (.11), Cons (-.01) , <i>TaxRev</i> (.05)	1.54(11.26)	.37(24.26)
Low	.01	.63	<i>MCI</i> (-.59), <i>GMT</i> (-.02), <i>CBGov</i> (-.07), <i>LCBI</i> (1.52), <i>IID</i> (.15), <i>DD</i> (.23), <i>FCD</i> (-.04), <i>FXD</i> (.36), <i>IT</i> (3.75), Cons (-.01) , <i>TaxRev</i> (.02)	1.42(8.25)	.45(13.33)

There is legal and (to a lesser extent) operational CBI effectiveness at the High level of Government Consumption. There is also a small negative coefficient for *TaxRev* at this High level. It is significant also that at the higher levels of Government Consumption, the coefficients support fiscal policy theory. In other words, an increase in government consumption (i.e. demand) tends to lead to an increase in economic output inflation. However, at lower levels of consumption, perhaps this does not hold true. In other words, there may exist a type of “breaking point” related to inflation as consumption increases.

b.) GDP Growth

Table 5.16.1b—GDP GARCH Results: Government Consumption

Level	Adj. R ²	DW	Significant Variables (Coefficients)	ARCH Coefficient (z-stat)	GARCH Coefficient (z-stat)
High	.03	2.34	<i>MCI</i> (-.23), <i>GMT</i> (.03), <i>CBGov</i> (-.01), <i>IID</i> (-.08), <i>FCD</i> (-.80), <i>FXD</i> (.33), <i>IT</i> (.26), Cons (.00)	.99(9.13)	.51(20.86)
Medium High	-.01	2.29	<i>MCI</i> (-.44), <i>GMT</i> (-.01), <i>CBGov</i> (-.05), <i>LCBI</i> (.22), <i>IID</i> (.01), <i>DD</i> (.65), <i>FCD</i> (.00), <i>FXD</i> (-.09), <i>IT</i> (.01), Cons (.00) , <i>TaxRev</i> (.01)	1.91(5.30)	.31(8.31)
Medium	-.02	3.22	<i>CBGov</i> (.03), <i>IID</i> (.01), <i>DD</i> (.03), <i>IT</i> (-.01), <i>TaxRev</i> (.00)	3.89(2.12)	.17(4.99)
Medium Low	-.01	2.62	<i>MCI</i> (.20), <i>GMT</i> (.00), <i>CBGov</i> (.00), <i>IID</i> (.03), <i>DD</i> (.03), <i>FCD</i> (.00), <i>FXD</i> (.20), Cons (.00) , <i>TaxRev</i> (.00)	12.65(2.07)	.03(.95)
Low	-.01	3.09	<i>MCI</i> (.20), <i>GMT</i> (.00), <i>CBGov</i> (-.05), <i>LCBI</i> (.08), <i>IID</i> (.10), <i>DD</i> (.05), <i>FCD</i> (-.05), <i>FXD</i> (.01), Cons (-.01) , <i>TaxRev</i> (.00)	7.67(5.04)	.12(3.59)

Legal and Operational CBI effectiveness is observable at the Low level of Government Consumption. Inflation Targeting is positively correlated with GDP growth at the High and Medium High levels. Government Consumption, although statistically significant, seems to play a miniscule role in economic growth. Also striking is the strongly positive IT coefficient in the High level, which then becomes nil at later levels.

c.) Unemployment

Table 5.16.1c—Unemployment GARCH Results: Government Consumption

Level	Adj. R ²	DW	Significant Variables (Coefficients)	ARCH Coefficient (z-stat)	GARCH Coefficient (z-stat)
High	.03	2.34	<i>MCI</i> (-.23), <i>GMT</i> (.03), <i>CBGov</i> (-.01), <i>IID</i> (-.08), <i>FCD</i> (-.80), <i>FXD</i> (.34), <i>IT</i> (.26), Cons (.00)	.99(9.13)	.51(20.86)
Medium High	.00	1.79	<i>MCI</i> (-.27), <i>GMT</i> (.01), <i>CBGov</i> (.01), <i>LCBI</i> (.20), <i>IID</i> (.01), <i>DD</i> (.04), <i>FCD</i> (.03), <i>FXD</i> (-.03), <i>IT</i> (-.47), Cons (.02) , <i>TaxRev</i> (-.01)	1.22(3.34)	.65(9.54)
Medium	-.01	1.91	<i>GMT</i> (-.01), <i>CBGov</i> (-.07), <i>LCBI</i> (.33), <i>IID</i> (-.01), <i>DD</i> (-.41), <i>FCD</i> (.03), <i>FXD</i> (-.01), <i>IT</i> (-.24), Cons (.01)	4.15(5.98)	.23(5.61)
Medium Low	-.03	2.08	<i>MCI</i> (-.01), <i>GMT</i> (-.01), <i>CBGov</i> (.00), <i>LCBI</i> (.01), <i>IID</i> (.00), <i>DD</i> (.11), <i>FXD</i> (.02), Cons (.00)	2.04(1.22)	.70(10.14)
Low	-.04	2.35	<i>MCI</i> (.22), <i>GMT</i> (.00), <i>CBGov</i> (-.06), <i>LCBI</i> (.32), <i>IID</i> (.14), <i>DD</i> (-.20), <i>FCD</i> (.01), <i>FXD</i> (.07), <i>IT</i> (-.66), Cons (-.01)	3.58(5.19)	.47(6.60)

There is moderate evidence for operational CBI effectiveness (although in several cases contradicted by legal CBI). However, this effect tends to diminish with the level of consumption. There is also a negative IT coefficient, indicating again the possible benefits of Inflation Targeting. Additionally, in several cases, the Financial Crisis variable is positively correlated with Unemployment.

Conclusion- In terms of Inflation, there is some evidence of legal CBI effectiveness in High Government Consumption countries (other results are mixed). For GDP growth, countries in the Low category show evidence of effectiveness both operationally and legally (other results are also mixed). There are mixed results for Unemployment.

H17- There is stronger evidence of CBI effectiveness among countries operating under a higher level of Operational Independence.

Table 5.17.1a—Inflation GARCH Results: Operational Independence

Level	Adj. R ²	DW	Significant Variables (Coefficients)	ARCH Coefficient (z-stat)	GARCH Coefficient (z-stat)
High	.20	.74	<i>MCI (.58), CBGov (-.05), LCBI (.94), IID (.38), DD (-.44), FCD (.47), FXD (.35), IT (.38), Cons (-.02), TaxRev (-.01)</i>	1.24(7.03)	.42(17.27)
Medium High	.07	1.20	<i>MCI (.24), GMT (-.01), CBGov (.01), LCBI (.62), DD (.25), FCD (-.14), FXD (1.12), IT (4.19), TaxRev (-.02)</i>	1.71(7.60)	.37(14.17)
Medium	-.02	.52	<i>MCI (-.69), GMT (-.01), LCBI (.82), IID (.38), DD (-.04), FXD (.26), Cons (.00), TaxRev (.01)</i>	1.45(8.80)	.38(20.61)
Medium Low	.16	.77	<i>MCI (.32), GMT (.04), CBGov (-.02), LCBI (.62), IID (-.15), DD (1.93), FCD (-.16), FXD (.52), IT (-.92), Cons (-.02), TaxRev (-.02)</i>	1.33(7.71)	.39(15.26)
Low	-.11	.62	<i>CBGov (.00), IID (.00), DD (.01), FXD (-.20), IT (.84), Cons (.00)</i>	.97(11.09)	.49(38.63)

The most benefit in terms of Inflation seems to come at the High and Medium High levels of Operational Independence. It is noteworthy that the Legal CBI variable has a positive coefficient in all subsamples where it appears statistically significant, illustrating the chasm that can (and many times does) exist between operational and legal CBI. Also noteworthy is the positive coefficients for FXD in four of the five subsamples, indicating the possible relationship between a floating exchange rate and inflation levels.

b.) GDP Growth

Table 5.17.1b—GDP GARCH Results: Operational Independence

Level	Adj. R ²	DW	Significant Variables (Coefficients)	ARCH Coefficient (z-stat)	GARCH Coefficient (z-stat)
High	-.01	3.13	<i>MCI</i> (.27), <i>GMT</i> (-.02), <i>CBGov</i> (-.02), <i>LCBI</i> (.31), IID (.03), DD (.83), FXD (.68), IT (-1.11), Cons (-.01), TaxRev (-.01)	3.08(5.17)	.18(7.86)
Medium High	-.01	3.07	<i>MCI</i> (.26), <i>GMT</i> (.01), <i>CBGov</i> (.00), <i>LCBI</i> (-.12), DD (.04), FCD (-.06), FXD (.09), IT (.32), Cons (.00), TaxRev (-.01)	2.46(3.34)	.26(13.42)
Medium	-.01	2.63	<i>MCI</i> (-1.00), <i>GMT</i> (.00), <i>CBGov</i> (-.07), <i>LCBI</i> (.24), IID (.17), DD (.58), FCD (-.01), FXD (.43), Cons (-.03), TaxRev (.03)	3.28(5.80)	.17(4.81)
Medium Low	-.03	1.79	<i>MCI</i> (-.01), <i>GMT</i> (.02), <i>CBGov</i> (-.03), <i>LCBI</i> (-.11), IID (-.01), DD (.68), FCD (-.01), FXD (.01), TaxRev (-.01)	6.08(.92)	.09(1.06)
Low	-.02	2.81	<i>CBGov</i> (.00), IID (-.09), DD (.11), FXD (.21), IT (-.10), TaxRev (.00)	3.03(2.12)	.29(3.61)

Here it is reasonably clear that a highly operationally independent central bank is best suited for achieving economic growth (the Legal CBI coefficient is also positive in the High level, reinforcing this assumption). Also of note is the Foreign Exchange Rate Regime variable having a positive relationship with GDP growth in all cases, indicating the positive link between a floating exchange rate and economic growth (as was previously indicated, there is also a positive link between inflation levels and a floating exchange rate; these two relationships mutually confirm the validity of the other).

c.) Unemployment

Table 5.17.1c—Unemployment GARCH Results: Operational Independence

Level	Adj. R ²	DW	Significant Variables (Coefficients)	ARCH Coefficient (z-stat)	GARCH Coefficient (z-stat)
High	-.07	2.21	MCI (-.18), GMT (.00), CBGov (.02), LCBI (.10), DD (-.15), FCD (.00), FXD (.00), Cons (.00), TaxRev (.00)	1.23(4.57)	.70(45.72)
Medium High	-.02	2.42	MCI (.12), GMT (.01), CBGov (-.18), LCBI (-.07), IID (-.03), DD (.08), FCD (-.17), FXD (.08), IT (.05), Cons (-.01), TaxRev (.00)	2.38(4.76)	.69(17.19)
Medium	-.08	1.85	MCI (-.39), GMT (-.01), CBGov (.00), IID (.01), DD (.18), FCD (.00), FXD (.28), Cons (.00), TaxRev (.00)	2.51(16.35)	.51(48.21)
Medium Low	-.07	1.77	MCI (.52), GMT (.03), CBGov (.00), LCBI (-.51), IID (.12), DD (1.11), FCD (.13), FXD (-.09), IT (.29), Cons (.00), TaxRev (-.02)	1.40(5.09)	.61(30.23)
Low	-.03	1.87	MCI (-.33), GMT (-.01), CBGov (.01), LCBI (.03), IID (.01), DD (.00), FCD (.00), FXD (.00), IT (.00), Cons (.00), TaxRev (.00)	2.44(41.72)	.44(49.79)

Unemployment seems to best be contained by a medium range operationally independent central bank, although results are less clear.

Conclusion- For GDP growth, a highly operationally independent central bank seems best. For Unemployment, a central bank with a medium level of operational independence is preferable. For Inflation, there is no clear link.

For Hypotheses 18 through 20, five year periods will be compared to the overall time frame (Q1:1960- Q1:2009). This will be done to isolate the more immediate effects of Central Bank Independence. The variables of Inflation, GDP Volume Index and Unemployment will be examined once more.

H18- There is a beneficial “Signaling Effect” present with the announcement of Central Bank Independence.

GARCH regression statistics will be examined in an “event study” format for the three variables examined above—Inflation, GDP Volume Index, and Unemployment—

for the two years immediately prior to each respective country's central bank independence and the three years immediately afterwards, for a total window of five years. This will be done to determine if there is any type of “announcement” effect (any positive effect just from news being disseminated in the market). The full sample will be followed by the three regional samples in the same format as above. Each “event study” set of results will be compared to the results for the total time period.

Table 5.18.1a--Inflation GARCH Results: Event Study vs. Total Time Period

Region	<i>Event Study--Significant Variables (Coefficients)</i>	<i>Total Time Period—Significant Variables (Coefficients)</i>
Overall	<i>MCI (.37), GMT (.06), LCBI (.41), DD (-1.46), FCD (1.61), FXD (1.32), Cons (-.04), IT (-.88), TaxRev (.03)</i>	<i>MCI (.10), GMT (-.01), CBGov (-.10), LCBI (.60), IID (.07), DD (-.42), FXD (.33), IT (.25), Cons (.01)</i>
Asia	<i>GMT (-.10), CBGov (4.85), LCBI (.34), DD (-3.07), FCD (1.37), FXD (1.59), Cons (-.04), IT (-1.57)</i>	<i>MCI (-.32), GMT (.01), CBGov (-.08), IID (.57), DD (-1.42), FCD (1.64), FXD (.80), Cons (-.01), IT (.53)</i>
Latin America	<i>MCI (-2.32), GMT (.14), CBGov (-2.77), LCBI (.99), IID (.42), FXD (-2.83), TaxRev (.17)</i>	<i>MCI (-.21), GMT (.00), CBGov (-.36), LCBI (.74), IID (.04), DD (.29), FCD (-.01), FXD (.36), IT (.51), Cons (.00), TaxRev (.00)</i>
Developed West	<i>IT (-1.27)</i>	<i>GMT (.03), CBGov (-.13), LCBI (-.23), IID (.26), DD (.46), FXD (.07), IT (.47), TaxRev (-.01)</i>

It is apparent that there is no beneficial “Signaling Effect” regarding CBI and Inflation. For each of the three regions (Overall, Asia, and Latin America) where the Legal CBI variable appears significant, it is actually positively correlated with Inflation. In the other region, the Developed West, LCBI is not significant at all. Operational CBI variables are also positively related to Inflation, indicating no increase in *de facto* power on the part of the respective central bank to curb Inflation. The only possible beneficiary of a “Signaling Effect” here was Latin America, with a fairly substantial difference in the negative size of the MCI variable during the Event Study time frame.

Next, GDP Volume will be examined.

Table 5.18.1b--GDP GARCH Results: Event Study vs. Total Time Period

Region	<i>Event Study</i>--Significant Variables (Coefficients)	<i>Total Time Period</i>--Significant Variables (Coefficients)
Overall	<i>MCI</i> (1.97), <i>GMT</i> (.03), <i>CBGov</i> (.20), <i>LCBI</i> (-.03), <i>IID</i> (.20), <i>DD</i> (1.32), <i>FXD</i> (-.24), <i>Cons</i> (-.01), <i>IT</i> (.58), <i>TaxRev</i> (-.04)	<i>MCI</i> (.21), <i>GMT</i> (.00), <i>CBGov</i> (-.12), <i>LCBI</i> (.32), <i>IID</i> (.01), <i>DD</i> (.30), <i>FCD</i> (-.37), <i>FXD</i> (.35), <i>Cons</i> (.00), <i>IT</i> (-.31), <i>TaxRev</i> (-.01)
Asia	<i>MCI</i> (2.18), <i>GMT</i> (.08), <i>CBGov</i> (-2.89), <i>LCBI</i> (.06), <i>IID</i> (.17), <i>DD</i> (-1.55), <i>FCD</i> (-.73), <i>Cons</i> (.06), <i>IT</i> (.11)	<i>MCI</i> (-.45), <i>GMT</i> (.02), <i>LCBI</i> (.83), <i>IID</i> (-.15), <i>DD</i> (-1.15), <i>FCD</i> (-.32), <i>FXD</i> (.31), <i>Cons</i> (.01), <i>IT</i> (-1.03), <i>TaxRev</i> (.01)
Latin America	<i>MCI</i> (-.05), <i>GMT</i> (.00), <i>LCBI</i> (.02), <i>IID</i> (.03), <i>DD</i> (-.27), <i>FCD</i> (-.05), <i>FXD</i> (.07), <i>Con</i> (-.01), <i>IT</i> (4.95), <i>TaxRev</i> (.00)	<i>MCI</i> (.12), <i>GMT</i> (.01), <i>CBGov</i> (-.08), <i>LCBI</i> (-.07), <i>IID</i> (.09), <i>DD</i> (.15), <i>FCD</i> (-.03), <i>FXD</i> (.11), <i>Cons</i> (-.01), <i>IT</i> (.06), <i>TaxRev</i> (-.01)
Developed West	<i>MCI</i> (-.63), <i>CBGov</i> (-24.62), <i>LCBI</i> (.19), <i>FCD</i> (-.54), <i>FXD</i> (-1.26), <i>Cons</i> (.22), <i>IT</i> (-1.25)	<i>CBGov</i> (-.01), <i>LCBI</i> (.39), <i>IID</i> (-.08), <i>DD</i> (.31), <i>FCD</i> (-.58), <i>Cons</i> (.01), <i>IT</i> (.25), <i>TaxRev</i> (-.01)

In the Overall sample as well as in Asia, there is a positive relationship between operational independence (as expressed through MCI and the GMT) and an increase in the GDP Volume Index. Therefore, there is some evidence to support the possibility of an announcement type effect for this variable.

Next, Unemployment will be examined.

Table 5.18.1c--Unemployment GARCH Results: Event Study vs. Total Time Period

Region	<i>Event Study</i>--Significant Variables (Coefficients)	<i>Total Time Period</i>--Significant Variables (Coefficients)
Overall	<i>MCI</i> (1.02), <i>GMT</i> (.03), <i>DD</i> (.98), <i>TaxRev</i> (-.05)	<i>MCI</i> (.03), <i>CBGov</i> (-.03), <i>LCI</i> (-.04), <i>IID</i> (.00), <i>DD</i> (.23), <i>FCD</i> (.31), <i>FXD</i> (.00), <i>Cons</i> (.00), <i>IT</i> (-.21), <i>TaxRev</i> (.00)
Asia	<i>MCI</i> (1.92), <i>GMT</i> (.06), <i>LCBI</i> (-.14), <i>DD</i> (1.20), <i>TaxRev</i> (-.06)	<i>MCI</i> (.04), <i>GMT</i> (.00), <i>CBGov</i> (.01), <i>LCBI</i> (.00), <i>IID</i> (.01), <i>DD</i> (-.01), <i>FCD</i> (.00), <i>FXD</i> (.01), <i>Cons</i> (.00), <i>IT</i> (-.01), <i>TaxRev</i> (.00)
Latin America	<i>LCBI</i> (.30), <i>IID</i> (.42), <i>DD</i> (-1.66), <i>IT</i> (-.62)	<i>MCI</i> (.22), <i>GMT</i> (-.01), <i>CBGov</i> (-.10), <i>LCBI</i> (.34), <i>IID</i> (-.02), <i>DD</i> (.12), <i>FCD</i> (-.22), <i>FXD</i> (.06), <i>Cons</i> (-.01), <i>IT</i> (-.35), <i>TaxRev</i> (.00)
Developed West	<i>GMT</i> (-.08), <i>FCD</i> (1.27), <i>IT</i> (-.70)	<i>MCI</i> (-.01), <i>GMT</i> (.00), <i>CBGov</i> (.01), <i>IID</i> (-.03), <i>DD</i> (.03), <i>FCD</i> (.01), <i>FXD</i> (.12), <i>Cons</i> (.01), <i>IT</i> (-.11), <i>Tax Rev</i> (-.01)

Although there is a small inverse relationship between Unemployment and LCBI in Asia, it is much smaller than the large positive coefficients seen elsewhere (with the

exception of the small negative GMT coefficient in the Developed West). Thus, there seems to be no announcement effect for Unemployment.

Conclusion- There is some evidence for a CBI “Signaling Effect” that spurs economic growth through an increase in operational CBI, but little support for other variables.

5.19 Pre-CBI vs. Post CBI Time Frame Country Study

H19- Among countries that experienced a financial crisis both before and after having an independent central bank, post-CBI time frames fared better than pre-CBI time frames.

The five-year period from the year of financial crisis forward will be examined.

Table 5.19.1a—Inflation Results: Crisis Before vs. After CB Status Change

<i>Independent CB Financial Crisis Period— Significant Variables (Coefficients)</i>	<i>Dependent CB Financial Crisis Period— Significant Variables (Coefficients)</i>
<i>GMT (-.51), CBGov (5.98), DD (12.03), FCD (1.30), FXD (8.65), Cons (-.15), TaxRev (-.16)</i>	<i>MCI (-.78), GMT (.13), CBGov (.11), IID (-.01), FCD (.00), FXD (-.32), Cons (.00), TaxRev (-.03)</i>

Both conditions saw an inverse relationship between operational CBI. However, there was actually a more strongly positive (as expressed by the coefficient) Financial Crisis variable during the Independent CB crisis period. There is also a much stronger positive relationship between the foreign exchange rate regime and inflation during the independent CB periods.

Table 5.19.1b—GDP Results: Crisis Before vs. After CB Status Change

<i>Independent CB Financial Crisis Period— Significant Variables (Coefficients)</i>	<i>Dependent CB Financial Crisis Period— Significant Variables (Coefficients)</i>
<i>MCI (-.61), GMT (.13), CBGov (.96), IID (-2.02), DD (-2.64), FCD (-.34), FXD (1.24), Cons (-.03), TaxRev (-.03)</i>	<i>MCI (-.66), GMT (.07), CBGov (-.16), IID (-.14), FCD (-.12), FXD (.30), Cons (.10), TaxRev (-.08)</i>

Again, neither country status fared well with regards to economic/GDP growth in the period after a financial crisis. However, the Financial Crisis variable is actually more strongly negative in the Independent CB Financial Crisis period, detracting again from the strength of the hypothesis.

Table 5.19.1c—Unemployment Results: Crisis Before vs. After CB Status Change

<i>Independent CB Financial Crisis Period— Significant Variables (Coefficients)</i>	<i>Dependent CB Financial Crisis Period— Significant Variables (Coefficients)</i>
FCD (.55) , FXD (-.79)	IID (-.71), FXD (.43)

There is more meager evidence when examining Unemployment. However, it is of note that the Financial Crisis Dummy variable is quite strongly positively related to Unemployment during the Independent CB phase. Also, a floating exchange rate regime seemed to ease Unemployment during the independent CB period, while it was actually positive during the dependent CB period.

Conclusion- There appears to be little difference in how a country fared during times of financial crisis with an independent or dependent central bank.

5.20 Independent vs. Dependent Central Banks During Times of Financial Crisis

H20- Among those countries that experienced a financial crisis, those countries that had an independent central bank at the time of crisis fared better than those that had a dependent central bank.

Once again the macroeconomic variables of Inflation, GDP Volume Index and Unemployment will be examined. The two time frames of 1997 (Asian Financial Crisis) and 1995 (Latin American Financial Crisis) plus the immediate five years afterward will be observed.

Table 5.20.1a—Inflation Results: Asian Crisis- Independent vs. Dependent CB

<i>Independent CB Financial Crisis Period— Significant Variables (Coefficients)</i>	<i>Dependent CB Financial Crisis Period— Significant Variables (Coefficients)</i>
<i>MCI (3.02)</i> , FCD (1.12) , IT (1.15)	<i>MCI (11.32)</i> , <i>GMT (1.33)</i> , <i>CBGov (-36.58)</i> , IID (-9.18), DD (8.17), FCD (2.22) , Cons (.49)

It appears that Asian countries that had an independent central bank during the 1997 crisis may have fared better than those with a dependent central bank, as shown by the lower MCI and FCD coefficients (they both are still positive, however).

Table 5.20.1b—GDP Results: Asian Crisis- Independent vs. Dependent CB

<i>Independent CB Financial Crisis Period— Significant Variables (Coefficients)</i>	<i>Dependent CB Financial Crisis Period— Significant Variables (Coefficients)</i>
<i>MCI (11.61), GMT (.72), CBGov (-113.22), IID (-3.91), FXD (-5.73), TaxRev (.91)</i>	<i>MCI (23.39), GMT (2.06), CBGov (-50.81), FCD (1.06), Cons (.89), IT (6.99)</i>

Several items here are worthy of note. First, the operational CBI variable (MCI) coefficient is much higher under the dependent than independent side. Second, the Financial Crisis Dummy variable is not present in the GDP results for independent central banks. Third, the Inflation Targeting variable is strongly positive under the dependent CB side, as is TaxRev under the independent CB side.

Table 5.20.1c—Unemployment Results: Asian Crisis- Independent vs. Dependent CB

<i>Independent CB Financial Crisis Period— Significant Variables (Coefficients)</i>	<i>Dependent CB Financial Crisis Period— Significant Variables (Coefficients)</i>
<i>CBGov (7.39), IT (.42)</i>	<i>CBGov (1.41), Cons (-.03)</i>

It would appear that excessive central bank governor turnover is much more strongly related to unemployment under an independent central bank regime than a dependent one.

Next, Latin America will be examined through the lens of the 1995 Latin American Financial Crisis.

Table 5.20.2a—Inflation Results: Latin American Crisis- Independent vs. Dependent CB

<i>Independent CB Financial Crisis Period— Significant Variables (Coefficients)</i>	<i>Dependent CB Financial Crisis Period— Significant Variables (Coefficients)</i>
<i>MCI (2.03), CBGov (6.77), IID (-1.04), FCD (.84), FXD (9.84), Cons (-.36), IT (-1.81), TaxRev (-.18)</i>	<i>MCI (-4.81), CBGov (4.90), FXD (4.09), Cons (-.21), IT (-1.73)</i>

It is curious to note that operational independence (as noted by a negative MCI coefficient and smaller CBGov coefficient) seemed to be more effective at curbing inflation under a dependent central bank regime. Moreover, the exchange rate regime

variable is much more strongly positive under the independent CB status than the dependent central bank type regime.

Table 5.20.2b—GDP Results: Latin American Crisis- Independent vs. Dependent CB

<i>Independent CB Financial Crisis Period— Significant Variables (Coefficients)</i>	<i>Dependent CB Financial Crisis Period— Significant Variables (Coefficients)</i>
<i>GMT (.74), CBGov (-5.19), DD (-8.91), FXD (-5.08), Cons (-.03)</i>	Unavailable

Table 5.20.2c—Unemployment Results: Latin American Crisis- Independent vs. Dependent CB

<i>Independent CB Financial Crisis Period— Significant Variables (Coefficients)</i>	<i>Dependent CB Financial Crisis Period— Significant Variables (Coefficients)</i>
IT (1.01)	FXD (-3.31)

It is difficult to conclude that Latin American countries that had an independent central bank at the time of crisis fared better than its dependent central bank neighbors.

From these statistics, it is somewhat difficult to assess the effectiveness of central bank independence as it relates to the most recent regional currency crisis of 1995 and 1997.

Conclusion- There is little support that an independent central bank assisted in enduring a financial crisis more than its dependent central bank neighbors (the only support is for Inflation during the Asian Crisis).

5.21 Summary of Hypotheses Support

The following table is a summary of the support for the twenty hypotheses put forth in this section.

Table 5.21.1- Hypothesis Support Summary

H1 (Inflation) – Strong (via operational independence, especially post-CBI; legal CBI benefits are limited, however)
H2 (Money Supply) – Moderate (mainly through the CBGov variable)
H3 (Interest Rates) – Fair (primarily through operational independence post-CBI)
H4 (Foreign Exchange Rates) – Strong (via operational CBI and to a lesser extent, legal CBI)
H5 (Direct Investment in the Economy) – Strong (via operational CBI and to a lesser extent, legal CBI)
H6 (Time Deposits) – No Support
H7 (Unemployment) – Limited (evidence found only for Developed West; IT beneficial)
H8 (Exports) – Moderate (both operational and legal CBI; IT also favorable)
H9 (Economic Growth) – Fairly Strong (both operational and legal CBI)
H10 (Corruption Levels) – Moderate (operational independence effectiveness at low levels of corruption and moderately effective at medium levels of corruption)
H11 (Legal System) – Moderate (in favor of English Common Law)
H12 (Inflation Targeting) – Partial (mainly for GDP growth)
H13 (Level of Constraints) – Moderate (although legal and/or operational support does appear at other levels)
H14 (Economic Development) – Little Support (most of the benefit occurs in the mid-level stages of development)
H15 (Tax Revenue) – Partial (benefits can be seen at any level of Tax Revenue)
H16 (Government Consumption) – Partial (Inflation and to a lesser extent Unemployment)
H17 (Operational Independence) – Partial (more support for GDP growth)
H18 (“Signaling Effect”) – Partial (more support for GDP growth through an increase in operational independence)
H19 (Pre vs. Post CBI Time Frame) – No Support
H20 (Independent vs. Dependent Central Bank at Time of Crisis) – Little (Inflation during the Asian Crisis).

As can be seen from the above table, it is apparent that central bank independence seems to be quite effective at creating a stable macroeconomic backdrop from which a country can potentially prosper. Central bank independence seems to be effective at curbing inflation and supporting the exchange rate, perhaps ultimately assisting with in the economic growth of the country. However, it is also clear that operational independence is a much stronger tool than purely legal independence.

Chapter 6- Conclusion

The idea of central bank independence has been at the forefront of financial market liberalization since the late 1980s and is still quite relevant (de Haan et al, 2008). Many countries around the world, both developed and developing, have granted autonomy to their central banks in order to achieve monetary and fiscal objectives. There are several general motives put forth in the literature as to what macroeconomic and political events drove countries to grant this authority. However, the results in the early literature on the subject, with some exceptions, seemed to come to a common conclusion that an increase in central bank independence would lead to a lowering in inflation rates. However, much of the early work was done in the early to mid-1990s, before many of the developing and developed world's central banks had achieved independence.

This study has been an attempt to update and expand upon empirical studies using a current database, more appropriate methodology and an increased perspective of the potential gains from central bank independence. One of the main objectives of this study has been to examine the nuances of operational vs. legal central bank independence to quantify the true benefits of central bank independence. One primary objective of this study has been to examine countries that achieved Central Bank Independence in the 1990s and analyze the results of this independence; given the fact that now sufficient post-independence data (ten to fifteen years or more) is available. Some of the unique contributions of this study are that it not only looks at Inflation (which is the macroeconomic variable most associated with CBI), but also such factors as Interest Rates, Direct Investment, Unemployment and Exchange Rates. It examines the carefully chosen survey of countries from an overall, regional (Asian, Latin American, and

Developed West), and categorical level. It also examines such issues as Operational vs. Legal Central Bank Independence (or “de jure” and “de facto” independence) and attempts to quantify these levels of autonomy. It also has included and attempted to quantify such pertinent issues as Corruption levels, the Constraint Level (i.e. “checks and balances”), the legal system of a country, its level of economic development and fiscal stance, all of which have tested the validity and practical implications of central bank independence during both “ordinary” and “extraordinary” times.

A summary of the major findings of this study are as follows:

Table 6.1- Summary of Major Findings

1. Although CBI perhaps does not directly lead to growth in many cases, it does seem to help lay the foundation for it to occur (through an increase in direct investment, stable inflation and foreign exchange rates, lower unemployment, etc.).
2. Operational independence appears to be as equally significant (if not perhaps more so in some cases) as legal independence in relation to CBI’s potential outcomes.
3. Central bank independence seems to be more strongly related to the above hypotheses in countries of Asia and Latin America (countries with “middle levels” of political constraints), although support for the benefits of CBI in the Developed West does exist.
4. Legal systems, the system of “checks and balances”, corruption levels, and economic development are all factors in determining the success of CBI.
5. Inflation Targeting has been an effective way to stabilize the economy as well; however, it only appears to work in countries that have firmly adopted it and/or given it adequate time to operate.
6. Financial crises are detrimental to all aspects of an economy and can provoke damage in all regions and country categories.
7. There is a good deal of Causality (both unidirectional and bidirectional) regarding the measures of CBI (both operational and legal) and the macroeconomic dependent variables.
8. Overall, Central Bank Governor Turnover does appear to be a relatively accurate proxy of CBI (with more turnover indicating less independence); however, this proxy in and of itself must be used with caution (and perhaps could even be a positive event in developed countries).
9. There does appear to be a good bit of hysteresis in several variables, including Inflation, Exports, GDP Growth and Unemployment.
10. Overall, it seems that CBI has been effective, but is perhaps just one element (albeit a powerful one) among a more comprehensive package of reforms.

There seems to be a general indication that central bank independence, although a relatively powerful tool, is not a panacea. Even after the passing of laws creating an independent central bank in a given country, it by no means will serve to cure all monetary and fiscal ills of that country. Additionally, it will not prevent financial crises from occurring or help it weather the crisis to any significant degree. However, this is not to say that central bank independence is not a worthy endeavor. Although by itself, CBI does not appear to have the power to cure high levels of inflation (see Cukierman, 1994), it does seem to assist in maintaining lower rates of inflation (partially through the control of the Money Supply) and does assist somewhat in attaining other macroeconomic goals. However, these benefits are experienced to different degrees depending on the region and other aspects of a country's legal system, corruption levels and levels of constraints (i.e. system of "checks and balances"). In this regard, this study concurs somewhat with the "middle constraint" theory of Acemoglu et al (2008). In addition, many of these countries seem to be the traditionally politically left leaning countries that need CBI for purposes of international credibility (Hicks, 2004).

Although CBI by itself may have a limited effect on many macroeconomic outcomes, it is probable that CBI is part of a wider range of reforms, all seeking to stabilize and modernize the economy of the respective country. Perhaps in this way, successful CBI may serve as a key part (although only a part) of a package of reforms or, in a similar way, may "pave the way" for reforms to come. Again, the practical implications of these results for both academics and policymakers is that the concept of Central Bank Independence has continued to evolve and is still a very relevant issue for the economy of the first decade of the 21st century and beyond. Its benefits can be moved

beyond simply a potential measure to counteract Inflation, but to perhaps influence the entire spectrum of a nation's economy. Although its benefits are not seen "overnight", it is an unambiguous and distinct measure for a nation to take to improve the long term prospects of an economy and its population. Through the broader set of macroeconomic variables chosen here, as well as the more narrow set focused on in later hypotheses, it is clear that the gains from Central Bank Independence can be felt by a nation's residents (through more stable prices, greater economic growth and less unemployment) and be a definite step for improving the country's overall level of welfare.

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Appendices

Table A1-- Central Bank Legislation Dates

Country	Initial Legal Reform	Last Amended
Argentina	1992	
Brazil		1988
Chile	1989	
China	2003	
England	1998	
India	1949	1999
Indonesia	1999	2004
Japan	1997	2001
Korea	1998	
Malaysia	1989	1994
Mexico	1955	1993
New Zealand	1989	2003
Paraguay	1995	
Portugal	1998	2001
Philippines	1993	
Peru	1993	
Spain	1994	1998
Uruguay	1995	
United States	1959	2000
Venezuela	1992	
Thailand		2000

A2-- Regional Groupings of Sample Countries

Asia	Latin America	Developed West
China	Argentina	England
India	Brazil	New Zealand
Indonesia	Chile	Portugal
Japan	Mexico	Spain
Korea	Paraguay	United States
Malaysia	Peru	
Philippines	Uruguay	
Thailand	Venezuela	

Table A3-- Country and Crisis Years (Beginning Date)

Country	Year of Crisis
Argentina	1982, 1995, 2001, 2008
Brazil	1982, 1995, 1999, 2008
Chile	1982, 2008
China	1997, 2008
England	1972, 2008
India	1997, 2008
Indonesia	1997, 2008
Japan	1997, 2008
Korea	1997, 2008
Malaysia	1997, 2008
Mexico	1982, 1995, 2008
New Zealand	1992, 2008
Paraguay	1987, 2008
Paraguay	1997, 2008
Portugal	1992, 2008
Philippines	1984, 2008
Philippines	1997, 2008
Peru	1998, 2008
Spain	1992, 2008
Uruguay	1982, 2002, 2008
United States	2008
Venezuela	1984, 1987, 1995, 2008
Thailand	1997, 2008

A4-- Legal Systems

English Common Law	French Civil Law	Germanic Law
England	Argentina	Japan
India	Brazil	Korea
Malaysia	Chile	
New Zealand	Indonesia	
Thailand	Mexico	
United States	Paraguay	
	Peru	
	Philippines	
	Portugal	
	Spain	
	Uruguay	
	Venezuela	

A5-- Inflation Targeting

High	Medium	Low
England	United States	Uruguay
Chile	Spain	Paraguay
Korea	Portugal	Malaysia
Brazil	Peru	Venezuela
Thailand	Philippines	India
Mexico	Japan	China
New Zealand	Indonesia	Argentina

A6-- Corruption

Low	Medium	High
England	Uruguay	China
United States	Malaysia	Argentina
Chile	Korea	Philippines
Japan	Peru	India
Portugal	Brazil	Venezuela
Spain	Mexico	Indonesia
New Zealand	Thailand	Paraguay

A7-- Constraints

High	Medium High	Medium	Medium Low	Low
New Zealand	Japan	Argentina	China	Indonesia
England	United States	Brazil	Peru	Paraguay
		Chile	Venezuela	
		Korea		
		India		
		Malaysia		
		Mexico		
		Philippines		
		Portugal		
		Spain		
		Uruguay		

A8-- Operational Independence

High	Medium High	Medium	Medium Low	Low
United States	Philippines	Argentina	China	Korea
Peru	Indonesia	Chile	Spain	Brazil
Malaysia	Portugal	England	Uruguay	Japan
Mexico	Paraguay	Thailand	New Zealand	India
	Venezuela			

A9-- Government Consumption (as a % of GDP)

High	Medium High	Medium	Medium Low	Low
England	Brazil	Uruguay	Venezuela	Philippines
New Zealand	Spain	China	Thailand	Argentina
United States	Japan	Chile	India	Mexico
Portugal	Malaysia	Korea	Peru	Indonesia
				Paraguay

A10-- Tax Revenue (as a % of GDP)

High	Medium High	Medium	Medium Low	Low
England	Uruguay	Brazil	Chile	Peru
New Zealand	Portugal	Venezuela	Thailand	India
United States	Japan	Korea	Indonesia	Argentina
Spain	Malaysia	Mexico	Philippines	Paraguay
				Chine

A11-- Country Development

High	Medium High	Medium	Medium Low	Low
United States	Spain	Argentina	Malaysia	Paraguay
New Zealand	Venezuela	Chile	Peru	Indonesia
Japan	Portugal	Uruguay	Brazil	China
England	Korea	Mexico	Thailand	India
				Philippines

A12-- Central Bank Status During Financial Crises

Country	Year of Crisis	CB Status
Argentina	1982	Dep
Argentina	1995	Ind
Argentina	2001	Ind
Brazil	1982	Dep
Brazil	1995	Semi
Brazil	1999	Semi
Mexico	1982	Dep
Mexico	1995	Ind
Paraguay	1987	Dep
Paraguay	1997	Ind
Philippines	1984	Dep
Philippines	1997	Ind
Uruguay	1982	Dep
Uruguay	2002	Ind
Venezuela	1984	Dep
Venezuela	1987	Dep
Venezuela	1995	Ind